

## SOLUTIONS CHEMISTRY

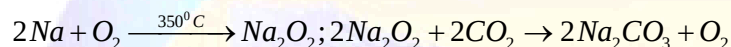
1. D.

Due to the larger size of hydrated  $\text{Li}^+$  ion conductivity decreases.

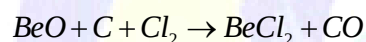
2. A

As metallic character of element attached to oxygen atom increases, the difference between the electronegativity values of element and oxygen increases and thus basic character of oxides increases and vice versa, Hence, the increasing correct order of basic nature is  $\text{Al}_2\text{O}_3 < \text{MgO} < \text{Na}_2\text{O} < \text{K}_2\text{O}$

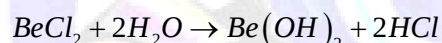
3. C



4. D

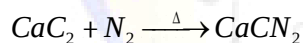
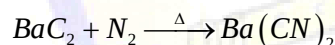


(y)

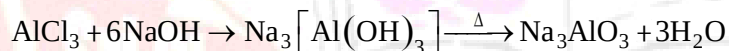
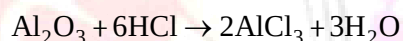


(z)

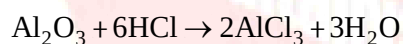
5. C



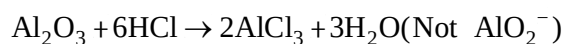
6. A)



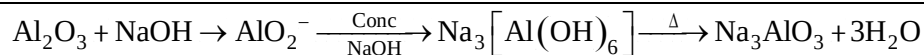
B)



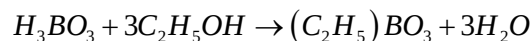
C)



D)



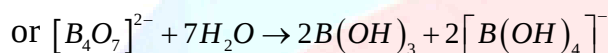
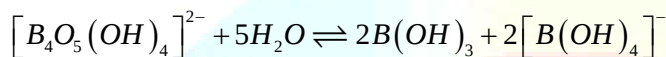
7. C



Methyl borate

(green flame)

8. C



9. ABD

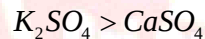
The solubility of the alkali metal carbonates increases down the group from Li to Cs. This is because of the fact that down the group with increasing size of cation the lattice energy as well as hydration energy also decrease but the change in lattice energy is more as compare to that of hydration energy. lithium hydrogen carbonate is not obtained in the solid form while all other elements form solid hydrogen carbonates.

b)  $\text{K}_2\text{CO}_3$  cannot be prepared by solvay process because intermediate formed,  $\text{KHCO}_3$  is appreciable soluble in water.

c)  $\text{Li}_2\text{CO}_3$  and  $\text{MgCO}_3$  both are not thermally stable.

d) Fact ref. NCERT

10. ABD



Water soluble

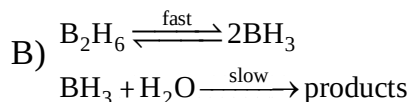
Insoluble

11. AC

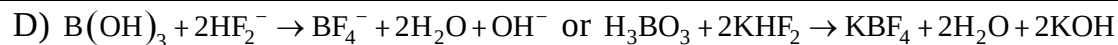
Conceptual

12. ABCD

A) Boron do not contain vacant d-orbitals will Al contain vacant d-orbitals.



C) Due to steric hindrance in 2,6-dimethyl pyridine.



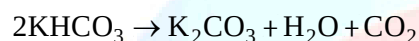
13. 2

Out of s-block elements Be and Mg do not impart any colour to flame.

14. 7

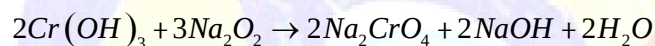
8 moles of a mixture of  $Li_2CO_3$  and  $KHCO_3$  contain 3:1 ratio means it contains 6 moles of  $Li_2CO_3$  and 2 moles  $KHCO_3$ .

On heating,  $Li_2CO_3 \rightarrow Li_2O + CO_2$

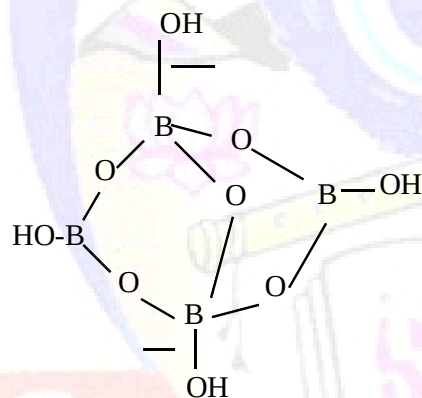


6 moles of  $CO_2$  will be evolved on heating 6 moles of  $Li_2CO_3$  and 1 mol. of  $CO_2$  is evolved from 2 moles of  $KHCO_3$

15. 3



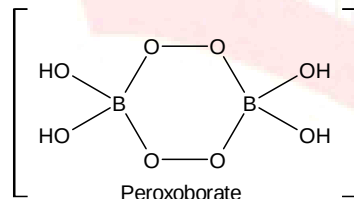
16. 7



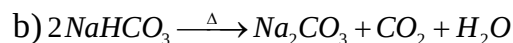
17. 6



18. 8

19.  $(a \rightarrow r, s); (b \rightarrow r); (c \rightarrow pqrs); (d \rightarrow q)$

a) solvay process is used for the manufacture of  $\text{NaHCO}_3$  and  $\text{Na}_2\text{CO}_3$ ; the raw material is  $\text{NaCl}$



c)  $\text{Na}_2\text{O} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH}$ ;  $\text{NaOH}$  being basic turns red litmus blue.

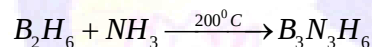
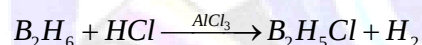
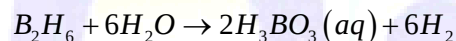
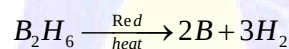
$\text{Na}_2\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2\text{O}_2$ ;  $\text{NaOH}$  being basic turns red litmus blue but  $\text{H}_2\text{O}_2$  being agent bleaches coloured litmus.

$\text{NaHCO}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NaOH} + \text{H}_2\text{CO}_3$  solution is alkaline and turns red litmus blue.

d)  $\text{Na}_2\text{O}_2 + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3$ ;  $2\text{Na}_2\text{O}_2 + 2\text{CO} \rightarrow 2\text{Na}_2\text{CO}_3 + \text{O}_2$ , Hence, it absorbs  $\text{CO}$  and  $\text{CO}_2$  and liberates oxygen.

20.  $(a \rightarrow pq); (b \rightarrow p, s); (c \rightarrow p, r); (d \rightarrow p)$

Reactions are  $\rightarrow$



(Borazole)

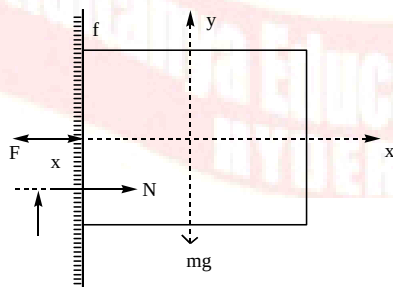
(Inorganic benzene borazine)

## PHYSICS

21. Compare the standard moment of inertias of the bodies

$$22. \quad I_A = \frac{(8\pi r^2 t) r^2}{2}, I_B = \left(8\pi 16r^2 \frac{t}{4}\right) \frac{16r^2 / e}{2}, I_B = 64I_A$$

23. The different forces on the block are shown in figure.



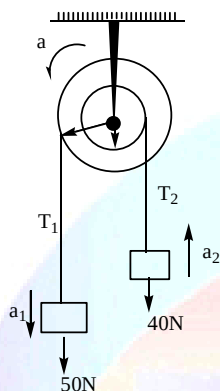
For equilibrium,  $f = mg$

And normal reaction,  $N = F$

Torque due to  $F = 0$  ( $F$  passes through centre)

As the body is in equilibrium and hence, torque due to friction about centre of mass is equal to torque due to normal reaction about centre of mass.

24. From figure shown



$$50 - T_1 = 5(2\alpha)$$

$$T_2 - 40 = 4(\alpha)$$

$$T_1(2) - T_2(1) = 4\alpha$$

From (1), (2) and (3),

$$2(50 - 10\alpha) - (40 + 4\alpha) = 4\alpha$$

$$\Rightarrow 100 - 20\alpha - 40 - 4\alpha = 4\alpha$$

$$\Rightarrow 28\alpha = 60$$

$$\Rightarrow \alpha = \frac{60}{28} = 2.1 \text{ m/s}^2$$

25. For equilibrium of rod considering limiting friction we use

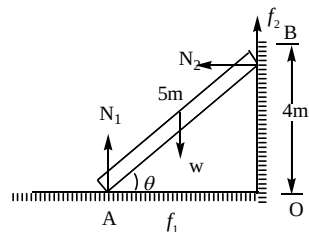
$$f_1 = N_2$$

$$\mu N_1 = N_2$$

$$-w + N_1 + f_2 = 0$$

$$N_1 + \mu N_2 = w$$

$$N_1 + \mu^2 N_1 = W$$



Balancing all torque about point A, we use

$$-w\left(\frac{3}{2}\right) + f_2(3) + N_2(4) = 0$$

$$\Rightarrow \frac{-3W}{2} + 3f_2 + 4N_2 = 0$$

$$\Rightarrow 3\mu N_2 + 4N_2 = \frac{3}{2}(w)$$

$$\Rightarrow \mu N_1(3\mu^2 + 4\mu) = \frac{3}{2}N_1(1 + \mu^2)$$

$$2(3\mu^2 + 4\mu) = 3 + 3\mu^2$$

$$6\mu^2 + 8\mu = 3 + 3\mu^2$$

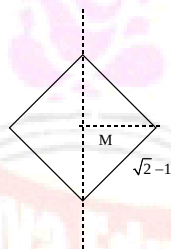
$$3\mu^2 + 8\mu - 3 = 0$$

$$\Rightarrow 3\mu^2 + 9\mu - \mu - 3 = 0$$

$$3\mu(\mu + 3) - 1(\mu + 3) = 0$$

$$\mu = \frac{1}{3} \text{ (As } \mu \neq -3)$$

26. As



$$I = \frac{1}{4} I_{\text{full square}}$$

$$= \frac{1}{4} M_{\text{full}} l_{\text{full}}^2 / 12$$

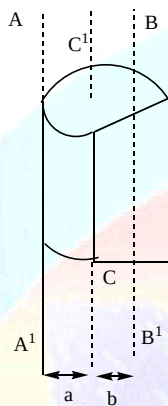
$$I = \frac{1}{4} (4m) \frac{(\sqrt{2}l)^2}{12}$$

$$I = \frac{MI^2}{6}$$

27. Using parallel axis theorem

$$I_{CC^1} = I_B - Mb^2$$

$$I_{AA^1} = I_{CC^1} + Ma^2$$



28.  $I = (24M)R^2 + 24\left(\frac{MI^2}{3}\right)$

$$= 24MI^2 + 8MI^2$$

$$I = 32MI^2$$

29. Applying net torque =  $\sum \vec{r}_i \times \vec{F}_i$

30. B, C

From figure we have

$$I_y = \frac{1}{2}mR^2 \text{ \& } I_x = I_z = \frac{3}{4}mR^2$$

31. Case 1:

$$FR = I\alpha$$

$$40 \times 0.25 = 4\alpha$$

$$\alpha = \frac{5}{2} \text{ rad/s}^2$$

$$\omega_1 = \sqrt{2\alpha\theta}$$

$$= \sqrt{2\alpha \frac{l}{r}}$$

$$= \sqrt{2 \times \frac{5}{2} \times \frac{5}{0.25}}$$

$$= 10 \text{ rad/s}$$

Case 2:

$$\text{Mass of the block is } m = \frac{40}{10} = 4 \text{ kg}$$

For conservation of energy ,

$$[\text{Loss in GPE of 4 kg mass}] = [\text{Gain in KE of 4kg mass}] + \{\text{Gain in RKE of Cylinder}\}$$

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$\text{As } v = R\omega ,$$

$$mgh = \frac{1}{2}m(R\omega)^2 + \frac{1}{2}I\omega^2$$

$$(4)(10)(5) = \frac{1}{2}(4)\left(\frac{1}{4}\right)^2 \omega^2 + \frac{1}{2}(4)\omega^2$$

$$\Rightarrow \omega_2 = \frac{40}{\sqrt{17}} \text{ rad / s}$$

32. As the sphere is rotating uniformly the tangential acceleration of any particle on sphere will be zero and all particles will only have normal acceleration.

33.  $N2L = \text{block}$

$$T - 50 \times 10 = 50 \times 2 \Rightarrow T = 600 \text{ N}$$

$$8 \times \frac{2}{0.25} = F \times 0.10 - 0.25 \times 600$$

$$\Rightarrow 0.10F = 8 + 150$$

34. Just after the thread is burnt

$$\text{Spring force} = 600 \times 0.01 = 6 \text{ N remains same}$$

$$\text{Torque equation} = I\alpha = 6 \times 0.60 + 20 \times 0.10 (\text{c.w})$$

$$\Rightarrow 2\left(\frac{12}{12} + 0.01\right)\alpha = 5.6$$

$$\Rightarrow \alpha = \frac{5.6 \times 12}{2 \times 1.12}$$

$$\text{Applying } N2L = 6 + 20 - N = 2 \times 0.10 \times \frac{5.6 \times 12}{2 \times 1.12}$$

$$\Rightarrow N = 20$$

35. Torque equation (discus)

$$(I_1 + I_2)\alpha = r_1 T$$

$$N 2L = \text{block}$$

$$mg - T = mr_1 \alpha$$

36. A rod of length  $l$  is pivoted about an end. Find the moment of inertia of the rod about this axis if the linear mass density of rod varies as  $\rho = ax^2 + b \text{ kg/m}$

As mass of the rod varies with its length, here we cannot use the expression

$$\frac{Ml^2}{3} \text{ which is only used for uniformly distributed mass along the length of rod pivoted}$$

at an end. In this case we consider an elemental length  $dx$  from the axis at a distance  $x$ . Let its mass be  $dm$ , where

$$dm = \rho dx$$

$$= (ax^2 + b) dx$$

During rotation of rod this  $dm$  revolves in a circle of radius  $x$ , hence its moment of inertia  $dl$  is given as

$$dl = dm x^2$$

$$= (ax^2 + b) x^2 dx$$

The moment of inertia of the whole rod is given by integrating the above expression

$$\text{within limits from } 0 \text{ to } l \text{ as } I = \int_0^l (ax^2 + b) x^2 dx$$

$$\text{Or } = \left[ \frac{ax^5}{5} + \frac{bx^3}{3} \right]_0^l$$

$$\text{Or } = \frac{al^5}{5} + \frac{bl^3}{3}$$

$$37. \text{ i) } \frac{Ml^2}{3} \alpha_1 = mg \frac{l}{2} \cos 53^\circ$$

$$\text{ii) } -\frac{Ml^2}{3} \alpha_1 = mg \frac{l}{2} \cos 53^\circ - ma_0 \frac{l}{2} \sin 53^\circ$$

$$\Rightarrow a_0 \sin 53^\circ = 2g \cos 53^\circ$$

$$a_0 = 2g \frac{3}{4} = \frac{3}{2}g$$

$$38. \quad I_3 = I_1 = I_4 = \frac{2}{3}MR^2$$

$$I_2 = \frac{2}{3}MR^2 + MR^2 = \frac{5}{3}MR^2$$

$$\Rightarrow I_1 + I_2 + I_3 + I_4 = 3\frac{2}{3}MR^2 + \frac{5}{3}MR^2$$

$$= \frac{11}{3}MR^2 = \frac{11}{2}I_1$$

$$39. \quad dm = \lambda R d\theta$$

$$dI_z = R^2 \sin^2 \theta dm \Rightarrow I_x = \int R^2 \sin^2 \theta dm$$

$$dI_y = R^2 \cos^2 \theta dm \Rightarrow I_y = \int R^2 \cos^2 \theta dm$$

$$dI_z = R^2 dm \Rightarrow I_z = \int R^2 dm$$

40. Applying rotational equation and then applying torque equation about the respective pivots

### MATHS

$$41. \quad \text{Given } L = \left(1 - \frac{4}{3^2}\right) \left(1 - \frac{4}{4^2}\right) \left(1 - \frac{4}{5^2}\right) \dots$$

$$= \left[ \left(1 - \frac{2}{3}\right) \left(1 - \frac{2}{4}\right) \left(1 - \frac{2}{5}\right) \left(1 - \frac{2}{6}\right) \dots \right] \times \left[ \left(1 + \frac{2}{3}\right) \left(1 + \frac{2}{4}\right) \left(1 + \frac{2}{5}\right) \dots \right]$$

$$= \left[ \frac{1}{3} \times \frac{2}{4} \times \frac{3}{5} \times \frac{4}{6} \times \dots \right] \times \left[ \left(\frac{5}{3}\right) \left(\frac{6}{4}\right) \left(\frac{7}{5}\right) \dots \right]$$

$$= \frac{1}{3} \times \frac{2}{4} = \left(\frac{1}{6}\right)$$

(Remembering terms in two braces are reciprocal of each other)

$$\text{Also, } M = \frac{2^3-1}{2^3+1}, \frac{3^3-1}{3^3+1}, \frac{4^3-1}{4^3+1} \dots$$

$$\left( \text{Now } \frac{x^3-1}{x^3+1} = \frac{(x-1)(x^2+x+1)}{(x+1)(x^2-x+1)} \right)$$

$$M = \left( \frac{2-1}{2+1}, \frac{3-1}{3+1}, \frac{4-1}{4+1}, \frac{5-1}{5+1} \dots \right) \times$$

$$\left( \frac{2^2+2+1}{2^2-2+1} \right) \left( \frac{3^2+3+1}{3^2-3+1} \right) \left( \frac{4^2+4+1}{4^2-4+1} \right)$$

$$M = \lim_{n \rightarrow \infty} \frac{1 \times 2}{(n+1)(n)} \frac{n^2 + n + 1}{3} = \frac{2}{3};$$

$$N = \prod_{n=1}^{\infty} \frac{\left(1 + \frac{1}{n}\right)^2}{\left(1 + \frac{2}{n}\right)} = \lim_{n \rightarrow \infty} \prod \frac{(n+1)^2 / n^2}{(n+2)/n}$$

$$= \lim_{n \rightarrow \infty} \prod_{r=1}^n \left( \frac{(r+1)^2}{r(r+2)} \right)$$

$$= \lim_{n \rightarrow \infty} \left( \prod_{r=1}^n \frac{r+1}{r} \cdot \frac{r+1}{r+2} \right)$$

$$= \lim_{n \rightarrow \infty} \left( \frac{2}{1} \times \frac{3}{2} \times \frac{4}{3} \times \dots \times \frac{n+1}{n} \right) \times \left( \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{n+1}{n+2} \right)$$

$$= \lim_{n \rightarrow \infty} \left( \frac{2(n+1)}{n+2} \right) = 2$$

$$L^{-1} + M^{-1} + N^{-1} = 6 + \frac{3}{2} + \frac{1}{2} = 8$$

$$42. \quad \lim_{n \rightarrow \infty} \left[ \left\{ 4 \sum_{r=0}^n (r+1)(r+2)(r+3) \right\}^{\frac{1}{4}} - n \right]$$

$$\lim_{n \rightarrow \infty} \left[ \left\{ \frac{4(n+1)(n+2)(n+3)(n+4)}{4} \right\}^{\frac{1}{4}} - n \right]$$

$$\lim_{n \rightarrow \infty} \left[ \left\{ (n+1)(n+2)(n+3)(n+4) \right\}^{\frac{1}{4}} - n \right]$$

$$\lim_{n \rightarrow \infty} \left[ n \left\{ \left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \left(1 + \frac{3}{n}\right) \left(1 + \frac{4}{n}\right) \right\}^{\frac{1}{4}} - n \right]$$

$$\lim_{n \rightarrow \infty} \left[ \frac{\left\{ \left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \left(1 + \frac{3}{n}\right) \left(1 + \frac{4}{n}\right) \right\}^{1/4} - 1}{\frac{1}{n}} \right]$$

$$\lim_{x \rightarrow 0} \left[ \frac{\{(1+x)(1+2x)(1+3x)(1+4x)\}^{1/4} - 1}{x} \right]$$

$$\lim_{x \rightarrow 0} \left[ \frac{(1+10x+35x^2+50x^3+24x^4)^{1/4} - 1}{x} \right] = \frac{10}{4} = \frac{5}{2} \text{ (By L.H.Rule)}$$

43.  $2 + f(x)f(y) = f(x) + f(y) + f(xy); y \rightarrow x$

$$2 + [f(x)]^2 = 2f(x) + f(x^2)$$

put  $x=1$ ;

$$\Rightarrow 2 + f(1)^2 = 3f(1)$$

$$\Rightarrow (f(1))^2 - 3f(1) + 2 = 0$$

$$\Rightarrow (f(1)-1)(f(1)-2) = 0$$

$$\Rightarrow f(1)=1 \text{ or } f(1)=2, \text{ Re placing } y \text{ by } \frac{1}{x}$$

$$\Rightarrow 2 + f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + f(1)$$

$$\Rightarrow 2 + f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) + f(1)$$

If  $f(1)=2$

$$\Rightarrow f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right)$$

$$\Rightarrow f(x) = 1 \pm x^2 \text{ but } f(2)=5$$

$$\Rightarrow 1 \pm (2)^n$$

$$\Rightarrow n=2 \text{ and } f(x) = 1 + x^2$$

If  $f(1)=1$ , Then  $2 + f(x)(1) = f(x) + 1 + f(x)$

$$\Rightarrow f(x) = 1 \forall x \in \mathbb{R}$$

But  $f(2)=5$

$$\Rightarrow f(1) \neq 1$$

$f(x) = 1 + x^2$  which is a continuous function

$$\lim_{x \rightarrow 5} f(5) = 1 + 5^2 = 26$$

44.  $\lim_{x \rightarrow \infty} (a^x + e^x)^{\frac{1}{x}} = a, a > 0$

$$a = \lim_{x \rightarrow \infty} (a^x + e^x)^{1/x}$$

$$\Rightarrow \ln a = \lim_{x \rightarrow \infty} \frac{\ln(a^x + e^x)}{x}$$

This gives as  $\infty / \infty$  format.

Hence L.H rule, we get  $\lim_{x \rightarrow \infty} Lna = \left[ \frac{a^x \ln a + e^x}{a^x + e^x} \right]$

$$\lim_{x \rightarrow \infty} Lna = \left[ \frac{a^x \ln a + e^x}{a^x + e^x} \right]$$

$$\Rightarrow a \in (e, \infty)$$

$$\begin{aligned} 45. \quad \sum_{r=1}^n t_r &= \sum_{r=1}^n \frac{r}{1-3r^2+r^4} = \sum_{r=1}^n \frac{r}{(r^2-1)^2-r^2} \\ &= \sum_{r=1}^n \frac{1}{2} \frac{(r^2+r-1)-(r^2-r-1)}{(r^2-r-1)(r^2+r-1)} \\ &= \frac{1}{2} \sum_{r=1}^n \left[ \frac{1}{r^2-r-1} - \frac{1}{r^2+r-1} \right] = \frac{1}{2} \left[ -1 - \frac{1}{(n^2+n-1)} \right] \\ &= \frac{1}{2} \left[ \frac{-n^2-n+1-1}{n^2+n-1} \right] = \frac{1}{2} \left[ \frac{-n^2-n}{n^2+n-1} \right]; \lim_{n \rightarrow \infty} \sum_{r=1}^n t_r = -\frac{1}{2} \end{aligned}$$

$$46. \quad \text{Here } \left| \frac{3}{\pi} \tan^{-1} 2x \right| > 1$$

$$\Rightarrow |\tan^{-1}(2x)| > \frac{\pi}{3} \Rightarrow |2x| > \sqrt{3}$$

$$\text{For } |2x| > \sqrt{3}, \left| \left( \frac{3}{\pi} \tan^{-1} 2x \right) \right| > 1$$

$$\Rightarrow \left( \frac{3}{\pi} \tan^{-1} 2x \right)^{2\pi} \rightarrow \infty \text{ as } n \rightarrow \infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{\left( \frac{3}{\pi} \tan^{-1}(2x) \right)^{2\pi} + 5} = 0$$

$$\Rightarrow f(x) = 0 \text{ for } |2x| > \sqrt{3}$$

$$47. \quad \lim_{x \rightarrow 0} \left[ f(x) + \ln \left( 1 - \frac{1}{e^{f(x)}} \right) - \ln(f(x)) \right]$$

$$\lim_{x \rightarrow 0} \left[ f(x) + \ln(e^{f(x)} - 1) - \ln e^{f(x)} - \ln(f(x)) \right]$$

$$\lim_{x \rightarrow 0} \left[ \ln(e^{f(x)} - 1) - \ln f(x) \right] = \lim_{x \rightarrow 0} \left[ \ln \left( \frac{e^{f(x)} - 1}{f(x)} \right) \right]$$

$$= 0 \text{ if } f(x) \rightarrow 0 \text{ as } x \rightarrow 0$$

As  $f(x)$  is defined  $\forall x \in R$

$$f(0) = 0$$

48.  $\lim_{x \rightarrow 0} \left( \sin^2 \left( \frac{\pi}{2 - ax} \right) \right)^{\sec^2 \frac{\pi}{2 - bx}}$

$$\text{Let } k = \lim_{x \rightarrow 0} \left( \sin^2 \left( \frac{\pi}{2 - ax} \right) \right)^{\sec^2 \frac{\pi}{2 - bx}}$$

Taking  $\ln$  both sides,  $\ln k$

$$= \sec^2 \frac{\pi}{2 - bx} \ln \left( \sin^2 \left( \frac{\pi}{2 - ax} \right) \right)$$

$$= \ln k = \lim_{x \rightarrow 0} \left[ \frac{\ln \left( \sin^2 \left( \frac{\pi}{2 - ax} \right) \right)}{\cos^2 \left( \frac{\pi}{1 - bx} \right)} \right]$$

$$\text{By L.H rule, we get } \ln k = -\frac{a^2}{b^2}$$

$$\Rightarrow k = e^{\frac{a^2}{b^2}}$$

49. a, d

$$f(1+0) = \lim_{h \rightarrow 0} \{1 + h - 1\} - \{1 + h\} = \lim_{h \rightarrow 0} \{h - 1\} = -1$$

$$f(1-0) = \lim_{h \rightarrow 0} \{1 - h - 1\} - \{1 - h\} = \lim_{h \rightarrow 0} \{h - 0\} = 0$$

50. b, c

Since the greatest integer function is discontinuous (sensitive) at integral values of  $x$ , for a given limit to exist both left and right-hand must be equal.

$$L.H.L = \lim_{x \rightarrow 1^-} (2 - x + a[x - 1] + b[1 + x])$$

$$= 2 - 1 + a(-1) + b(1) = 1 - a + b$$

$$R.H.L = \lim_{x \rightarrow 1^+} (2 - x + a[x-1] + b[1+x])$$

$$= 2 - 1 + a(0) + b(2) = 1 + 2b$$

On comparing, we have  $-a=b$

51. a,b,c,d

$$\text{We have } \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0^+} \frac{\tan^2 \{x\}}{x^2 - [x]^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{\tan^2 x}{x^2} = 1 \quad \dots\dots 1$$

$$\text{Also } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0} \sqrt{\{x\} \cot \{x\}} = \sqrt{\cot 1} \quad \dots\dots 2$$

$$(\because x \rightarrow 0, [x] = -1 \Rightarrow \{x\} = x + 1 \Rightarrow \{x\} \rightarrow 1)$$

$$\text{Also, } \cot^{-1} \left( \lim_{x \rightarrow 0^+} f(x) \right)^2 = \cot^{-1}(\cot 1) = 1$$

52. a,b,c

53. Given that  $\lim_{x \rightarrow 0} \frac{\cos^2 x - \cos x = e^x \cos x + e^x - \frac{x^3}{2}}{x^n}$

$$= \lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x) - \frac{x^3}{2}}{x^n}$$

$$= \lim_{x \rightarrow 0} \frac{\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots - 1\right) \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right) - \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots\right) - \frac{x^3}{2}}{x^n}$$

$$= n=4$$

54. Conceptual

55.  $\lim_{x \rightarrow 0^+} \left[ \frac{\lim_{n \rightarrow \infty} \sum_{k=1}^n \left[ k^2 (\sin x)^x \right]}{n^3} \right] = \frac{1}{3} \lim_{x \rightarrow 0^+} (\sin x)^x$

$$L=1/3$$

56.  $f(n, \theta) = \frac{2^n \tan \frac{\theta}{2^n}}{\tan \theta}$

$$\therefore \lim_{n \rightarrow \infty} f(n, \theta) = \frac{\theta}{\tan \theta}$$

$$57. \lim_{x \rightarrow \infty} \left( f(x) + \frac{3f(x)-1}{f^2(x)} \right) = 3$$

$$\text{or } \left( \lim_{x \rightarrow \infty} f(x) + \frac{3 \lim_{x \rightarrow \infty} f(x) - 1}{\left( \lim_{x \rightarrow \infty} f(x) \right)^2} \right) = 3$$

$$\text{or } \left( y + \frac{3y-1}{y^2} \right) = 3$$

$$\text{or } y^2 - 3y^2 + 3y - 1 = 0$$

$$\text{or } (y-1)^3 = 0$$

$$\text{or } y = 1$$

$$58. m=0, n=2$$

$$59. p) \frac{\tan x}{x} > 1 \text{ and } \frac{\tan x}{x} \rightarrow 1 \text{ for } x \rightarrow 0$$

$$\Rightarrow \frac{\tan x}{x} = 1 + h; h \rightarrow 0$$

$$\Rightarrow 200 \frac{\tan x}{x} = 200 + 200h; h \rightarrow 0$$

$$\Rightarrow \left[ 200 \frac{\tan x}{x} \right] = 200$$

$$\text{Also } \frac{\sin x}{x} < 1 \text{ and } \frac{\sin x}{x} \rightarrow 1 \text{ for } x \rightarrow 0$$

$$\Rightarrow \frac{\sin x}{x} = 1 - h; h \rightarrow 0^+$$

$$\Rightarrow 200 \left( \frac{\sin x}{x} \right) = 200 - 200h; h \rightarrow 0$$

$$\Rightarrow \left[ 200 \frac{\sin x}{x} \right] = 199$$

$$\lim_{x \rightarrow 0} \left( \left[ 200 \frac{\tan x}{x} \right] + \left[ 200 \frac{\sin x}{x} \right] \right) = 399$$

$$q) 100 \frac{\tan x}{x} = 100(1+h) = 100 + 100h; h \rightarrow 0$$

$$\left\{ 100 \frac{\tan x}{x} \right\} = 100 \text{ as } h \rightarrow 0$$

$$\text{Also } \left\{ 100 \frac{\sin x}{x} \right\} = \{100(1-h)\} = \{100 - 100h\}$$

$$\{99 + (1 - 100h)\} = 1 - 100h \rightarrow 1 \text{ as } h \rightarrow 0$$

$$\lim_{x \rightarrow 0} \left\{ 100 \frac{\tan x}{x} \right\} + \left\{ 100 \frac{\sin x}{x} \right\} = 0 + 1 = 1$$

$$\text{r) } \frac{\sin x}{x} = 1 - h; h \rightarrow 0^+ \text{ as } x \rightarrow 0^+$$

$$\Rightarrow \left[ \frac{\sin x}{x} \right] = [1 - h] = 0 \text{ and } \frac{x}{\tan^{-1} x} > 1 \text{ for } x \rightarrow 0$$

$$\Rightarrow \frac{x}{\tan^{-1} x} = 1 + h; h \rightarrow 0^+$$

$$\Rightarrow \left[ \frac{x}{\tan^{-1} x} \right] = 1$$

$$\lim_{x \rightarrow 0} \left( 100 \left[ \frac{\sin x}{x} \right] + 200 \left[ \frac{x}{\tan^{-1} x} \right] \right) = 0 + 200 = 200$$

$$\text{s) As } x \rightarrow 0, \frac{\sin^{-1} x}{x} > 1 \text{ and } \frac{\sin^{-1} x}{x} \rightarrow 1 \text{ as } x \rightarrow 0$$

$$\Rightarrow \frac{\sin^{-1} x}{x} = 1 + h; h \rightarrow 0^+$$

$$\left\{ \frac{\sin^{-1} x}{x} \right\} = h \rightarrow 0 \text{ as } h \rightarrow 0^+$$

$$\text{Also } \frac{\tan^{-1} x}{x} < 1 \text{ and } \frac{\tan^{-1} x}{x} \rightarrow 1 \text{ as } x \rightarrow 0$$

$$\Rightarrow 200 \frac{\tan^{-1} x}{x} = 200(1-h); h \rightarrow 0^+$$

$$\Rightarrow \left[ 200 \frac{\tan^{-1} x}{x} \right] = [200 - 200h] = 199$$

$$60. (p) \rightarrow d; (q) \rightarrow c; (r) \rightarrow a; (s) \rightarrow b$$

$$\phi(x) = \frac{a_0 x^m + a_1 x^{m+1} + \dots + a_k x^{m+k}}{b_0 x^n + \dots + b_l x^{l+e}}$$

$$\phi(x) = \left( \frac{a_0 + a_1 x + \dots + a_k x^k}{b_0 + b_1 x + \dots + b_l x^l} \right) \cdot x^{(m-n)}$$

$$\text{p) } m > n$$

$$\Rightarrow \lim_{x \rightarrow 0} (\phi(x))$$

$$\Rightarrow \frac{a_0}{b_0} (0)^{m-n} = 0 (m > n)$$

$$(p) \rightarrow (d)$$

$$q) m = n \Rightarrow \lim_{x \rightarrow 0} \phi(x) = \frac{a_0}{b_0}$$

$$(q) \rightarrow c$$

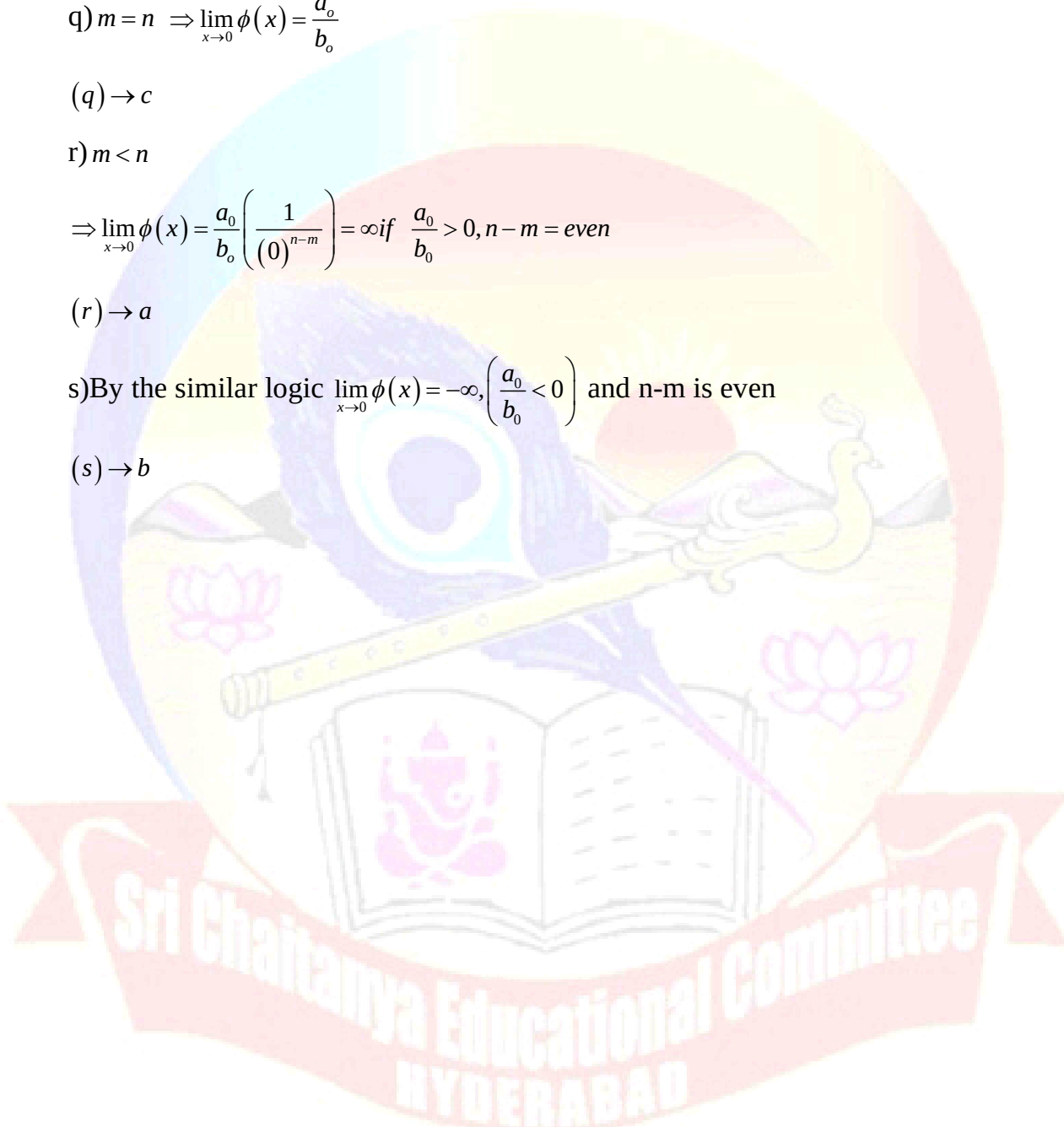
$$r) m < n$$

$$\Rightarrow \lim_{x \rightarrow 0} \phi(x) = \frac{a_0}{b_0} \left( \frac{1}{(0)^{n-m}} \right) = \infty \text{ if } \frac{a_0}{b_0} > 0, n-m = \text{even}$$

$$(r) \rightarrow a$$

$$s) \text{ By the similar logic } \lim_{x \rightarrow 0} \phi(x) = -\infty, \left( \frac{a_0}{b_0} < 0 \right) \text{ and } n-m \text{ is even}$$

$$(s) \rightarrow b$$



# Success



## JEE ADVANCED WEEKEND TEST



### Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

Date: 23-09-17

Max. Marks: 180

Name of the Student: \_\_\_\_\_

I.D. NO:

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

## 2014\_PAPER-I

### IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

### OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your roll number.

### DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

**23-09-17\_Jr.Super60-I\_Jee Adv\_WAT-19\_Weekend Syllabus**

**MATHS:** SYLLABUS COVERED FROM 18-09-17 TO 21-09-17

**PHYSICS:** SYLLABUS COVERED FROM 18-09-17 TO 22-09-2017

**CHEMISTRY: GROUP 13:** Preparation, properties and uses and aluminium, alumina, aluminium chloride and alums, **Group 14, Water, heavy water,  $H_2O_2$**

**JEE-ADVANCE-2014-P1-Model**

Time: 07:30 AM to 10:30 AM

**IMPORTANT INSTRUCTIONS**

Max Marks: 180

**PHYSICS:**

| Section                 | Question Type                          | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|-------------------------|----------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 1 – 10)   | Questions with Multiple Correct Choice | 3         | 0          | 10        | 30          |
| Sec – II(Q.N : 11 – 20) | Questions with Integer Answer Type     | 3         | 0          | 10        | 30          |
| <b>Total</b>            |                                        |           |            | <b>20</b> | <b>60</b>   |

**CHEMISTRY:**

| Section                 | Question Type                          | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|-------------------------|----------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 21 – 30)  | Questions with Multiple Correct Choice | 3         | 0          | 10        | 30          |
| Sec – II(Q.N : 31 – 40) | Questions with Integer Answer Type     | 3         | 0          | 10        | 30          |
| <b>Total</b>            |                                        |           |            | <b>20</b> | <b>60</b>   |

**MATHEMATICS:**

| Section                 | Question Type                          | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|-------------------------|----------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 41 – 50)  | Questions with Multiple Correct Choice | 3         | 0          | 10        | 30          |
| Sec – II(Q.N : 51 – 60) | Questions with Integer Answer Type     | 3         | 0          | 10        | 30          |
| <b>Total</b>            |                                        |           |            | <b>20</b> | <b>60</b>   |

Sec: Jr. Super60-I

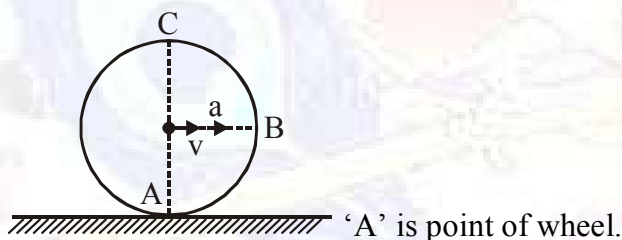
space for rough work

**2**

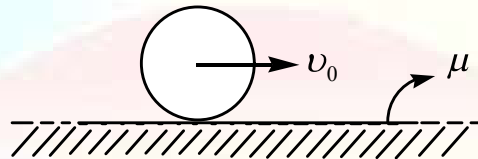
**PHYSICS****MaxMarks:60****SECTION-I****(One or More options Correct Type)**

This section contains 10 multiple choice questions. Each question has four choices (A) (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

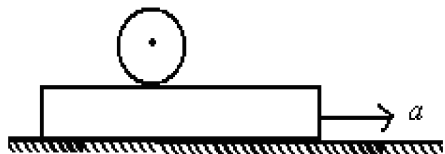
1. A wheel is rolling without slipping on a horizontal plane. At a certain instant, it has velocity ' $v$ ' and acceleration ' $a$ ' of centre of mass as shown in the figure. At that instant, acceleration of



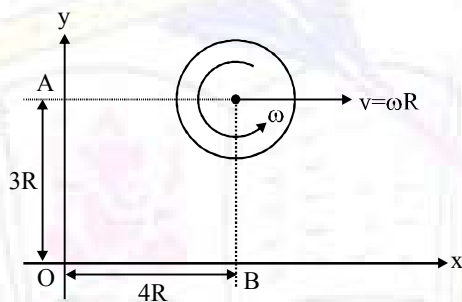
- A) A is vertically upwards  
 B) B may be vertically downwards  
 C) C cannot be horizontal  
 D) some point on the rim may be horizontal in forward direction.
2. A sphere is thrown along a rough surface as shown. Mass of the sphere is  $M$  and radius is  $R$ . If  $t$  is total time taken to start pure rolling after thrown, then:



- A) speed of the sphere when it starts pure rolling is  $\frac{5v_0}{7}$
- B) work done by friction is greater in first half i.e. 0 to  $\frac{t}{2}$  compare to later half i.e.,  $\frac{t}{2}$  to  $t$
- C) work done by friction is zero after time  $t$
- D) the net work done on the sphere is  $\frac{1}{4}Mv_0^2$
3. An uniform solid cylinder is placed on a horizontal board. The coefficient of friction between them is 0.3. The board is imparted a constant acceleration of  $a = 6m/s^2$  in a horizontal direction at right angles of the cylindrical axis. Then (take  $g = 10m/s^2$ )

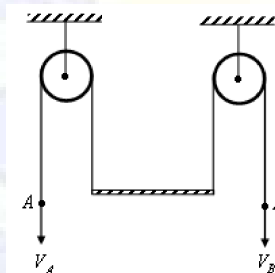


- A) The acceleration of the cylinder axis in the absence of slipping will be  $1.5 \text{ m/s}^2$ .
- B) The acceleration of the cylinder axis in the absence of slipping will be  $2 \text{ m/s}^2$
- C) The limiting value of 'a' for which there will be no slipping is  $12 \text{ m/s}^2$
- D) The limiting value of 'a' for which there will be no slipping is  $9 \text{ m/s}^2$
4. A disc of mass  $m$  and radius  $R$  moves in the  $x$ - $y$  plane as shown in the fig. The angular momentum of the disc at the instant shown is:



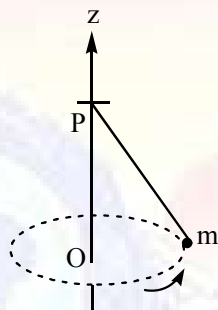
- A)  $\frac{5}{2}mR^2\omega$  about O
- B)  $\frac{7}{2}mR^2\omega$  about O
- C)  $\frac{1}{2}mR^2\omega$  about A
- D)  $4mR^2\omega$  about A

5. A thin uniform rod of mass 5 kg and length 1 m is held in horizontal position with the help of strings attached to ends of rod. Other ends of strings are held by some external agent. Now end A is pulled down with speed  $V_A = 3t$  and end B is pulled down with speed  $V_B = t$  where  $t$  is time in s. Choose the correct choice (s) at time  $t = 0$ .



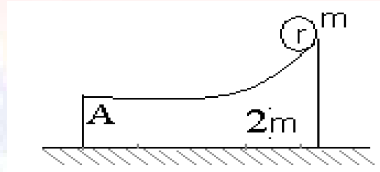
- A) Angular acceleration of rod is  $2 \text{ rad/s}^2$   
 B) Tension in left string is  $\frac{185}{6} \text{ N}$   
 C) Acceleration of rod is  $1 \text{ m/s}^2$   
 D) Tension in right string is  $\frac{170}{3} \text{ N}$
6. A small mass  $m$  is attached to a massless string whose other end is fixed at P as shown in the figure. The mass is undergoing circular motion in the  $x$ - $y$  plane with centre at  $O$

and constant angular speed  $\omega$ . If the angular momentum of the system calculated about  $O$  and  $P$  are denoted by  $\vec{L}_O$  and  $\vec{L}_P$  respectively, then



- A)  $\vec{L}_O$  and  $\vec{L}_P$  do not vary with time  
 B)  $\vec{L}_O$  varies with time while  $\vec{L}_P$  remains constant  
 C)  $\vec{L}_O$  remains constant while  $\vec{L}_P$  varies with time  
 D)  $\vec{L}_O$  and  $\vec{L}_P$  both vary with time.
7. A sphere of mass  $m$  and radius ' $r$ ' is released from rest from the top of a wedge of mass ' $2m$ ' initially at rest on a smooth horizontal surface as shown. There is sufficient friction between the sphere and the wedge so that the sphere rolls on the wedge without slipping. The sphere leaves the wedge horizontally at A after descending through a vertical displacement  $h$ . Let  $V_1$ ,  $V_2$  and  $\omega$  be the magnitudes of the velocity

of the centre of sphere, velocity of wedge and the angular velocity of the sphere at the instant the sphere leaves the wedge. Choose the correct statements.



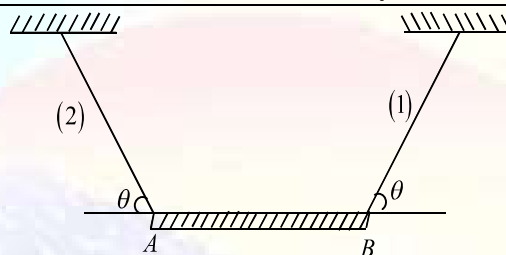
A)  $V_1 + V_2 = r\omega$

B) Total mechanical energy of sphere and wedge system is conserved during the process

C) The ratio of total kinetic energy of the sphere and to the wedge at the instant the sphere leaves the wedge is 19:5

D)  $V_1 = \sqrt{\frac{5gh}{6}}$

8. A uniform rod AB of length 'L' mass M is suspended by two massless identical threads. Initially the rod AB is in equilibrium in the horizontal position and the system is in vertical plane as shown in fig. Immediately after the thread (1) breaks, (take g is acceleration due to gravity)



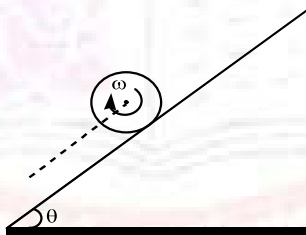
A) Tension in thread (2) is  $\frac{Mg \sin \theta}{1 + 3 \sin^2 \theta}$

B) Angular acceleration of the rod about its centre of mass is  $\frac{6g \sin^2 \theta}{L(1 + 3 \sin^2 \theta)}$

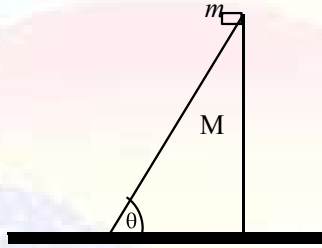
C) The end A of the rod has acceleration component in vertically upward direction

D) Centre of mass of the rod has horizontal acceleration component towards leftward

9. A solid sphere is given an angular velocity  $\omega$  about a horizontal axis and kept on a rough fixed incline plane. Then choose the correct statement.



- A) If  $\mu = \tan \theta$ , then the sphere will be in linear equilibrium for some time and after that pure rolling down the plane will start.
- B) If  $\mu = \tan \theta$ , then the sphere will move up the plane and frictional force acting all the time will be  $2mg \sin \theta$ .
- C) If  $\mu = \frac{\tan \theta}{2}$ , then there will never be pure rolling (consider inclined plane to be long enough.)
- D) If incline plane is not fixed and it is on smooth horizontal surface then linear momentum of the system (wedge and sphere) can be conserved in horizontal direction.
10. Friction between the wedge shown and the horizontal floor is sufficient to prevent the wedge from sliding. The mass of the wedge is  $M$  and its angle of inclination is  $\theta$ . A small block of mass  $m$  is just placed near the top of the wedge and released. Friction coefficient between the wedge and the block is  $\mu$ .



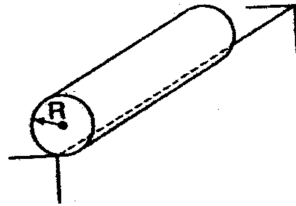
- A) If  $\mu > \tan \theta$ , the wedge will not topple for any value of  $m$ .
- B) If  $\mu < \tan \theta$ , the wedge will topple for  $m > \frac{M}{3 \sin \theta (\sin \theta - \mu \cos \theta)}$ .
- C) The wedge will topple for any value of  $m$  irrespective of  $\mu$ .
- D) If  $\mu < \tan \theta$ , the wedge will not topple for  $m > \frac{M}{3 \sin \theta (\sin \theta - \mu \cos \theta)}$ .

**SECTION-II**  
**(Integer Value Correct Type)**

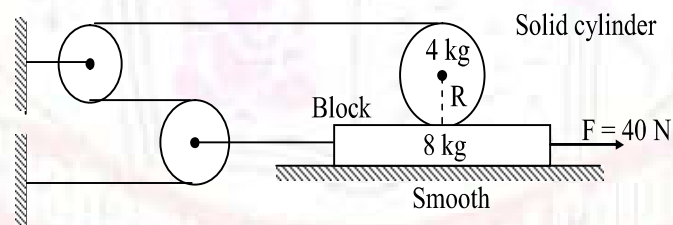
This section contains 10 questions. The answer to each question is a single digit integer, ranging from 0 to 9 (both inclusive).

11. A rectangular rigid fixed block has a long horizontal edge. A solid homogeneous cylinder of radius  $R$  is placed horizontally at rest with its length parallel to the edge such that the axis of the cylinder and the horizontal edge are in same vertical plane shown in the figure. There is sufficient friction present at the edge, so that a very small

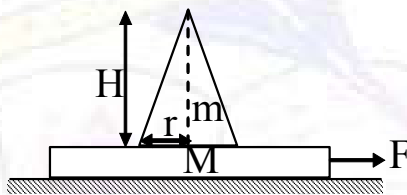
displacement causes the cylinder to roll off the edge without slipping. Then the ratio of the translational to rotational kinetic energies of the cylinder when its center of mass is in horizontal line with the edge is.....



12. An ideal string is wrapped several times on a solid cylinder of mass 4 kg and radius 1 m. The pulleys are ideal and the surface between block and ground is smooth. Cylinder rolls without slipping on the block. If the torque acting on the cylinder is  $\frac{10x}{9}$  N-m, then find the value of x.



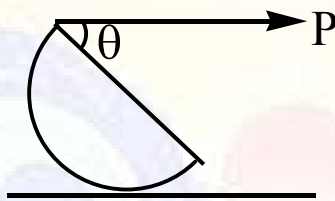
13. A uniform rod of length  $2\ell$  and mass  $m$  is suspended from one end by an inextensible vertical string and other end lies on smooth ground. The angle made by rod with vertical is  $\theta = \sin^{-1}(1/\sqrt{3})$ . If  $N_1$  and  $N_2$  represents the contact force from ground on rod just before and just after cutting the string then find the ratio of  $N_1 / N_2$ .
14. A solid cone of mass ( $m = 1$  kg) is placed on a plank of mass  $M = 1$  kg which is placed on a smooth horizontal plane. The coefficient of friction between cone and the plank is  $1/4$ . If a horizontal force  $F$  is applied on the plank, then find the maximum value of  $F$  for which the cone is in equilibrium with respect to plank.  
(Take  $g = 10 \text{ m/s}^2$  and  $H/r = 20$ )



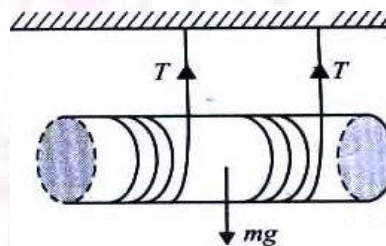
15. The solid semi-cylinder of mass  $m$  and radius  $R$  is rolled slowly through an angle  $\theta$  by the horizontal force  $P$ . If the coefficient of static friction is  $\mu_s$ , the angle  $\theta$  at which

the cylinder begins to slip on the horizontal surface as P gradually increased is

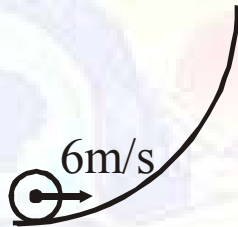
$$\theta = \sin^{-1} \left( \frac{3\pi\mu_s}{n - 3\pi\mu_s} \right) \text{ Find 'n'.$$



16. A solid cylinder of mass  $m = 4 \text{ kg}$  and radius  $R = 10 \text{ cm}$  has two ropes wrapped around it, one near each end. The cylinder is held horizontally by fixing the two free ends of the cords to the hooks on the ceiling and both the cords are exactly vertical. The cylinder is released to fall under gravity. The linear acceleration of the cylinder is  $\frac{n}{n+1}g$  then  $n = ?$ .

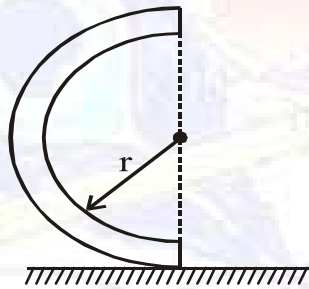


17. A disc of radius 0.1 m rolls without sliding on a horizontal surface with a velocity of 6 m/s. It then ascends a smooth continuous track as shown in figure. The height up to which it will ascend is  $\frac{x}{5}$  m. Value of x is? ( $g = 10 \text{ m/s}^2$ )



18. A constant power is supplied to a rotating disc. Angular velocity ( $\omega$ ) of disc varies with number of rotations (n) made by the disc as  $\omega \propto n^{1/x}$ . Value of x is

19. A semicircular ring of mass  $m$  and radius  $r$  is released from rest in the position shown with its lower edge resting on a horizontal surface. The minimum coefficient of static friction  $\mu_s$  which is necessary to prevent any initial slipping of the ring is  $\frac{x}{10}$ . Value of  $x$  is



20. A binary star consists of two stars A (mass  $2.2M_s$ ) and B (mass  $11M_s$ ), where  $M_s$  is the mass of the sun. They are separated by distance  $r$  and are rotating about their centre of mass, which is stationary. What is the ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass?

**CHEMISTRY****Max Marks: 60****SECTION-I****(One or More options Correct Type)**

This section contains 10 multiple choice questions. Each question has four choices (A) (B) (C) and (D) out of which **ONE or MORE** are correct.

21. The correct property(ies) about Boron and Aluminium is/are:
- A) Maximum Covalency of Boron is 4 and that of aluminium is 6
  - B) Aluminium dissolves in dilute alkali and liberates  $H_2$  but Boron does not
  - C) Oxide of Boron is acidic whereas that of aluminium is amphoteric
  - D) Halides of Boron exist as monomers whereas the halides of other boron family members exist as dimers.
22. Identify the correct statement(s) regarding hydrogen gas
- A) Unlike  $H_2$ ,  $D_2$  does not show spin isomerism.
  - B) Orthohydrogen differs from parahydrogen w.r.t their b.p., sp. Heat and thermal conductivity
  - C) As the temperature is increased from absolute zero some of the para form gradually changes to ortho form.
  - D) Parahydrogen has lower internal energy than orthohydrogen.

23. The correct statement(s) is/are
- A)  $I.E_1$  of Pb >  $I.E_1$  of Sn
  - B)  $CO_2$ ,  $SiO_2$  &  $GeO_2$  are acidic,  $SnO_2$  &  $PbO_2$  are amphoteric
  - C)  $SiF_6^{2-}$  is known but  $SiCl_6^{2-}$  is not known
  - D)  $C \gg Si > Ge \approx Sn$  (Catenation)
24. The correct statement(s) is/are:
- A) The only solid bicarbonates known are those of the group-1 metals and of  $NH_4^+$
  - B) Carbon monoxide is quantitatively estimated by  $I_2O_5$
  - C)  $CH_4 > SiH_4 > GeH_4 > PbH_4$  (Thermal stability)
  - D) Graphite > diamond > fullerenes (Thermal stability)
25. Which of the following methods can be used for the preparation of anhydrous aluminium chloride?
- A) Heating  $AlCl_3 \cdot 6H_2O$
  - B) Heating a mixture of alumina and coke in a current of dry chlorine gas
  - C) Passing dry HCl gas over heated aluminium powder
  - D) Passing dry chlorine gas over heated aluminium

26. Select the correct statement about silicates?

- A) Cyclic silicate having three silicon atoms contains six  $Si-O-Si$  linkages
- B) On an average over all  $2\frac{1}{2}$  oxygen atoms per tetrahedron unit are shared in double chain silicate
- C)  $(Si_2O_5)_n^{2n-}$  is the repeating unit or basic unit of double chain silicate
- D)  $SiO_4^{4-}$  units polymerise to form silicate because Si atom has less tendency to form  $\pi$ -bond with oxygen

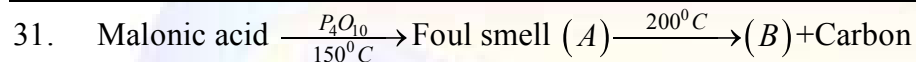
27. The correct statement(s) is/are

- A) Does not neutralise dilute  $HNO_3$  :  $SiO_2, PbO_2, CO$
- B) Reacts with HF :  $SiO_2, SnO$
- C) Solid at room temperature :  $SiO_2, LiHCO_3$
- D) May act as reducing agent :  $CO, SnO$

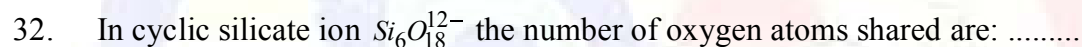
28. The correct statements among the following are
- A)  $H_2$  is more rapidly adsorbed on solid surface than  $D_2$ .
  - B) H-H bond dissociation energy is greater than D-D bond dissociation energy.
  - C) Deuterated compound can be easily prepared by exchange reaction of deuterium with hydrogen using heavy water rather than  $D_2$ .
  - D) The bond length is shortest in  $T_2$  among the isotopes of hydrogen.
29.  $SiO_2 + NaOH \rightarrow A \xrightarrow{H_2F_2} B \xrightarrow{H_2O} C + D$
- Hence
- A)  $A = Na_2SiO_3$
  - B)  $B = SiF_4$
  - C)  $C = H_2SiO_3; D = H_2SiF_6$
  - D)  $C = F_2; D = H_2O_2$
30. The correct statement(s) is/are about carbides:
- A)  $Be_2C$  and  $Al_4C_3$  are methanides
  - B)  $B_4C$  and  $SiC$  are covalent
  - C) Interstitial carbides retain many of the properties of metals
  - D) Interstitial carbides are less denser and harder than pure metals

**SECTION-II****(Integer Value Correct Type)**

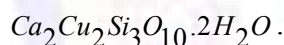
This section contains 10 questions. The answer to each question is a single digit integer, ranging from 0 to 9 (both inclusive).



The total number of sigma and Pi bonds present in the compound A are: .....



33. The number of oxygen atoms not acting as bridges (Si-O-Si) in the mineral kinoite



34. How many of the following species have  $sp^2$  hybrid atoms?

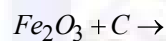
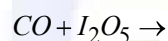
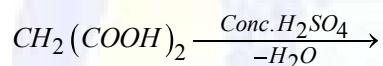
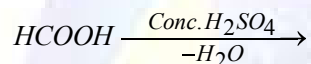
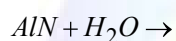
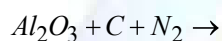
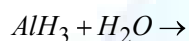
Diamond, Graphite,  $\text{C}_6\text{O}$ ,  $(\text{SiH}_3)_3\text{N}$ ;  $\text{AlF}_3$ ;  $\text{AlCl}_3$  (monomer);  $\text{Al}_2(\text{CH}_3)_6$ ;  $\text{SiC}$ ,

$\text{H}_3\text{BO}_3$ , layered BN, Borazine.

35. How many of the following compounds are amphoteric?



36. How many of the following reactions produce a paramagnetic gas?



37. In how many of the following reactions hydrogen gas is produced as one of the product.

(i) Electrolysis of acidulated water

(ii) Passing steam over red hot coke.

(iii) Reaction of aluminium with sodium hydroxide

(iv) Reaction of Zinc with dil HCl

(v) Pyrolysis of hydrocarbons

(vi) Reaction of sodium amalgam with water.

(vii) By dissolving hydrolith in water.

(viii) By heating a mixture of steam and hydro carbons

(ix) Electrolysis of brine.

38. How many of them are 3-D silicates?

Feldspars, Ultra marines, Zeolites, Pyroxenes, Amphiboles, Beryl, Zircon, Asbestos

39. X = Oxidation, state of carbon in CO

Y = Oxidation state of carbon in  $\text{COCl}_2$

They  $y - x$  is:

40. How many of the following reactions lead to the formation of CO as one of the products?



## SECTION-I

(One or More options Correct Type)

This section contains 10 multiple choice questions. Each question has four choices (A) (B) (C) and (D) out of which ONE or MORE are correct.

41. Let  $f(x) = g(x) \left( \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} \right)$  where  $g(x)$  is a continuous function, then  $\lim_{x \rightarrow 0} f(x)$  exists, if

A)  $g(x)$  is any polynomialB)  $g(x) = x^2$ C)  $g(x) = \sin x$ D)  $g(x) = 2 + 3x + 4x^3$ 

42. Let  $f: R \rightarrow R$  be a function given by

$$f(x) = \begin{cases} 5x & \text{if } x \in Q \\ x^2 + 6 & \text{if } x \in R - Q \end{cases} \quad \text{where } Q \text{ is set of rational numbers then}$$

A)  $f(x)$  is continuous for  $x \in R$ B)  $f(x)$  is continuous at finite number of values of  $x$ C)  $f(x)$  is continuous at  $x = 2$ D)  $f(x)$  is continuous at  $x = 0$

43. Let  $f(x)$  be a non constant polynomial function satisfying the relation
- $$f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2 \text{ for all real } x \text{ and } y \text{ and } f(0) = 1, \text{ suppose } f(4) = 65.$$
- Then, which of the following statement(s) is/are true
- A)  $f'(x)$  is a polynomial of degree 2
- B) Roots of  $f'(x) = 2x + 1$  are real
- C)  $xf'(x) = 3[f(x) - 1]$
- D)  $f'(-1) = 3$
44. Consider the function  $f: \mathbb{R} \rightarrow \mathbb{R}$  defined by  $f(x) = [2 \sin x]$  where  $[\bullet]$  is G.I.F, then which of the following is/are true
- A)  $f(x)$  is continuous at  $x = -\frac{\pi}{2}$
- B)  $f(x)$  is not differentiable at  $x = -\frac{3\pi}{2}$
- C)  $f(x)$  is continuous at  $x = -\pi$
- D)  $f(x)$  is not differentiable at  $x = \frac{\pi}{2}$

45. Let  $f: \left[0, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$  be defined by  $f(x) = \max\left\{\sin x, \cos x, \frac{3}{4}\right\}$  then which of the following is not true.

A)  $f(x)$  is continuous for all real  $x \in [0, \pi/2]$

B)  $f(x)$  is not differentiable at  $x = \frac{\pi}{4}$

C) Number of points where  $f(x)$  is not differentiable is 2

D) Number of points where  $f(x)$  is not continuous is 3

46. If  $f(x) = \begin{cases} \sin\left(\frac{\pi}{2}(x - [x])\right) & x < 5 \\ 5(b-1) & x = 5 \\ ab^2 \frac{|x^2 - 11x + 24|}{x-3} & x > 5 \end{cases}$  is continuous at  $x = 5$ ,  $a, b \in \mathbb{R}$ , then  $([\bullet])$  denotes

G.I.F)

A)  $a+b = \frac{773}{540}$

B)  $a-b = \frac{523}{540}$

C)  $a < b$

D)  $\frac{a}{b} = \frac{125}{648}$

47. Let  $f(x) = \begin{cases} x\left[\frac{1}{x}\right] + x[x] & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$  (where  $[x]$  denotes the greatest integer function).

Then the correct statement(s) is/are

- A) limit of  $f(x)$  exists for  $x = -1$   
 B)  $f(x)$  has discontinuity at  $x = 1$   
 C)  $f(x)$  has discontinuity at  $x = 2$   
 D)  $f(x)$  is discontinuous at all positive integers
48. The set of points of differentiability of the function  $f(x) = \begin{cases} x^n \sin\left(\frac{1}{x}\right) & x \neq 0 (n \geq 2) \\ 0 & x = 0 \end{cases}$  contains (Where  $R$  is set of real numbers)  
 A)  $(-\infty, 0)$       B)  $R - \{0\}$       C)  $(0, \infty)$       D)  $R$
49. Suppose that  $f: R \rightarrow R$  is continuous and satisfying the equation  $f(x) \cdot f(f(x)) = 1$ , for all real  $x$ . Let  $f(1000) = 999$ , then which of the following is/are true  
 A)  $f(500) = \frac{1}{500}$       B)  $f(199) = \frac{1}{199}$   
 C)  $f(x) = \frac{1}{x} \quad \forall x \in R - \{0\}$       D)  $f(1999) = \frac{1}{1999}$

50. Let  $f: \mathbb{R} \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$  be a function defined by  $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$  then which of the following is/are true
- A)  $f(x)$  is not differentiable at  $x = -1$  and  $x = 1$
- B)  $f(x)$  is continuous for all real  $x$
- C)  $f(x) = 2 \tan^{-1} x \quad \forall x \in \mathbb{R}$
- D)  $f(x)$  is differentiable  $\forall x \in \mathbb{R}$

**SECTION-II****(Integer Value Correct Type)**

This section contains 10 questions. The answer to each question is a single digit integer, ranging from 0 to 9 (both inclusive).

51. The number of points where  $f(x) = \left[\frac{x}{3}\right]$  in  $x \in (0, 30)$  is discontinuous is  $([\bullet])$  is G.I.F)
52. Let  $f$  be a function defined on  $(-1, 1)$  as  $f(x) = \begin{cases} \frac{2^{\sin^{-1} x} - 2^{\tan^{-1} x}}{2^{\tan x} - 2^{\sin x}} & \text{if } x \neq 0 \\ K & \text{if } x = 0 \end{cases}$  then the value of  $K$  such that  $f(x)$  is continuous at  $x = 0$  is

53. If  $f(x) = \begin{cases} \left( \sin \frac{2x^2}{a} + \cos \left( \frac{3x}{b} \right) \right)^{\frac{ab}{x^2}} & x \neq 0 \\ e^3 & x = 0 \end{cases}$  is continuous at  $x=0$  where  $a, b \in \mathbb{R} - \{0\}$  and

$K$  is minimum value of 'a' then  $|4K|$  is equal to

54. The function  $f(x)$  is discontinuous only at  $x=0$  such that  $f^2(x)=1 \quad \forall x \in \mathbb{R}$ . The total number of such functions is

55. Number of points where the function  $f(x) = \max(|\tan x|, \cos|x|)$  is non differentiable in the interval  $(-\pi, \pi)$  is

56. If  $f(x) = \begin{cases} x+2 & x < 0 \\ -x^2-2 & 0 \leq x < 1 \\ x & x \geq 1 \end{cases}$  then the number of points of discontinuity of  $|f(x)|$  is

57.  $f(x) = a|\sin x| + be^{|x|} + c|x|^3$  ( $a, b, c \in \mathbb{R}$ ). If  $f(x)$  is differentiable at  $x=0$ , then the value of  $a+b$  is equal to

58. Let  $g(x) = \frac{(x-1)^n}{\log_e \cos^m(x-1)}$ ;  $0 < x < 2$  where  $m \neq 0, n > 0$  are integer. Let  $P$  be the left hand derivative of  $|x-1|$  at  $x=1$  and  $\lim_{x \rightarrow 1^+} g(x) = P$ , then  $m+n =$

59. Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a continuous function satisfying  $f(0)=1$  and  $f(2x)-f(x)=x$  then the value of  $f(3)$  is

60. 
$$f(x) = \begin{cases} \frac{1 - \cos 3x \cos 9x \cos 27x \dots \cos(3^n x)}{1 - \cos \frac{x}{3} \cos \frac{x}{9} \cos \frac{x}{27} \dots \cos \left(\frac{x}{3^n}\right)} & \text{if } x \neq 0 \\ 3^{10} & \text{if } x = 0 \end{cases}$$

If  $f(x)$  is continuous at  $x=0$ ,

then the value of  $n$  is  $(n \in \mathbb{N})$



# Sri Chaitanya IIT Academy., India.

A.P., TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. SUPER60-I

JEE ADVANCED

DATE : 23-09-17

TIME : 07:30 AM to 10:30 AM

WAT-19

MAX MARKS : 180

## KEY SHEET

2014-P1

### PHYSICS

|    |      |    |      |    |     |    |    |    |    |
|----|------|----|------|----|-----|----|----|----|----|
| 1  | ABCD | 2  | ABC  | 3  | BD  | 4  | AC | 5  | AB |
| 6  | C    | 7  | ABCD | 8  | ABD | 9  | AD | 10 | AB |
| 11 | 6    | 12 | 8    | 13 | 1   | 14 | 4  | 15 | 4  |
| 16 | 2    | 17 | 9    | 18 | 3   | 19 | 4  | 20 | 6  |

### CHEMISTRY

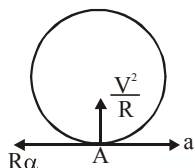
|    |     |    |     |    |      |    |      |    |     |
|----|-----|----|-----|----|------|----|------|----|-----|
| 21 | AC  | 22 | BCD | 23 | ABCD | 24 | ABCD | 25 | BCD |
| 26 | BCD | 27 | ABD | 28 | AC   | 29 | ABC  | 30 | ABC |
| 31 | 8   | 32 | 6   | 33 | 8    | 34 | 7    | 35 | 5   |
| 36 | 0   | 37 | 9   | 38 | 3    | 39 | 2    | 40 | 5   |

### MATHS

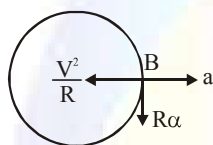
|    |     |    |      |    |      |    |     |    |    |
|----|-----|----|------|----|------|----|-----|----|----|
| 41 | BC  | 42 | BC   | 43 | ABCD | 44 | ABD | 45 | BD |
| 46 | ACD | 47 | ABCD | 48 | ABCD | 49 | AB  | 50 | AB |
| 51 | 9   | 52 | 1    | 53 | 1    | 54 | 6   | 55 | 4  |
| 56 | 1   | 57 | 0    | 58 | 4    | 59 | 4   | 60 | 4  |

**SOLUTIONS****PHYSICS**

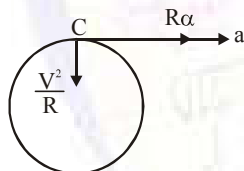
1. At A :

for a rolling wheel,  $a = R\alpha$  $\therefore$  (A) is correct.

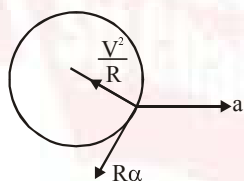
At B :

If  $\frac{V^2}{R} = a$  then  $a_B$  may be vertically downwards $\therefore$  (B) is correct.

At C :

 $\therefore$  (C) is correct.

Consider this

 $\therefore$  (D) is correct.

$$2. \quad \mu mgR = \frac{2}{5} mR^2 \alpha$$

$$\alpha = \frac{5}{2} \frac{\mu g}{R}$$

$$\omega = \alpha t; \quad v = v_0 - \mu g t$$

For pure rolling,  $V = R\omega$

$$V_0 - \mu g t = R \alpha t$$

$$t = \frac{2}{7} \frac{V_0}{\mu g}$$

$$\therefore V = V_0 - \frac{2}{7} V_0 = \frac{5}{7} V_0$$

3.  $ma_c = F_{fr}$

$$I \propto r F_{fr} = r m a_c$$

$$a - a_c = \alpha r = \frac{r m a_c}{I} r$$

$$\text{or, } a - a_c = \frac{m r^2 a_c}{\frac{m r^2}{2}} = 2 a_c$$



$$\text{or, } 3a_c = a, \quad a_c = \frac{a}{3} = \frac{6}{3} = 2 \text{ m/s}^2$$

$$\text{So, } f_{FR} = \frac{ma}{3} \text{ so maximum value of this force is equal } \mu mg$$

$$\text{Hence } \mu mg = \frac{ma_{lim}}{3} \text{ or, } a_{lim} = 3\mu g = 3 \times 0.3 \times 10 = 9 \text{ m/s}^2$$

4. (a)  $m(R\omega)(3R) - \frac{MR^2}{2}\omega$

$$= \frac{5}{2} m R^2 \omega$$

(c) Conceptual

5.  $a_C + \frac{L}{2}\alpha = a_1$

$$a_C - \frac{L}{2}\alpha = a_2$$

$$\Rightarrow \alpha = \frac{(a_1 - a_2)}{L} = 2 \text{ rad/s}^2$$

$$\alpha = \frac{(a_1 + a_2)}{2} = 2 \text{ m/s}^2$$

$$T_1 + T_2 - mg = m a_c$$

$$(T_1 - T_2) \frac{L}{2} - \frac{mL^2}{12} \alpha$$

$$\Rightarrow T_1 = 185/6 \text{ N and } T_2 = 175/5 \text{ N}$$

$$6. \quad \vec{L} = m(\vec{r} - \vec{r}_O) \times \vec{v}$$

$$\vec{L}_O = m(\vec{r} - \vec{r}_O) \times \vec{v} = mr\vec{e}_r \times m\vec{v} = m\omega r^2 \hat{z}$$

$$\vec{L} = m(\vec{r} - \vec{r}_P) \times \vec{v} = m(r\vec{e}_r - h\hat{z}) \times m\vec{v} = m\hbar\omega r\vec{e}_r + m\omega r^2 \hat{z}$$

$$7. \quad V_1 = 2V_2 \quad i \quad rw - V_1 = V_2 \quad \text{and by conservation of energy}$$

$$mgh = \frac{1}{2}mV_1^2 + \frac{1}{2}2mV_2^2 + \frac{1}{2}\frac{2}{5}mr^2\omega^2$$

$$\Rightarrow V_1 = \sqrt{\frac{5gh}{6}}$$

Non-conducting of mass M and radius R

$$8. \quad T \cos \theta = ma_{cm} x$$

$$Mg - T \sin \theta = ma_{cm} y$$

$$T \sin \theta \frac{L}{2} = I_{cm} \alpha$$

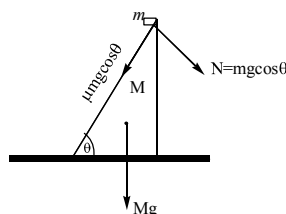
And apply thread constrained along length of the thread acceleration is zero.

$$\tau = \frac{mg \sin \theta}{1 + 3 \sin^2 \theta}$$

$$\alpha = \frac{6g \sin^2 \theta}{L(1 + 3 \sin^2 \theta)}$$

9. Due to torque of friction about CM,  $\omega$  eventually, decreases to zero, initially there is no translation. Friction is sufficient for pure rolling therefore after sometime pure rolling begin. There is no external force in  $x$  direction therefore momentum is conserved along  $x$  direction.

10.



$$N \sin \theta h > Mg \frac{b}{3} + \mu mg \cos \theta h \cos \theta$$

$$mg \cos \theta \cdot \sin \theta h > Mg \frac{b}{3} + \mu mgh \cos^2 \theta$$

$$3m \sin \theta (\sin \theta - \mu \cos \theta) > M$$

$$m > \frac{M}{3 \sin \theta (\sin \theta - \mu \cos \theta)}$$

11. from C.O.M.E,  $\frac{v^2}{R} = \frac{4}{3}g(1 - \cos \theta)$

From dynamics of circular motion  $mg \cos \theta = \frac{mv^2}{R}$

On solving above eqs.  $\theta = \cos^{-1} \frac{4}{7}$

$$K_R = \frac{mgR}{7}$$

When the cylinder loses its constant i.e., the frictional force vanishes, also torque due to gravitational force is zero. Hence its angular velocity is constant and rotational kinetic energy becomes constant, while its translational kinetic energy increases.

Applying conservation of energy at position (1) and position (3)

Decrease in gravitational P.E. = Gain in rotational K.E. + translational K.E.

$$K_T = (mgR) - \frac{mgR}{7} = \frac{6}{7}mgR$$

From (3) and (4), we have

$$\frac{K_T}{K_R} = \frac{\frac{6}{7}mgR}{\frac{mgR}{7}} \text{ or } \frac{K_T}{K_R} = 6$$

$$mgR + 0 = mgR \cos \theta + \frac{1}{2}I\omega^2 + \frac{1}{2}mv^2$$

But  $\omega = \frac{v}{R}$

And  $I = \frac{1}{2}mR^2$

Therefore,  $mgR = mgR \cos \theta + \frac{1}{2} \left( \frac{1}{2}mR^2 \right) \left( \frac{v^2}{R^2} \right) + \frac{1}{2}mv^2$

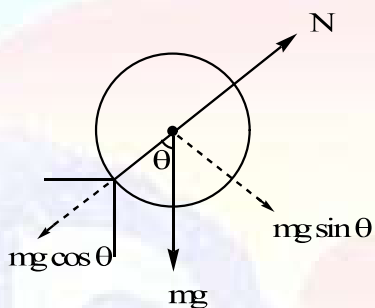
$$\text{Or } \frac{v^2}{R} = \frac{4}{3}g(1 - \cos \theta) \dots\dots\dots (1)$$

When the cylinder leaves the constant, normal reaction  $N=0$  and  $\theta = \theta_c$ . Hence

$$mg \cos \theta_c = \frac{mv^2}{R}$$

$$\text{Or } \frac{v^2}{R} = g \cos \theta_c \dots\dots\dots (2)$$

From eqns. (1) and (2),



$$\frac{4}{3}g(1 - \cos \theta_c) = g \cos \theta_c$$

$$\text{Or } \cos \theta_c = \frac{4}{7}$$

$$\text{Or } \theta_c = \cos^{-1}\left(\frac{4}{7}\right)$$

All the time it leaves the contact,

$$\cos \theta = \cos \theta_c = \frac{4}{7},$$

On substituting it in eqn. (1), we get

$$\therefore v = \sqrt{\frac{v}{2}gR\left(1 - \frac{4}{7}\right)} = \sqrt{\frac{4}{7}gR}$$

Therefore, speed of CM of cylinder just before it leaves the constant is  $\sqrt{\frac{4}{7}gR}$ .

(c) At the moment, when the cylinder leaves the constant,

$$v = \sqrt{\frac{4}{7}gR}$$

Therefore, rotational kinetic energy,

$$K_R = \frac{1}{2}I\omega^2$$

$$\text{Or } K_R = \frac{1}{2} \left( \frac{1}{2} m R^2 \right) \left( \frac{v^2}{R^2} \right) = \frac{1}{4} m v^2$$

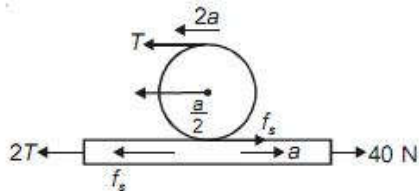
12. FBD of cylinder and block are as shown by Newton's laws

$$40 - 2T - f_s = 8a \quad (1)$$

$$T - f_s = 4a/2 \quad (2)$$

Subtracting equation (2) from equation (1)

$$40 - 3T = 6a$$



$$T = (40 - 6a)/3$$

Also, by  $\tau = I\alpha$ , we get

$$T \times R + f_s \times R = I \frac{3a}{2R}$$

$$\Rightarrow T + f_s = 3Ia/2R^2$$

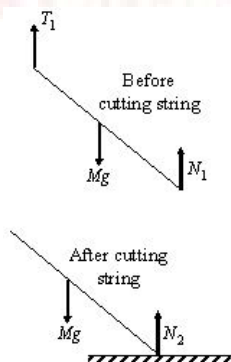
$$\Rightarrow (40 - 6a)/3 + (40 - 12a)/3 = 3Ia/2R^2 \Rightarrow a = 80/(18 + 9I/2R^2)$$

$$\text{As } \tau = I\alpha = I \times (3a/2R)$$

$$\therefore \tau = I\alpha = I \times (3a/2R)$$

$$\therefore \tau = \frac{3I}{2R} \frac{80}{\left(18 + \frac{9I}{2R^2}\right)} \Rightarrow \tau = \frac{3 \times 80}{18 + 9} = \frac{80}{9} \text{ N-m}$$

13. Before cutting string



$$N_1 = T = Mg/2$$

After cutting string

$$Mg - N_2 = Ma \quad (i)$$

$$N_2 \ell \sin \theta = Ma(2\ell)^2 / 12 \quad (ii)$$

By constraint that lower end will have acceleration only in horizontal direction.

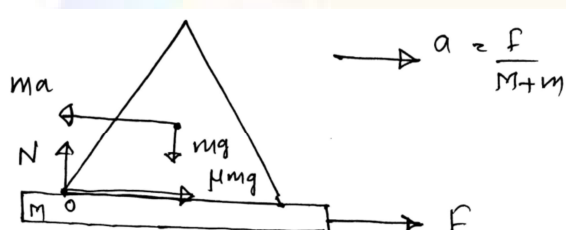
$$\ell a \sin \theta = a \quad (iii)$$

from (i), (ii) and (iii)

$$\Rightarrow N_2 = Mg / (3 \sin^2 \theta + 1)$$

$$\Rightarrow N_1 / N_2 = (3 \sin^2 \theta + 1)^{1/2} = 1$$

14.



$\tau$  about O = 0

$$ma \frac{H}{4} = mgr$$

$$\frac{F}{M+m} \cdot \frac{H}{4} = gr$$

$$F = 4N$$

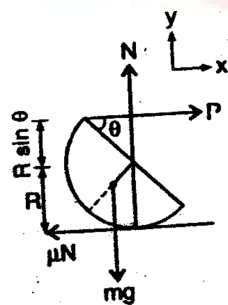
15.

$$P = \mu_s N$$

$$N = mg$$

$$\sum \tau = P(R + R \sin \theta) - mg \frac{4R}{3\pi} \sin \theta = 0$$

$$\text{Also } \sin \theta = \frac{3\pi\mu_s}{4 - 3\pi\mu_s}$$



16.  $mg - 2T = Ma$

$$2TR = IX = \frac{mr^2}{2} \left[ \frac{a}{R} \right]$$

$$T = \frac{Ma}{4} \quad \text{And } a = \frac{2a}{3}$$

17. Let  $m$  be the mass of the disc. Then translational energy of the disc is

$K_T = \frac{1}{2}mv^2$  When it ascends on a smooth track its rotational kinetic energy will remain same while translational kinetic energy will go on decreasing. At highest point.

$$K_T = mgh \quad \text{or} \quad \frac{1}{2}mv^2 = mgh \quad \text{or} \quad h = \frac{v^2}{2g} = \frac{(6)^2}{2 \times 19} = 1.8 \text{ m}.$$

18. Comparing this with linear motion,

$$\text{Power } P = F.v$$

$$\text{We have } P = \tau.\omega \quad \text{or} \quad I \cdot \left( \omega \frac{d\omega}{d\theta} \right) \cdot \omega = P$$

$$\text{or} \quad \omega^2 d\omega = \frac{P}{I} d\theta \quad \text{on integration, we find that } \omega \propto \theta^{1/3}$$

$$\text{or} \quad \omega \propto (n)^{1/3}.$$

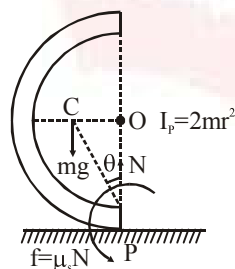
19. Centre of mass of semicircular ring lies at a distance of  $\frac{2r}{\pi}$  from the center.

$$OC = \frac{2r}{\pi}, \quad OP = r$$

$$\text{Hence, } CP = \sqrt{(OC)^2 + (OP)^2}$$

$$\text{or} \quad CP = 1.185r.$$

If there is no slipping, then the instantaneous axis of rotation passes through P.



Angular acceleration of ring about P is

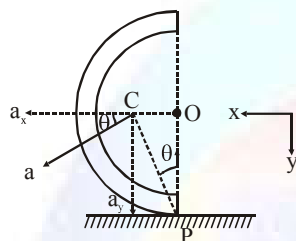
$$\alpha = \frac{\tau_P}{I_P} = \frac{(mg)(CO)}{(2mr^2)}$$

$$(mg)\left(\frac{2r}{\pi}\right) = \frac{g}{\pi r} \quad \dots(i)$$

Acceleration of center of mass,

$$a = (CP) (\alpha)$$

$$= (1.185r)\left(\frac{g}{\pi r}\right) = \left(\frac{1.185}{\pi}\right)g \quad \dots(ii)$$



From Newton's law,

$$\Sigma F_y = ma_y$$

$$\text{or } mg - N = ma_y$$

$$\text{or } mg - N = ma \sin \theta$$

$$\text{or } mg - N = m\left(\frac{1.185}{\pi}\right)g\left(\frac{2r}{\pi}\right)\left(\frac{1}{1.185r}\right)$$

$$= \frac{2}{\pi^2} mg = 0.2 mg$$

$$\text{or } N = mg - 0.2mg$$

$$\text{or } N = 0.8 mg \quad \dots(iii)$$

$$\Sigma F_x = ma_x$$

$$\text{or } \mu_s N = (m) (a \cos \theta)$$

$$\text{or } \mu_s (0.8mg) = (m)\left(\frac{1.185g}{\pi}\right)\left(\frac{r}{1.185r}\right)$$

$$\text{or } \mu_s = 0.398$$

Therefore, minimum value of static friction is 0.398.

$$20. \quad I_{system} = m_A r_A^2 + m_B r_B^2 = \left[ m_A m_B / (m_A + m_B) \right] r^2$$

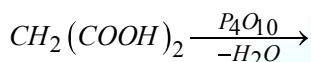
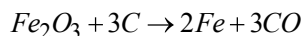
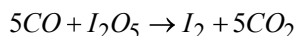
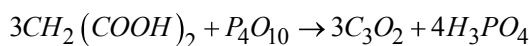
$$L_{system} = I_{system} \omega = \left[ m_A m_B / (m_A + m_B) \right] \omega r^2$$

$$I_B = m_B r_B^2 \left[ m_B m_A^2 / (m_A + m_B)^2 \right] r^2$$

$$L_B = I_B \omega = \left[ m_B m_A^2 / (m_A + m_B)^2 \right] \omega r^2 \quad \text{SS}$$

**CHEMISTRY**

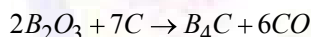
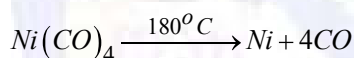
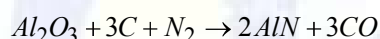
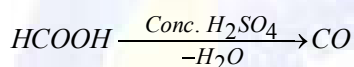
21. (b) Boron liberates hydrogen with alkali.
22. Both  $H_2$  and  $D_2$  show spin isomerism. Ortho- $H_2$  has more m.pt and b.pt than Para- $H_2$  both increase in temperature para convertes into Ortho. Due to opposite spins of protons Para  $H_2$  has lesser internal energy than Ortho- $H_2$
23. (a) Lanthanide contraction.  
(c) Due to large size of chlorine.
24. (b)  $5CO + I_2O_5 \rightarrow 5CO_2 + I_2$
25. Salts of aluminium suffer hydrolysis on heating.
26. Concept of silicates.  
(d) Silicon has less tendency to form multiple bonds with oxygen.
27. (a)  $SiO_2$  is acidic,  $PbO_2$  is amphoteric and CO is neutral.  
(c)  $LiHCO_3$  cannot exist in solid form.
28. Due to more rate of diffusion.  $H_2$  will be adsorbed rapidly than  $D_2$  on solid surface. Since O-D bond is polar while D-D bond is non-polar. So, exchange reactions with  $D_2O$  is easier. Atomic sizes of  $H_2$ ,  $D_2$  and  $T_2$  are same.
29.  $SiO_2 + NaOH \rightarrow Na_2SiO_3 \xrightarrow{H_2F_2} SiF_4 \xrightarrow{H_2O} H_2SiO_3 + H_2SiF_6$
30. Interstitial carbides are more denser and harder than pure metals.
31. A is  $C_3O_2$   $O=C=C=C=O$
32. In cyclic silicate ion  $Si_6O_{18}^{12-}$  the number of oxygen atoms shared is 6.
33. The number of monovalent oxygen atoms are present in the mineral kinoite  $Ca_2Cu_2Si_3O_{10} \cdot 2H_2O$  is 8.
34. Graphite,  $C_{60}$ ,  $(SiH_3)_3N$ ;  $AlCl_3$  (monomer),  $H_3BO_3$ , layered BN,  $B_3N_3H_6$  (Borazine) contain  $sp^2$  hybrid atoms.
35.  $Al_2O_3, SnO, SnO_2, PbO, PbO_2$
36.  $AlH_3 + 3H_2O \rightarrow Al(OH)_3 + 3H_2$   
 $Al_2O_3 + 3C + N_2 \rightarrow 2AlN + 3CO$   
 $AlN + 3H_2O \rightarrow Al(OH)_3 + NH_3$



37. All give Hydrogen gas.

38. Feldspars, Ultra marines, Zeolites

39.  $X = +2$ ,  $Y = +4$ .



### MATHS

41.  $\lim_{x \rightarrow 0^+} \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} = 1$  and  $\lim_{x \rightarrow 0^+} \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} = -1$ ,  $\lim_{x \rightarrow 0} f(x)$  exists, if  $\lim_{x \rightarrow 0} g(x) = 0$

42.  $f(x)$  is continuous at 'x' satisfying  $5x = x^2 + 6$  i.e.  $x = 2, 3$

43. Given  $f(x) \cdot f(y) = f(x) + f(y) + f(xy) - 2 \forall x, y \in R$

Put  $x = y = 1$  then  $f(1)^2 - 3f(1) + 2 = 0$

$$\Rightarrow f(1) = 1 \text{ (or) } f(1) = 2$$

If  $f(1) = 1$  then  $f(x) = 1 \forall x \in R$ .

A contradiction ( $\because$  degree of  $f(x)$  is positive)

$$\therefore f(1) \neq 1, \text{ hence } f(1) = 2$$

Replace 'y' with  $\frac{1}{x}$  then  $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right) (\because f(1) = 2)$

$\therefore f(x)$  must be in the form  $x^n + 1$  (or)  $x^n - 1$ .

$$\therefore f(4) = 65, f(x) = x^3 + 1 \Rightarrow f'(x) = 3x^2$$

44. Plotting the graph we get points of discontinuity at  $x = \frac{-3\pi}{2}, -\pi, 0, \frac{\pi}{2}$  and is continuous

at  $x = \frac{-\pi}{2}$ ,  $f(x)$  is not differentiable at  $x = \frac{-3\pi}{2}, \frac{\pi}{2}$

$$45. f(x) = \begin{cases} \cos x & 0 \leq x \leq \cos^{-1}(3/4) \\ 3/4 & \cos^{-1}(3/4) < x < \sin^{-1}(3/4) \\ \sin x & \sin^{-1}(3/4) \leq x \leq \pi/2 \end{cases}$$

$f(x)$  is continuous for all  $x$  and is not differentiable at 2 points

$$46. \lim_{x \rightarrow 5^-} f(x) = \lim_{x \rightarrow 5^+} f(x) = f(5)$$

$$1 = 5(b-1) = 3ab^2 \Rightarrow a = \frac{25}{108}, b = \frac{6}{5}$$

$$47. f(1^+) = \lim_{x \rightarrow 1^+} \left( x \left[ \frac{1}{x} \right] + x[x] \right) \\ = \lim_{x \rightarrow 1^+} (x(0) + x(1)) \\ = 1$$

$$f(1^-) = \lim_{x \rightarrow 1^-} \left( x \left[ \frac{1}{x} \right] + x[x] \right) \\ = \lim_{x \rightarrow 1^-} (x(1) + x(0)) \\ = 1$$

$$f(2^+) = \lim_{x \rightarrow 2^+} \left( x \left[ \frac{1}{x} \right] + x[x] \right) \\ = \lim_{x \rightarrow 2^+} (x(0) + x(2)) \\ = 4$$

$$f(2^-) = \lim_{x \rightarrow 2^-} \left( x \left[ \frac{1}{x} \right] + x[x] \right) \\ = \lim_{x \rightarrow 2^-} (x(0) + x(1)) \\ = 2$$

48. Clearly for  $x \neq 0$   $f(x)$  is differentiable.

At the point  $x = 0$ ,

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

$$= \lim_{x \rightarrow 0} \frac{x^n \sin(1/x)}{x}$$

$$= \lim_{x \rightarrow 0} x^{n-1} \sin(1/x) = 0$$

$f$  is differentiable at  $x=0$  also,  $\therefore f$  is differentiable  $\forall x$

49.  $f(1000)f(f(1000))=1$

$$\Rightarrow f(1000)f(999)=1$$

$$\Rightarrow 999f(999)=1$$

$$\therefore f(999) = \frac{1}{999}$$

The numbers 999 and  $\frac{1}{999}$  are in the range of  $f$ .

Hence, by intermediate value property of continuous function, function takes all values between 999 and  $\frac{1}{999}$ , then there exists

$$\alpha \in \left(\frac{1}{999}, 999\right) \text{ such that } f(\alpha) = 500$$

$$\text{Then } f(\alpha)f(f(\alpha))=1 \Rightarrow f(500) = \frac{1}{500}$$

$$\text{Similarly, } 199 \in \left(\frac{1}{999}, 999\right), \text{ thus } f(199) = \frac{1}{999}$$

But there is nothing to show that 1999 lies in the range of  $f$ . Thus (D) is not correct and (C) is also incorrect.

50.  $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} -\pi - 2\tan^{-1}x & \text{if } x < -1 \\ 2\tan^{-1}x & \text{if } -1 \leq x \leq 1 \\ \pi - 2\tan^{-1}x & \text{if } x > 1 \end{cases}$

$f(x)$  is continuous for all  $x$ , and is not differentiable  $x = -1, 1$

51.  $f(x) = \left\lceil \frac{x}{3} \right\rceil$  is not continuous at  $x = 3, 6, 9, 12, 15, 18, 21, 24, 27$

52.  $K = \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{2^{\sin^{-1}x} - 2^{\tan^{-1}x}}{2^{\tan x} - 2^{\sin x}}$

$$= \lim_{x \rightarrow 0} \frac{2^{\sin^{-1} x - \tan^{-1} x} - 1}{2^{\tan x - \sin x} - 1} \times \frac{2^{\tan^{-1} x}}{2^{\sin x}}$$

$$= \lim_{x \rightarrow 0} \frac{2^{\sin^{-1} x - \tan^{-1} x} - 1}{\sin^{-1} x - \tan^{-1} x} \times \frac{\sin^{-1} x - \tan^{-1} x}{\frac{2^{\tan x - \sin x} - 1}{\tan x - \sin x} \times (\tan x - \sin x)} \times \frac{2^{\tan^{-1} x}}{2^{\sin x}} = 1$$

53.  $\lim_{x \rightarrow 0} f(x) = f(0)$

$$\lim_{x \rightarrow 0} \frac{ab}{x^2} \left( \sin \frac{2x^2}{a} + \cos \frac{3x}{b} - 1 \right) = 3$$

$$ab \left( \frac{2}{a} - \frac{9}{2b^2} \right) = 3$$

$$4b^2 - 6b - 9a = 0$$

$$\therefore b \text{ is real, } D \geq 0 \Rightarrow a \geq -1/4$$

54. (i)  $f(x) = 1 \quad x \geq 0$

$$= -1 \quad x < 0$$

(ii)  $f(x) = -1 \quad x \geq 0$

$$= 1 \quad x < 0$$

(iii)  $f(x) = 1 \quad x > 0$

$$= -1 \quad x \leq 0$$

(iv)  $f(x) = -1 \quad x > 0$

$$= 1 \quad x \leq 0$$

(v)  $f(x) = 1 \quad \forall x \neq 0$

$$= -1 \quad x = 0$$

(vi)  $f(x) = -1 \quad \forall x > 0$

$$= 1 \quad x = 0$$

55. Draw the graph, we get the function is not differentiable and continuous at two points between  $x = -\pi/2$  and  $x = \pi/2$  also function is not continuous at  $x = \pi/2$  and  $x = -\pi/2$

$$56. \quad |f(x)| = \begin{cases} |x+2| & x < 0 \\ -x^2 - 2 & 0 \leq x < 1 \\ |x| & x \geq 1 \end{cases}$$

$$|f(x)| = \begin{cases} -x-2 & \text{if } x \leq -2 \\ x+2 & \text{if } -2 < x < 0 \\ x^2+2 & \text{if } 0 \leq x < 1 \\ x & \text{if } x \geq 1 \end{cases}$$

Clearly  $|f(x)|$  is discontinuous at  $x=1$

57.  $f(x) = a \sin x + be^x + cx^3$  if  $x \geq 0$

$$= -a \sin x + be^{-x} - cx^3 \text{ if } x < 0$$

Clearly  $f(x)$  is continuous at  $x=0$

$$f'(x) = \begin{cases} a \cos x + be^x + 3cx^2 & \text{if } x \geq 0 \\ -a \cos x - be^{-x} - 3cx^2 & \text{if } x < 0 \end{cases}$$

L.H.D=R.H.D

$$\Rightarrow -a - b = a + b \Rightarrow a + b = 0$$

58.  $\lim_{x \rightarrow 1^+} \frac{(x-1)^n}{\log_e \cos^m(x-1)} = -1$

$$\lim_{x \rightarrow 1^+} \frac{n(x-1)^{n-1}}{m \cdot \frac{-\sin(x-1)}{\cos(x-1)}} = -1$$

$$\Rightarrow \lim_{x \rightarrow 1^+} \frac{n(x-1)^{n-1}}{m \tan(x-1)} = 1$$

$$\Rightarrow \frac{n}{m} = 1, n=2, m+n=4$$

59.  $f(2x) - f(x) = x$

$$f(x) - f\left(\frac{x}{2}\right) = \frac{x}{2}$$

$$f\left(\frac{x}{2}\right) - f\left(\frac{x}{4}\right) = \frac{x}{4}$$

:

:

:

$$f\left(\frac{x}{2^{n-1}}\right) - f\left(\frac{x}{2^n}\right) = \frac{x}{2^n}$$

Adding  $f(2x) - f\left(\frac{x}{2^n}\right) = x + \frac{x}{2} + \frac{x}{4} + \dots + \frac{x}{2^n}$

$$\lim_{n \rightarrow \infty} \left[ f(2x) - f\left(\frac{x}{2^n}\right) \right] = \lim_{n \rightarrow \infty} x + \frac{x}{2} + \frac{x}{4} + \dots + \frac{x}{2^n}$$

$$f(2x) - 1 = \frac{x}{1 - \frac{1}{2}}$$

$$f(2x) = 2x + 1$$

$$f(x) = x + 1$$

60. Conceptual



# Success



## JEE ADVANCED WEEKEND TEST



### Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM to 10:30 AM

Date: 07-10-17

Max.Marks:198

Name of the Student: \_\_\_\_\_

I.D. NO:

|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|

### 2012\_PAPER-II

#### IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your roll number.

#### DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

### 07-10-17\_Jr.Super60-I\_Jee Adv\_WAT-20\_Weekend Syllabus

**MATHS:** SYLLABUS COVERED FROM 03-10-17 TO 06-10-17

**PHYSICS:** SYLLABUS COVERD FROM 03-10-17 TO 06-10-2017

**CHEMISTRY:** Detection of nitrogen, sulphur, phosphorus and halogens, Estimation of

carbon, hydrogen, nitrogen, halogens, sulphur, phosphorus, Hybridisation of carbon; Sigma and pi-bonds; Shapes of simple organic molecules

**JEE-ADVANCE-2012-P2-Model**

Time: 07:30 AM to 10:30 AM

**IMPORTANT INSTRUCTIONS**

Max Marks: 198

**PHYSICS:**

| Section                  | Question Type                                                        | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|----------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 1 – 8)     | Questions with Single Correct Choice                                 | 3         | -1         | 8         | 24          |
| Sec – II(Q.N : 9 – 14)   | Questions with Comprehension Type<br>(3 Comprehensions : 2+2+2 = 6Q) | 3         | -1         | 6         | 18          |
| Sec – III(Q.N : 15 – 20) | Questions with Multiple Correct Choice                               | 4         | 0          | 6         | 24          |
| <b>Total</b>             |                                                                      |           |            | <b>20</b> | <b>66</b>   |

**CHEMISTRY:**

| Section                  | Question Type                                                      | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|--------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 21 – 28)   | Questions with Single Correct Choice                               | 3         | -1         | 8         | 24          |
| Sec – II(Q.N : 29 – 34)  | Questions with Comprehension Type<br>(3 Comprehensions : 3+3 = 6Q) | 3         | -1         | 6         | 18          |
| Sec – III(Q.N : 35 – 40) | Questions with Multiple Correct Choice                             | 4         | 0          | 6         | 24          |
| <b>Total</b>             |                                                                    |           |            | <b>20</b> | <b>66</b>   |

**MATHEMATICS:**

| Section                  | Question Type                                                        | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|----------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 41 – 48)   | Questions with Single Correct Choice                                 | 3         | -1         | 8         | 24          |
| Sec – II(Q.N : 49 – 54)  | Questions with Comprehension Type<br>(3 Comprehensions : 2+2+2 = 6Q) | 3         | -1         | 6         | 18          |
| Sec – III(Q.N : 55 – 60) | Questions with Multiple Correct Choice                               | 4         | 0          | 6         | 24          |
| <b>Total</b>             |                                                                      |           |            | <b>20</b> | <b>66</b>   |

Sec: Jr. Super60-I

space for rough work

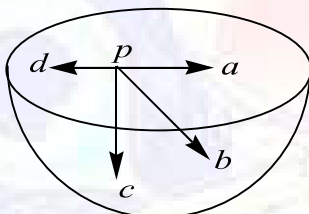
**2**

**SECTION – I**  
**(SINGLE CORRECT CHOICE TYPE)**

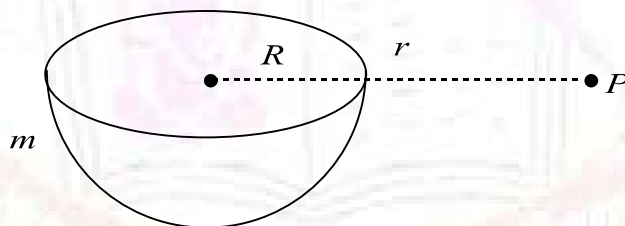
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

**Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.**

1. What is direction of gravitational field at a point  $P$ .  $P$  is on a plane containing biggest circle of uniform hollow hemisphere as shown.

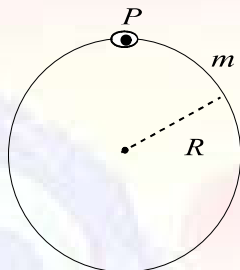


- A) a                      B) b                      C) c                      D) d
2. Let magnitude of gravitational field at a distance  $r$  from centre of uniform hemisphere of mass  $m$  at point  $P$  shown in figure be  $E$ . ( $R$  is radius of hemisphere)

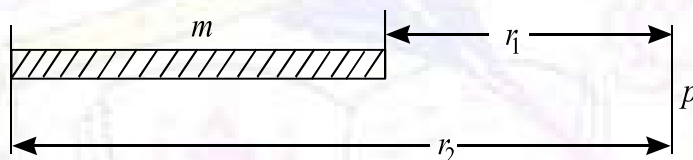


- A)  $E = \frac{Gm}{r^2}$                       B)  $E = \frac{Gm}{2r^2}$                       C)  $E > \frac{Gm}{r^2}$                       D)  $E = \frac{Gm}{(r-R)^2}$

3. Gravitational field at a point  $p$  in a small hole made in uniform thin spherical shell of mass  $m$  and radius  $R$  is approximately



- A)  $\frac{Gm}{R^2}$       B)  $\frac{Gm}{2R^2}$       C) 0      D)  $\frac{Gm}{3R^2}$
4. Gravitational field at point  $P$  due to rod of mass  $m$  is



- A)  $\frac{Gm}{r_1 r_2}$       B)  $\frac{Gm}{\left(\frac{r_1 + r_2}{2}\right)^2}$       C)  $\frac{Gm}{(r_1 + r_2)^2}$       D)  $Gm \left( \frac{1}{r_1} + \frac{1}{r_2} \right)^2$
5. A wire of uniform cross sectional area ' $a$ ' length  $l$  is having density which varies linearly from  $p_1$  at one end to  $p_2$  at other end. Both ends of wire are connected and

wire is transformed into a circular loop. Length of wire doesn't change. The component of gravitational field along the axis of circular loop made of given wire at a distance  $x$  from the centre of loop is

A)  $\frac{G(p_1 + p_2)al}{2x^2}$

B)  $\frac{G(p_1 + p_2)lax}{\left[\left(\frac{l}{2\pi}\right)^2 + x^2\right]^{\frac{3}{2}}}$

C)  $\frac{G(p_1 + p_2)lax}{2\left[\left(\frac{l}{2\pi}\right)^2 + x^2\right]^{\frac{3}{2}}}$

D)  $\frac{G\sqrt{p_1 p_2}lax}{2\left[\left(\frac{l}{2\pi}\right)^2 + x^2\right]^{\frac{3}{2}}}$

6. Gravitational field due to very long hollow cylinder of mass  $m$ , length  $l$ , and radius  $R$  at a point on the axis of cylinder just outside cylinder is ( $l \gg R$ )

A)  $\frac{2Gm}{Rl}$

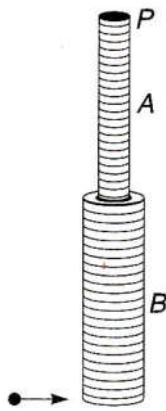
B)  $\frac{Gm}{Rl}$

C)  $\frac{Gm}{R^2}$

D)  $\frac{3Gm}{Rl}$

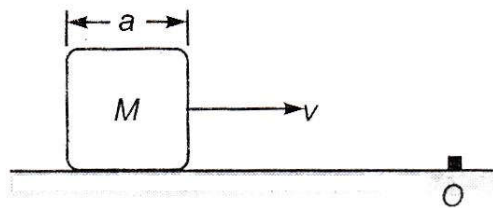
7. Two uniform rods  $A$  and  $B$  of length  $0.6\text{ m}$  each of masses  $0.01\text{ kg}$  and  $0.02\text{ kg}$  respectively are rigidly joined end to end. The combination is pivoted at the lighter end,  $P$  as show in figure. Such that it can freely rotate about point  $P$  in a vertical plane. A small object of mass  $0.05\text{ kg}$  moving horizontally, hits the lower end of the

combination and sticks to it. What should be the velocity of the object, so that the system could be raised to the horizontal position



- A) 6.3 m/s      B) 8.2 m/s      C) 4 m/s      D) 9 m/s

8. A cubical block of side  $a$  moving with velocity  $v$  on a horizontal smooth plane as shown. It hits a ridge at point  $O$ . The angular speed of the block after it hits  $O$  is



- A)  $3v/4a$       B)  $3v/2a$       C)  $\sqrt{3}v/\sqrt{2}a$       D) Zero

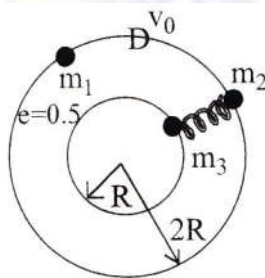
**SECTION - II**  
**(COMPREHENSION TYPE)**

This section contains **6 multiple choice questions** relating to three paragraphs with two questions on each paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE is correct**.

**Marking scheme +3 for correct answer, 0 if not attempted and -1 in all other cases.**

**Paragraph for Questions 9 and 10**

Three particles each of mass  $m$  can slide on fixed frictionless plane as shown in figure. Particle  $m_1$  moves with velocity  $V_0$  and hits particle  $m_2$ , the coefficient of restitution being  $e = \frac{1}{2}$ . Assuming that  $m_2$  and  $m_3$  are at rest initially and lie along a radial line before impact and the spring is initially unstretched. Consider only one collision between  $m_1$  and  $m_2$



9. The maximum velocity of  $m_3$  is

A)  $\frac{3}{5}V_0$

B)  $\frac{3}{10}V_0$

C)  $\frac{V_0}{2}$

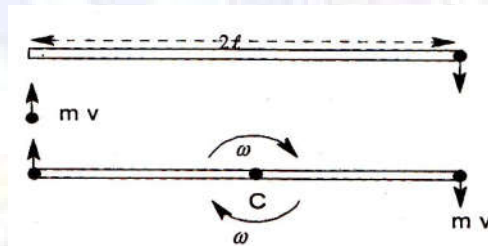
D)  $\frac{V_0}{5}$

10. The maximum stretch of the spring is

- A)  $\frac{3}{4}V_0\sqrt{\frac{m}{5k}}$       B)  $\frac{V_0}{4}\sqrt{\frac{m}{k}}$       C)  $\frac{V_0}{4}\sqrt{\frac{m}{3k}}$       D)  $\frac{3V_0}{4}\sqrt{\frac{m}{3k}}$

**Paragraph for Questions 11 and 12**

Two skaters, each of mass  $50\text{kg}$ , approach each other along parallel paths separated by  $3\text{m}$ . They have equal and opposite velocities of  $10\text{ms}^{-1}$ . The first skater carries a long light pole  $3\text{m}$  long and the second skater grabs the end of it as he passes on the frictionless surface and the two skaters are connected by the pole



11. The angular velocity of the pole after the skaters are connected is

- A)  $\frac{22}{3}\text{rads}^{-1}$       B)  $\frac{20}{3}\text{rads}^{-1}$       C)  $\frac{17}{3}\text{rads}^{-1}$       D)  $\frac{14}{3}\text{rads}^{-1}$

12. By pulling on the pole, the skaters reduce their distance apart to  $1\text{m}$ . The angular velocity now is

- A)  $45\text{rads}^{-1}$       B)  $50\text{rads}^{-1}$       C)  $55\text{rads}^{-1}$       D)  $60\text{rads}^{-1}$

**Paragraph for Questions 13 and 14**

A pair of stars rotates about a common center of mass. One of the stars has a mass  $M$  and the other has mass  $m$  such that  $M = 2m$ . The distance between the centres of the stars is  $d$  ( $d$  being large compared to the size of either star).

13. The period of rotation of the stars about their common centre of mass (in terms of  $d$ ,  $m$ ,  $G$ .) is

A)  $\sqrt{\frac{4\pi^2}{Gm}}d^3$       B)  $\sqrt{\frac{8\pi^2}{Gm}}d^3$       C)  $\sqrt{\frac{2\pi^2}{3Gm}}d^3$       D)  $\sqrt{\frac{4\pi^2}{3Gm}}d^3$

14. The ratio of the angular momentum of the two stars about their common centre of mass ( $L_m/L_M$ ) and the ratio of kinetic energies of the two stars ( $K_m/K_M$ ) is

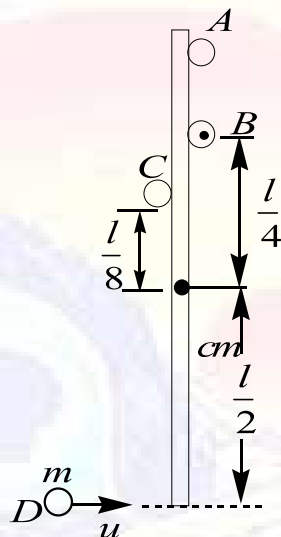
A) 1 & 1      B) 2 & 2      C) 2 & 9      D) 9 & 9

**SECTION – III**  
**(MULTIPLE CORRECT CHOICE TYPE)**

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

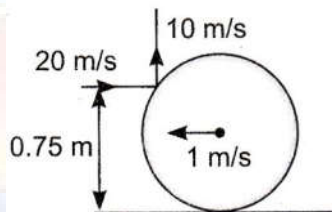
**Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.**

15. Rod inclined at an angle  $\theta$  with vertical strikes the horizontal frictionless ground. ( $\theta \neq 0$ ). The angular velocity of rod before striking is zero & linear velocity  $v$  is vertically downwards. Which of the following statements is / are correct?
- A) Angular momentum of rod is conserved during collision about point of impact on ground.
- B) For elastic collision the centre of rod rebounds back with velocity  $v$ .
- C) For elastic collision the centre of mass of rod rebounds back with velocity  $v$ .
- D) For perfectly inelastic collision the velocity of centre of mass just after collision becomes zero.
16. Mass of all the balls and uniform rod of length  $l$  is  $m$  and the system of balls  $A, B, C$  and rod was initially at rest on horizontal frictionless ground.  $A, B, C$  are touching rod but not sticking to it. Ball  $D$  collides with one end of rod perpendicularly. Which of the following is correct statement about direction of motion of small balls  $A, B, C, D$  & rod shown just after collision of ball  $D$  with rod?



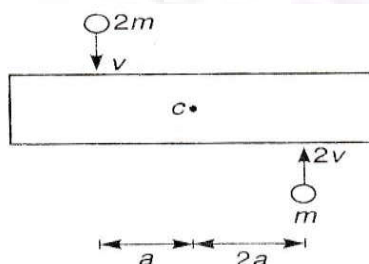
- A)  $A, B$  moves towards right just after collision
- B)  $C$  moves left just after collision
- C)  $D$  moves towards right just after collision
- D) Centre of mass of Rod moves towards right just after collision

17. A thin ring of mass  $2\text{ kg}$  and radius  $0.5\text{ m}$  is rolling without slipping on a horizontal plane with velocity  $1\text{ m/s}$ . A small ball of mass  $0.1\text{ kg}$ , moving with velocity  $20\text{ m/s}$  in the opposite direction, hits the ring at a height of  $0.75\text{ m}$  and goes vertically up with velocity  $10\text{ m/s}$ . Immediately after the collision



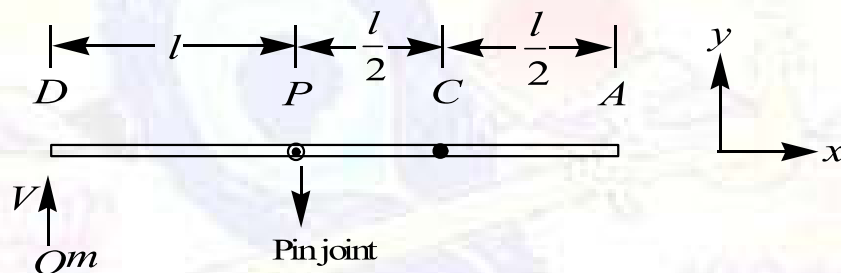
- A) The ring has pure rotation about its stationary CM
- B) The ring comes to a complete stop
- C) Friction between the ring and the ground is to the left
- D) There is no friction between the ring and the ground

18. A uniform bar of length  $6a$  and mass  $8m$  lies on a smooth horizontal table. Two point masses  $m$  and  $2m$  moving in the same horizontal plane with speed  $2v$  and  $v$  respectively, strike the bar [as shown in the figure] and stick to the bar after collision. Denoting angular velocity (about the centre of mass), total energy and centre of mass velocity by  $\omega$ ,  $E$  and  $v_c$  respectively, we have after collision



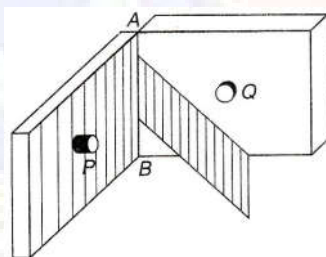
- A)  $v_c = 0$
- B)  $\omega = \frac{3v}{5a}$
- C)  $\omega = \frac{v}{5a}$
- D)  $E = \frac{3}{5}mv^2$

19. Two identical uniform rods connected end to end by a frictionless pin joint are placed along  $x$ -axis on a frictionless horizontal surface. These are free to rotate with respect to each other about vertical axis ( $z$ -axis) passing through pin joint. A particle of mass  $m$  moving with velocity  $v$  along  $y$ -axis strikes rod at one end as shown. The direction of velocities of different points on rod just after collision will be



- A)  $C$  along  $\hat{j}$     B)  $C$  along  $-\hat{j}$     C)  $A$  along  $-\hat{j}$     D)  $A$  along  $\hat{j}$
20. Two heavy metallic plates are joined together at  $90^\circ$  to each other. A laminar sheet of mass  $30\text{ kg}$  is hinged at the line  $AB$  joining the two heavy metallic plates. The hinges are frictionless. The moment of inertia of the laminar sheet about an axis parallel to  $AB$  and passing through its centre of mass is  $1.2\text{ kg-m}^2$ . Two rubber obstacles  $P$  and  $Q$  are fixed, one on each metallic plate at a distance  $0.5\text{ m}$  from the line  $AB$ . This

distance is chosen, so that the reaction due to the hinges on the laminar sheet is zero during the impact. Initially the laminar sheet hits one of the obstacles with an angular velocity  $1 \text{ rad/s}$  and turns back. If the impulse on the sheet due to each obstacles is  $6 \text{ N-s}$ . The location of centre of mass of laminar sheet from  $AB$  is  $x$  & angular velocity of laminar sheet just after impact is  $\omega$ . Values of  $x$  and  $\omega$  are?



- A)  $\omega = 0.1 \text{ rad/s}$     B)  $\omega = 1 \text{ rad/s}$     C)  $x = 0.1 \text{ m}$     D)  $x = 0.2 \text{ m}$

**CHEMISTRY****Max. Marks: 66****SECTION – I  
(SINGLE CORRECT CHOICE TYPE)**

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

**Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.**

21. In Lassaigne's test, the organic compound is fused with a piece of sodium metal in order to
- A) Increase the volatility of the compound  
B) Decrease the melting point of the compound  
C) Convert the ionic compound into a mixture of covalent compounds  
D) Convert the covalent compound into a mixture of ionic compounds
22. During the test for halogens, sodium extract is heated with  $\text{HNO}_3$ . The reason for this is
- A) To decompose  $\text{NaCN}$  and  $\text{Na}_2\text{S}$       B) To make  $\text{Na}_2\text{S}$  soluble  
C) To make  $\text{NaCN}$  soluble      D) to make silver halides insoluble
23. An organic compound was analyzed by Duma's method. 0.546g of an organic compound gave 38ml of nitrogen at 350K and 732 mm pressure. Calculate the percentage of nitrogen in the compound (vapour pressure of water at 350K is 32 mm)
- A) 6.25%      B) 25%      C) 12.5%      D) 15%

**Sec: Jr. Super60-I****space for rough work****15**

24. Which of the following is the most electronegative carbon?
- A)  $SP$  hybridised  
B)  $SP^2$  hybridized  
C)  $SP^3$  hybridised  
D) all have equal electronegative
25. Allyl isocyanide [  $CH_2CHCH_2NC$  ] has
- A) 9 sigma and 4 pi bonds  
B) 8 sigma and 5 pi bonds  
C) 8 sigma, 3 pi and 4 non-bonded electrons  
D) 9 sigma, 3 pi and 2 non-bonded electrons
26. Lassaigne's test for the detection of nitrogen will fail in case of
- A)  $NH_2CONH_2$   
B)  $H_2NCONHNH_2.HCl$   
C)  $H_2NNH_2.HCl$   
D)  $C_6H_5NHNH_2.HCl$
27. The number of tetrahedral bond angles present in methane?
- A) 4  
B) 2  
C) 6  
D) 8
28. In the Lassaigne's test for nitrogen in an organic compound, the Prussian blue color is obtained due to the formation of
- A)  $Na_4[Fe(CN)_6]$   
B)  $Fe_4[Fe(CN)_6]_3$   
C)  $Fe_2[Fe(CN)_6]$   
D)  $Fe_3[Fe(CN)_6]_4$

**SECTION - II**  
**(COMPREHENSION TYPE)**

This section contains **6 multiple choice questions** relating to two paragraphs with three questions on each paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE is correct**.

**Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.**

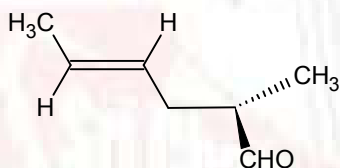
**Paragraph for Questions 29 and 30:**

During the estimation of nitrogen in an organic compound by Kjeldhal's method, the ammonia evolved from 0.5g of the compound, was absorbed in 50ml of 0.5 M  $H_2SO_4$ . The residual acid required 60ml of 0.5M  $NaOH$  solution

29. What volume of  $H_2SO_4$  is used by  $NH_3$  in the process?  
 A) 50ml                      B) 10ml                      C) 20 ml                      D) 30 ml
30. The percentage of nitrogen in the compound is  
 A) 2.8                      B) 56                      C) 18.6                      D) 14

**Paragraph for Questions 31 and 32:**

Answer questions 31 and 32 based on the following structure.



31. How many pure atomic orbital's of carbon & hydrogen are involved in bond formation in the above molecule?
- A) 12                      B) 9                      C) 15                      D) 16
32. How many  $sp^3$  hybridized carbon orbital's are present in above molecule?
- A) 4                      B) 12                      C) 8                      D) 16

**Paragraph for Questions 33 and 34:**

N, S, halogens and phosphorous present in an organic compound are detected by Lassaigne's test, by fusing the compound with *Na* metal. The compounds formed are extracted from fused by boiling it with distilled water. This extract is known as sodium fusion extract or Lassaigne's extract.

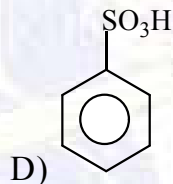
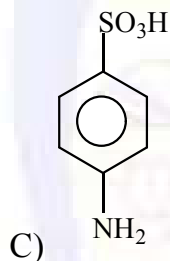
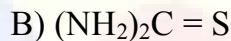
33. Violet color in the detection of sulphur with sodium nitroprusside is due to the formation of
- A)  $Na_3[Fe(CN)_5NOS]$                       B)  $Na_4[Fe(CN)_5NOS]$   
 C)  $Na_4[Fe(CN)_4S]$                       D)  $Na_2[Fe(CN)_5S]$
34. An organic compound containing *N*, *S*, and *O* as extra elements is fused with metallic sodium and then extracted with distilled water. Which species is not present in the Lassaigne's extract?
- A)  $NO_3^-$                       B)  $C\bar{N}$                       C)  $\bar{S}CN$                       D)  $S^{2-}$

**SECTION – III**  
**(MULTIPLE CORRECT CHOICE TYPE)**

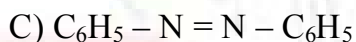
This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

**Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.**

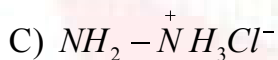
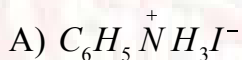
35. Which of the following compounds may give blood red coloration while performing Lassaigne's test for nitrogen?



36. Kjeldahl's method cannot be used for estimation of nitrogen in



37. Which of the following compounds will give white precipitate with  $\text{AgNO}_3$ ?



38. Which has at least one  $sp^2$  hybridized carbon atom?
- A)  $\text{CH}_2 = \bar{\text{C}}\text{H}$                       B)  $\text{H} - \text{C} \equiv \bar{\text{C}}$
- C)  $\text{H}_2\text{C} = \text{C} = \text{CH}_2$                   D)  $\text{O} = \text{C} = \text{C} = \text{C} = \text{O}$
39. Which of the following is the correct statement regarding ethene?
- A) All the atoms are in the same plane
- B)  $\pi$  Bond provides the most reactive centre in the molecule.
- C)  $SP^2$  Hybridized carbon atoms are present
- D) One  $\sigma_{\text{C}-\text{C}}$  bond is present
40. Which of the following is correct?
- A) Oxidation state of Fe in sodium Nitro prusside is 2+
- B) AgI is in yellow color due to polarization effect
- C) AgBr is a yellow ppt. and is sparingly soluble in excess  $\text{NH}_4\text{OH}$
- D) Yellow ppt is the detection of phosphorus when an organic compound is heated with  $\text{Na}_2\text{O}_2$  and then boiled with con.  $\text{HNO}_3$ , followed by the addition of Ammonium molybdate is  $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$

**MATHS:****Max. Marks: 66****SECTION – I  
(SINGLE CORRECT CHOICE TYPE)**

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

**Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.**

41. If  $y$  is a function of  $x$  and  $\ln(x+y) = 2xy$ , then  $y'(0) =$   
 A) 1                      B) -1                      C) 2                      D) 0
42. If  $\alpha$  be a repeated root of a quadratic equation  $f(x) = 0$  and  $A(x), B(x), C(x)$  be polynomials of degree 3, 4, 5 respectively, then  

$$F(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$
 is  
 A) not divisible by  $(x-\alpha)$                       B) divisible by  $(x-\alpha)$ , not by  $(x-\alpha)^2$   
 C) divisible by  $(x-\alpha)^2$ , not by  $f(x)$                       D) divisible by  $f(x)$
43. Let  $P(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ . If  $|P(x)| \leq |e^{x-1} - 1|, \forall x \geq 0$ , then the maximum value of  $|a_1 + 2a_2 + 3a_3 + \dots + na_n|$  is  
 A)  $\frac{1}{n}$                       B) 1                      C)  $n$                       D)  $e$

44. Let  $g(x) = f^{-1}(x)$  and  $f$  is a bijective function such that  $f^{-1}(x) = \frac{1}{1+x^4}$ , then  $g^{-1}(x)$  is equal to

A)  $(1+g(x)^4)^{-1}$     B)  $(1+(f(x))^4)^{-1}$     C)  $1+(f(x))^4$     D)  $1+(g(x))^4$

45. If  $f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \dots + \frac{x^2}{2} + x + 1$  then

A)  $f^{-1}(1) = f^{-1}(0)$     B)  $100f^{-1}(1) = f^{-1}(0)$

C)  $f^{-1}(1) = f^{-1}(-1)$     D)  $f^{-1}(1) = 100f^{-1}(0)$

46. Suppose that  $f$  is differentiable function with the property

$f(x+y) = f(x) + f(y) + x^2y^2$  and  $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 100$ , then  $f'(0)$  is equal to

A) 100

B) 20

C) 30

D) none of these

47. If  $\sqrt{x+y} + \sqrt{y-x} = c$ , then  $\frac{d^2y}{dx^2}$  is

A)  $\frac{2}{c}$

B)  $\frac{-2}{c^2}$

C)  $\frac{2}{c^2}$

D)  $\frac{-2}{c}$

48. If  $\sqrt{x^2 + y^2} = a e^{\tan^{-1}\left(\frac{y}{x}\right)}$ ;  $(a > 0), (y(0) = a e^{\frac{\pi}{2}})$  then  $y^{(1)}(0) =$

A)  $\frac{a}{2} e^{\pi/2}$

B)  $a e^{-\pi/2}$

C)  $\frac{-2}{a} e^{-\pi/2}$

D)  $\frac{a}{2} e^{-\pi/2}$

**SECTION - II**  
**(COMPREHENSION TYPE)**

This section contains **6 multiple choice questions** relating to three paragraphs with two questions on each paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE is correct**.

**Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.**

**Paragraph for Questions 49 and 50**

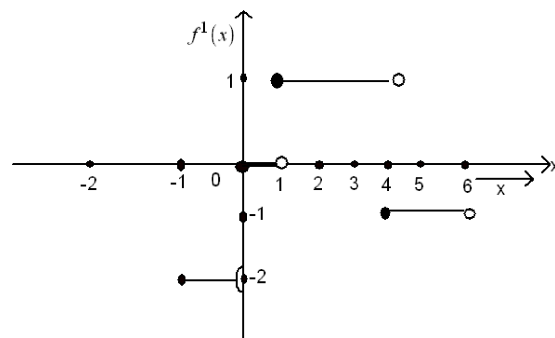
Let  $y = f(x)$  be a continuous function for  $-1 \leq x \leq 6$  such that  $f(0) = 0$  and

(1) The graph of  $f^1(x)$  is made of line segments joined such that left end point is included and right end point is excluded in each sub interval,

( $f^1(x)$  takes integral values)

(2) The graph starts at the point  $(-1, -2)$ .

(3) The derivatives of  $f(x)$ , where defined, agrees with the step pattern as shown here.



Using the above information answer the following.

49. The range of  $f(x)$  is \_\_\_\_\_  
 A)  $[-1, 3]$       B)  $(0, 3)$       C)  $[0, 3]$       D)  $(0, 3]$
50. Number of integral roots of the equation  $f(x) = 1$  is  
 A) Exactly 3      B) exactly 4      C) exactly 2      D) none

**Paragraph for Questions 51 and 52**

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuous and differentiable function such that

$$f(x+y) = f(x) \cdot f(y) \quad \forall x, y, f(x) \neq 0 \text{ and } f(0) = 1 \text{ and } f'(0) = 2.$$

$$\text{Let } g(xy) = g(x) \cdot g(y) \quad \forall x, y \text{ and } g'(1) = 2; g(1) \neq 0$$

51. Identify the correct option

A)  $f(2) = e^4$ ;  $g(3) = 9$

B)  $f(2) = 2e^2$ ;  $g(3) = 9$

C)  $f(1) < 4$ ;  $g(3) = 3$

D)  $f(3) > 729$ ;  $g(3) = 3$

52. The number of values of  $x$ , where  $f(x)=g(x)$ :

A) 0

B) 1

C) 2

D) 3

**Paragraph for Questions 53 and 54**

A function of the form  $x = f(t)$ ,  $y = g(t)$  where 't' is a parameter, is called as a parametric function, where  $f$ ,  $g$  are differentiable functions of 't'. Also it can be used that  $\left(\frac{dy}{dx}\right)\left(\frac{dx}{dt}\right) = \left(\frac{dy}{dt}\right)$ . Now answer the following questions on the basis of above information.

53. If  $x = 3 \cos \theta - \cos^3 \theta$ ,  $y = 3 \sin \theta - \sin^3 \theta$  then  $\frac{dy}{dx} = \text{-----}$  at  $\theta = \frac{\pi}{4}$

A)  $-\frac{2}{3}$

B)  $-1$

C)  $\frac{1}{2}$

D) 1

54.  $x = \sec \theta - \cos \theta$ ,  $y = \sec^6 \theta - \cos^6 \theta$ ;  $\theta \neq (2k+1)\pi/2, k \in \mathbb{Z}$ , then  $\left(\frac{dy}{dx}\right)^2 \left(\frac{x^2+4}{y^2+4}\right) = \text{-----}$

A) 36

B) 6

C) 30

D) None of these

**SECTION – III**  
**(MULTIPLE CORRECT CHOICE TYPE)**

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

**Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.**

55. A twice differentiable function  $f(x)$  is defined for all real numbers and satisfies the following conditions:

$$f(0)=2; \quad f'(0) = -5 \text{ and } f''(0)=3.$$

The function  $g(x)$  is defined by  $g(x) = e^{ax} + f(x) \forall x \in R$ , where 'a' is any constant.

If  $g'(0) + g''(0) = 0$  then 'a' can be equal to:

- A) 1                      B) -1                      C) 2                      D) -2

56. Let  $f(x)$  be a non constant polynomial satisfying the relation

$$f(x).f(y) = f(x) + f(y) + f(xy) - 2 \text{ for all real } x \text{ and } y \text{ and } f(0) \neq 2, \text{ suppose}$$

$$f(4) = 65. \text{ Then}$$

- A)  $f^1(x)$  Is a polynomial of degree 2  
B) Roots of  $f^1(x) = 2x + 1$  are real  
C)  $xf^1(x) = 3[f(x) - 1]$   
D)  $f^1(-1) = 3$

57. If  $\frac{d}{dx}\{f(x)\} = g(x)$  where  $g(x) = e^{-x} \cdot f(x) + e^x \cdot f(-x)$  then

(given  $f(0) = 0$ ).

A)  $g(x)$  Is an even function

B)  $f(x)$  is an odd function

C)  $g^1(0) = f^1(0) = g(0)$

D)  $g^1(0) \neq f(0)$

58. If  $y = e^{x \sin(x^3)} + (\tan x)^x$  then  $\frac{dy}{dx}$  may be equal to:

A)  $e^{x \sin(x^3)} \left[ 3x^3 \cos(x^3) + \sin(x^3) \right] + (\tan x)^x \left[ \ln \tan x + 2x \operatorname{cosec} 2x \right]$

B)  $e^{x \sin(x^3)} \left[ x^3 \cos(x^3) + \sin(x^3) \right] + (\tan x)^x \left[ \ln \tan x + 2x \operatorname{cosec} 2x \right]$

C)  $e^{x \sin(x^3)} \left[ \cos(x^3) + x^3 \sin(x^3) \right] + (\tan x)^x \left[ \ln \tan x + 2x \operatorname{cosec} 2x \right]$

D)  $e^{x \sin(x^3)} \left[ 3x^3 \cos(x^3) + \sin(x^3) \right] + (\tan x)^x \left[ \ln \tan x + \frac{x \sec^2 x}{\tan x} \right]$

59. Let  $g(x) = \ln f(x)$  where  $f(x)$  is twice differentiable positive function on  $(0, \infty)$

such that  $f(x+1) = x f(x)$  then for  $N = 1, 2, 3, \dots$ ;  $g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) = \dots$

A)  $-4 \left( 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right)$

B)  $4 \left( 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right)$

C)  $-4 \left( 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right)$

D)  $4 \left( 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right)$

60. If  $f(x-y)$ ,  $f(x)f(y)$  and  $f(x+y)$  are in A.P  $\forall x, y \in \mathbb{R}$ ;  $f(0) \neq 0$  then

A)  $f(4) = f(-4)$

B)  $f(2) + f(-2) = 0$

C)  $f'(4) + f'(-4) = 0$

D)  $f'(2) = f'(-2)$



# Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. SUPER60-I

JEE ADVANCED

DATE : 07-10-17

TIME : 07:30 AM to 10:30 AM

WAT-20

MAX MARKS : 198

## KEY SHEET

2012-P2

### PHYSICS

|    |    |    |    |    |     |    |    |    |    |
|----|----|----|----|----|-----|----|----|----|----|
| 1  | C  | 2  | C  | 3  | B   | 4  | A  | 5  | C  |
| 6  | B  | 7  | A  | 8  | A   | 9  | B  | 10 | A  |
| 11 | B  | 12 | D  | 13 | D   | 14 | D  | 15 | A  |
| 16 | CD | 17 | AC | 18 | ACD | 19 | BD | 20 | BC |

### CHEMISTRY

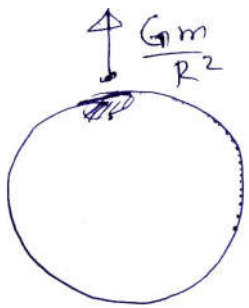
|    |     |    |     |    |    |    |      |    |      |
|----|-----|----|-----|----|----|----|------|----|------|
| 21 | D   | 22 | A   | 23 | A  | 24 | A    | 25 | D    |
| 26 | C   | 27 | C   | 28 | B  | 29 | C    | 30 | B    |
| 31 | C   | 32 | D   | 33 | B  | 34 | A    | 35 | BC   |
| 36 | ABC | 37 | BCD | 38 | AC | 39 | ABCD | 40 | ABCD |

### MATHS

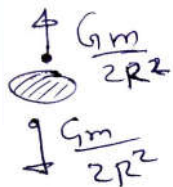
|    |      |    |    |    |    |    |   |    |    |
|----|------|----|----|----|----|----|---|----|----|
| 41 | A    | 42 | D  | 43 | B  | 44 | D | 45 | D  |
| 46 | A    | 47 | C  | 48 | C  | 49 | C | 50 | C  |
| 51 | A    | 52 | B  | 53 | B  | 54 | A | 55 | AD |
| 56 | ABCD | 57 | AB | 58 | AD | 59 | A | 60 | AC |

**SOLUTIONS****PHYSICS**

1. Conceptual
2. Conceptual
3. Net Gravitational field



Field due to local surface



Field at hole = (Net Gravitational field) – (Field due to local surface)

$$= \left( \frac{Gm}{2R^2} \right)$$

4.



$$E = \int \frac{Gdm}{r^2} = \int \frac{G\lambda dr}{r^2}$$

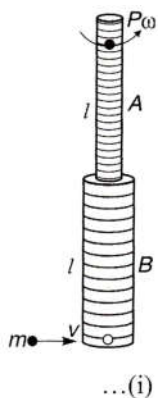
$$= G\lambda \left( \frac{1}{r} - \frac{1}{r_2} \right) = \frac{G\lambda(r_2 - r_1)}{r_1 r_2} \quad E = \frac{Gm}{r_1 r_2}$$

5.  $m = \frac{(p_1 + p_2)al}{2} R = \frac{l}{2\pi}$

$$E = \frac{Gm}{(R^2 + x^2)^{\frac{3}{2}}}$$

6. Conceptual

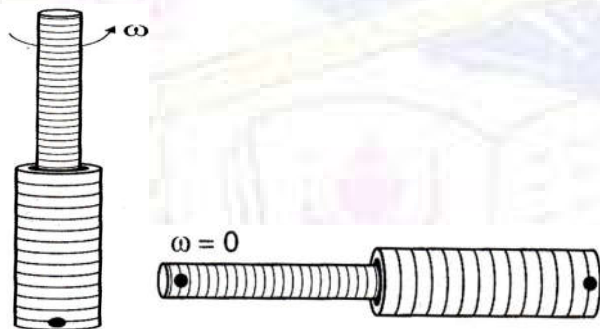
7. System is free to rotate but not free to translate. During collision, net torque on the system (rod A + rod B + mass  $m$ ) about point P is zero. Therefore, angular momentum of system before collision = angular momentum of system just after collision (about P). Let  $\omega$  be the angular velocity of system just after collision, then



$L_i = L_f \Rightarrow mv(2l) = I\omega$  Here,  $I$  = moment of inertia of system about p

$$= m(2l)^2 + m_A \left( \frac{l^2}{3} \right) + m_B \left[ \frac{l^2}{12} + \left( \frac{l}{2} + l \right)^2 \right] \text{ Given, } l = 0.6m, m = 0.05\text{ kg}, m_A$$

$= 0.01\text{ kg}$  and  $m_B = 0.02\text{ kg}$ . Substituting the values, we get  $I = 0.09\text{ kg} - m^2$



Therefore, from Eq. (i)

$$\omega = \frac{2mvl}{I} = \frac{(2)(0.05)(v)(0.6)}{0.09}$$

$$\omega = 0.67v$$

Now, after collision, mechanical energy will be conserved. Therefore, decrease in rotational KE = increase in gravitational PE

$$\text{or } \frac{1}{2} I \omega^2 = mg(2l) + m_A g \left( \frac{l}{2} \right) + m_B g \left( l + \frac{1}{2} \right)$$

$$\text{or } \omega^2 = \frac{gl(4m + m_A + 3m_B)}{I}$$

$$= \frac{(9.8)(0.6)(4 \times 0.05 + 0.01 + 3 \times 0.02)}{0.09}$$

$$= 17.64 (\text{rad} / \text{s})^2$$

$$\therefore \omega = 4.2 \text{ rad} / \text{s}$$

Equation Eqs. (ii) and (iii), we get

$$v = \frac{4.2}{0.67} \text{ m} / \text{s} \text{ or } v = 6.3 \text{ m} / \text{s}$$

8. Conservation of angular momentum

$$\left( \frac{ma^2}{6} + \frac{ma^2}{2} \right) \omega = \left( \frac{mva}{2} \right)$$

$$a \left( \frac{1}{3} + 1 \right) \omega = v$$

$$\omega = \frac{3v}{4a}$$

9. Conservation of angular momentum during collision of  $m_1$  &  $m_2$

$$m_1 V_0 (2R) = (m_1 V_1) 2R + (m_2 V_2) 2R$$

$$V_0 = V_1 + V_2 \text{ ----- (1)}$$

$$V_2 - V_1 = \frac{V_0}{2} \text{ ----- (2)}$$

$$2V_2 = \frac{3V_0}{2} \quad V_2 = \frac{3V_0}{4}$$

By conservation of angular momentum when the masses will have same  $\omega$

$$m \left( \frac{3V_0}{2} \right) R = 5mR^2 \omega$$

$$\frac{3V_0}{10} = R\omega$$

$$= (\text{maximum speed of } m_3)$$

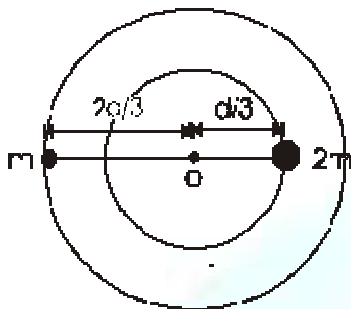
10.  $\frac{1}{2} kx^2 + \frac{1}{2} (5mR^2) \omega^2 = \frac{1}{2} m \left( \frac{3V_0}{4} \right)^2$

$$x = \sqrt{\frac{mV_0^2}{80k}}$$

11. Conceptual

12. Conceptual

13. Let the angular speed of revolution of both stars be  $\omega$  about the common centre, that is, centre of mass of system.



The centripetal force on star of mass  $m$  is

$$m\omega^2 \frac{2d}{3} = \frac{Gm(2m)}{d^2}. \text{ Solving we get } T = \sqrt{\frac{4\pi^2}{3Gm}} d^3$$

14. The ratio of angular momentum is simply the ratio of moment of inertia about center of mass of system.

$$\frac{L_m}{L_M} = \frac{I_m}{I_M \omega} = \frac{m \left( \frac{2d}{3} \right)^2}{2m \left( \frac{d}{3} \right)^2} = 2$$

Similarly, The ratio of kinetic energy is simply the ratio of moment of inertia about center of mass of system.

$$\frac{K_m}{K_M} = \frac{\frac{1}{2} I_m \omega^2}{\frac{1}{2} I_M \omega^2} = \frac{m \left( \frac{2d}{3} \right)^2}{2m \left( \frac{d}{3} \right)^2} = 2$$

15. Conceptual  
16. Conceptual  
17. The data is incomplete. Let us assume that friction from ground on ring is not impulsive during impact. From linear momentum conservation in horizontal direction.

We have

$$(-2 \times 1) + (0.1 \times 20) = (0.1 \times 0) + (2 \times v) \quad \begin{matrix} \text{---ve} & \text{+ve} \\ \longleftarrow & \longrightarrow \end{matrix}$$

Here,  $v$  is the velocity of CM of ring after impact. Solving the above equation, we have  $v = 0$  Thus CM becomes stationary.

$\therefore$  Correct option is (a).

Linear impulse during impact

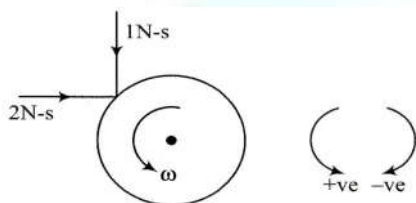
(i) In horizontal direction

$$J_1 = \Delta p = 0.1 \times 20 = 2N - s$$

(ii) In vertical direction

$$J_2 = \Delta p = 0.1 \times 10 = 1N - s$$

Writing the equation (about CM)



Angular impulse = Change in angular momentum We have,

$$1 \times \left( \frac{\sqrt{3}}{2} \times \frac{1}{2} \right) - 2 \times 0.5 \times \frac{1}{2} = 2 \times (0.5)^2 \left[ \omega - \frac{1}{0.5} \right]$$

Solving this equation  $\omega$  comes out to be positive or  $\omega$  anti-clockwise. So just after collision rightwards slipping is taking place. Hence, friction is leftwards. Therefore, option (c) is also correct.  $\therefore$  Correct options are (a) and (c)

18.  $P_i = 0$

$$\therefore P_f = 0 \text{ or } v_C = 0$$

$$L_i = L_f \text{ or } (2mv)a + (2mv)(2a) = I\omega \dots\dots\dots(i)$$

$$\text{Here, } I = \frac{(8m)(6a)^2}{12} + m(2a)^2 + (2m)(a^2) = 30ma^2$$

$$\text{Substituting in Eq. (i), we get } \omega = \frac{v}{5a}$$

$$\text{Further, } E = \frac{1}{2} I \omega^2 = \frac{1}{2} \times (30ma^2) \left( \frac{v}{5a} \right)^2 = \frac{3mv^2}{5}$$

$\therefore$  Correct options are (a), (c) and (d).

19. Conceptual

20. Let  $r$  be the perpendicular distance of CM from the line AB and  $\omega$  the angular velocity of the sheet just after colliding with rubber obstacle for the first time.

Obviously the linear velocity of CM before and after collision will be

$v_i = (r)(1 \text{ rad/s}) = r$  and  $v_f = r\omega$ .  $v_i$  and  $v_f$  will be in opposite directions.

Now, linear impulse on CM

= change in linear momentum of CM

$$\text{or } 6 = m(v_f + v_i) = 30(r + r\omega)$$

$$\text{or } r(1 + \omega) = \frac{1}{5} \quad \dots\dots\dots(i)$$

Similarly, angular impulse about AB = change in angular momentum about AB

Angular impulse = Linear impulse

× Perpendicular distance of impulse from AB

$$\text{Hence, } 6(0.5 \text{ m}) = I_{AB}(\omega + 1)$$

(Initial angular velocity = 1 rad/s)

$$\text{or } 3 = [I_{CM} + Mr^2](1 + \omega)$$

$$\text{or } 3 = [1.2 + 30r^2](1 + \omega) \quad \dots\dots\dots(ii)$$

Solving Eqs. (i) and (ii) for r, we get

$$r = 0.4 \text{ m and } r = 0.1 \text{ m}$$

But at  $r = 0.4 \text{ m}$ ,  $\omega$  comes out to be negative (-0.5 rad/s)

Which is not acceptable. Therefore,

(a)  $r$  = distance of CM from AB = 0.1m

(b) Substituting  $r = 0.1 \text{ m}$  in Eq. (i), we get  $\omega = 1 \text{ rad/s}$  ie, the angular velocity with which sheet comes back after the first impact is 1 rad/s.

(c) Since, the sheet returns with same angular velocity of 1 rad/s, the sheet will never come to rest.

## CHEMISTRY

21. To convert the covalent compound into a mixture of ionic compounds

22.  $\text{NaCN}$  and  $\text{Na}_2\text{S}$  decomposes into  $\text{HCN}$  and  $\text{H}_2\text{S}$

23.  $V_{N_2} = 38 \text{ ml}$

$$P = 732 - 32 = 700$$

$$V_{N_2} \text{ at STP} = \frac{700 \times 38}{350} \times \frac{273}{760} \text{ ml}$$

$$22400ml \text{ --- } 28gm N_2$$

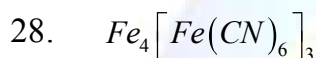
$$\frac{2 \times 38 \times 273}{760} \text{ --- } = \frac{28}{22400} \times \frac{2 \times 38 \times 273}{760} gm$$

$$\% \text{ Nitrogen} = \frac{28 \times 2 \times 38 \times 273 \times 100}{22400 \times 760 \times 0.546} = 6.25\%$$

24.  $SP > SP^2 > SP^3$



27. 6



29. / 30. The residual acid requires,  $60 \times 0.5 \text{ meq of } NaOH = 30meq NaOH$

The no.of meq of  $H_2SO_4$  that reacts with  $NH_3 = (50 \times 1 - 30) = 20meq$

no.of meq of  $NH_3$  is 20 meq

$$\text{Weight of } NH_3 = 20 \times 10^{-3} \times 17g$$

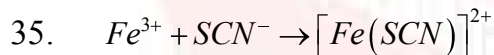
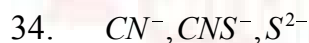
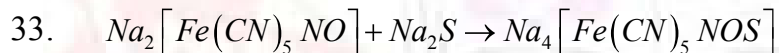
$$= 0.34g$$

Weight of Nitrogen in 0.34g  $NH_3$  is 0.28g

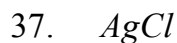
$$\% \text{ by weight of } N = \frac{0.28}{0.5} \times 100 = 56\%$$

31. Conceptual

32. Conceptual



36. Nitro compounds, Pyridine and azo compounds



38. Conceptual

39. NCERT - 327

40. NCERT

**MATHS**

41.  $\frac{1}{x+y}(1+y^1)=2(xy^1+y)$  Put  $y=0$

42.  $F^1(\alpha)=0 \Rightarrow F(x)$  is divisible  $f(x)$

43.  $|P^1(x)| = |a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1}| \leq |e^{x-1}|$

Put  $x=1 \Rightarrow |a_1 + 2a_2 + \dots + na_n| \leq 1$

44.  $f(g(x))=x \Rightarrow f^1(g(x))g^1(x)=1$

$$\therefore g^1(x) = \frac{1}{f^1(g(x))} = 1 + (g(x))^4$$

45. Conceptual

46. Observe  $f(0)=0$  and  $\lim_{x \rightarrow 0} \frac{f'(x)}{1} = 100 \Rightarrow f'(0)=100$

Now using first principle

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) + f(h) + x^2h^2 - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h)}{h} + 0$$

$$f'(0)=100$$

47. Conceptual

48. Take 'log' and differentiate twice and use  $y'(0)=-1$

49.  $f(x) = -2x + a : -1 \leq x < 0$

$$= b : 0 \leq x < 1$$

$$= x + c : 1 \leq x < 4$$

$$= -x + d : 4 \leq x < 6$$

50.  $a = b = 1 + x, 4 + c = d - 4$

$$f(0) = 0 \Rightarrow a = 0 = b, c = -1, d = 7$$

$$\therefore f(x) = -2x : -1 < x < 0$$

$$= 0 : 0 \leq x < 1$$

$$= x - 1 : 1 \leq x < 4$$

$$= 7 - x : 4 \leq x \leq 6$$

51. &amp; 52.

$$f(x) = e^{2x}$$

$$g(x) = x^2$$

$$53. \quad \frac{\frac{dy}{dx}}{\frac{d\theta}{dx}} = \frac{dy}{d\theta} = -\cot^2 \theta$$

$$54. \quad \frac{dx}{d\theta} = \tan \theta (\sec \theta + \cos \theta) = \tan \theta \sqrt{x^2 + 4}$$

$$\frac{dy}{dx} = 6 \tan \theta (\sec^6 \theta + \cos^6 \theta) = 6 \tan \theta \sqrt{y^2 + 4}$$

$$\frac{dy}{dx} = \frac{6\sqrt{y^2 + 4}}{\sqrt{x^2 + 4}} \Rightarrow \left(\frac{dy}{dx}\right)^2 \frac{(x^2 + 4)}{(y^2 + 4)} = 36$$

$$55. \quad g'(x) = ae^{ax} + f'(x) \Rightarrow g'(0) = a - 5$$

$$g''(x) = a^2 e^{ax} + f''(x) \Rightarrow g''(x) = a^2 + 3$$

$$a = -2, 1$$

$$56. \quad \text{Put } y = \frac{1}{x} \text{ to get } f(x) = 1 + x^3$$

$$57. \quad g(-x) = g(x) \text{ and } f'(x) = g(x)$$

$\therefore g(x)$  is even and  $f$  is odd

$$58. \quad e^{x \sin(x^3)} \left[ 3x^3 \cos(x^3) + \sin(x^3) \right] + (\tan x)^x \left[ \ln \tan x + 2x \operatorname{cosec} 2x \right]$$

$$e^{x \sin(x^3)} \left[ 3x^3 \cos(x^3) + \sin(x^3) \right] + (\tan x)^x \left[ \ln \tan x + \frac{x \sec^2 x}{\tan x} \right]$$

$$59. \quad g(x+1) - g(x) = \ln x \quad \text{differentiable twice and put } x = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots, \frac{2N-1}{2}$$

$$60. \quad 2f(x)f(y) = f(x-y) + f(x+y)$$

$\therefore f(x) = \cos x$  is suitable function

Since  $f(0) \neq 0$

# Success



## JEE ADVANCED WEEKEND TEST



### Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

**Sec: Jr. Super60-I**

**Time: 07:30 AM to 10:30 AM**

**Date: 14-10-17**

**Max.Marks: 186**

Name of the Student: \_\_\_\_\_

I.D. NO:

|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|

### 2016\_PAPER-I

#### IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.**  
Darken the appropriate bubble under each digit of your roll number.

#### DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

**14-10-17\_Jr.Super60-I\_Jee Adv\_WAT-21\_Weekend Syllabus**

**MATHS:** SYLLABUS COVERED FROM 09-10-17 TO 13-10-17

**PHYSICS:** SYLLABUS COVERED FROM 09-10-17 TO 13-10-2017

**CHEMISTRY:** ORGANICCHEMISTRY-Nomenclature of Hydrocarbons, Nomenclature of mono and bi-functional organic compounds including common names, Nomenclature of Aromatic compounds (Benzene), Structural isomerism (excluding Tautomerism)

**JEE-ADVANCE-2016-P1-Model**

Time: 07:30 AM to 10:30 AM

**IMPORTANT INSTRUCTIONS**

Max Marks: 186

**PHYSICS:**

| Section                  | Question Type                                               | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|-------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 1 – 5)     | Questions with Single Correct Choice                        | 3         | -1         | 5         | 15          |
| Sec – II(Q.N : 6 – 13)   | Questions with Multiple Correct Choice (Partial Marking +1) | 4         | -2         | 8         | 32          |
| Sec – III(Q.N : 14 – 18) | Questions with Integer Answer Type                          | 3         | 0          | 5         | 15          |
| <b>Total</b>             |                                                             |           |            | <b>18</b> | <b>62</b>   |

**CHEMISTRY:**

| Section                  | Question Type                                               | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|-------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 19 – 23)   | Questions with Single Correct Choice                        | 3         | -1         | 5         | 15          |
| Sec – II(Q.N : 24 – 31)  | Questions with Multiple Correct Choice (Partial Marking +1) | 4         | -2         | 8         | 32          |
| Sec – III(Q.N : 32 – 36) | Questions with Integer Answer Type                          | 3         | 0          | 5         | 15          |
| <b>Total</b>             |                                                             |           |            | <b>18</b> | <b>62</b>   |

**MATHEMATICS:**

| Section                  | Question Type                                               | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|-------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 37 – 41)   | Questions with Single Correct Choice                        | 3         | -1         | 5         | 15          |
| Sec – II(Q.N : 42 – 49)  | Questions with Multiple Correct Choice (Partial Marking +1) | 4         | -2         | 8         | 32          |
| Sec – III(Q.N : 50 – 54) | Questions with Integer Answer Type                          | 3         | 0          | 5         | 15          |
| <b>Total</b>             |                                                             |           |            | <b>18</b> | <b>62</b>   |

Sec: Jr. Super60-I

space for rough work

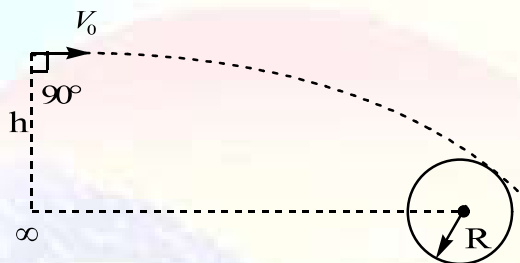
**2**

**SECTION – I**  
**(SINGLE CORRECT ANSWER TYPE)**

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.**

1. The gravitational potential energy of interactions of a system of six identical particles, each of mass  $m$  placed at the vertices of a regular hexagon of side 'a' is [PE = 0 at infinite separation]
- A)  $-\frac{Gm^2}{a}\left(1+\frac{1}{\sqrt{3}}+\frac{1}{4}\right)$       B)  $-\frac{3Gm^2}{a}\left(1+\frac{1}{\sqrt{3}}+\frac{1}{4}\right)$
- C)  $-\frac{6Gm^2}{a}\left(1+\frac{1}{\sqrt{3}}+\frac{1}{4}\right)$       D)  $-\frac{2Gm^2}{a}\left(1+\frac{1}{\sqrt{3}}+\frac{1}{4}\right)$
2. An unpowered spacecraft has to hit a far off planet of mass  $M$ , radius  $R$ . Maximum entrance area ( $\pi h^2$ ) from which 'the spacecraft can be projected so as to hit the planet. Size of the spacecraft is negligible compared to planet



A)  $\pi R^2 \left( 1 + \frac{GM}{RV_0^2} \right)$

B)  $\pi R^2 \left( 1 + \frac{2GM}{RV_0^2} \right)$

C)  $4\pi R^2 \left( 1 + \frac{2GM}{RV_0^2} \right)$

D)  $4\pi R^2 \left( 1 - \frac{2GM}{RV_0^2} \right)$

3. If a satellite is revolving around a planet of mass  $M$  in an elliptic orbit of semi-major axis  $a$ , the orbital speed of the satellite when it is at a distance  $r$  from the focus.

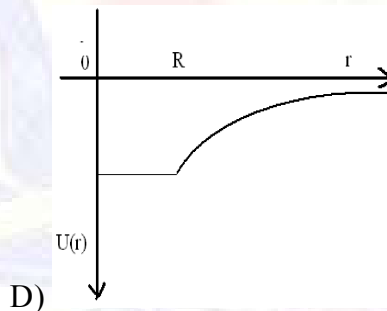
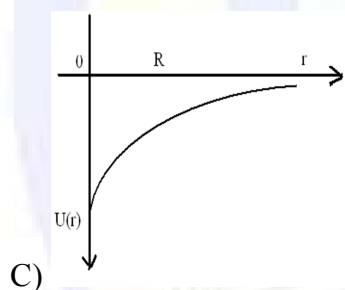
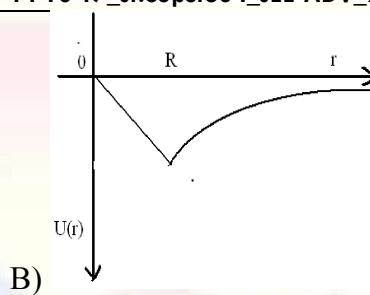
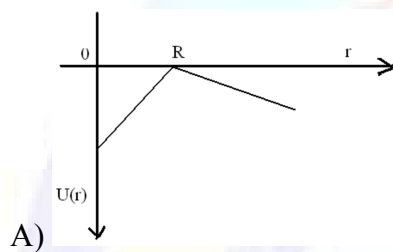
A)  $V^2 = GM \left[ \frac{2}{r} - \frac{1}{a} \right]$

B)  $V^2 = GM \left[ \frac{2}{r^2} - \frac{1}{a} \right]$

C)  $V^2 = GM \left[ \frac{2}{r^3} - \frac{1}{a^2} \right]$

D)  $V^2 = G \left[ \frac{2}{r} - \frac{1}{a} \right]$

4. A shell of mass  $M$  and radius  $R$  has a point mass  $m$  placed at a distance  $r$  from the centre. The gravitational potential energy  $U(r)$  vs  $r$  will be [assume potential energy at infinity to be zero]



5. A uniform hollow sphere has internal radius  $a$  and external radius  $b$ . The ratio of gravitational potential at a point on its outer surface to that of a point on its inner surface is

A)  $\frac{2(b^3 - a^3)}{3b(b^2 - a^2)}$       B)  $\frac{2(b^3 + a^3)}{3b(b^2 + a^2)}$       C)  $\frac{a(a^2 - b^2)}{3b(a^2 + b^2)}$       D) 1

**SECTION – II**  
**(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

6. A satellite is orbiting the earth in a circular orbit of radius  $r$ . Its
- A) kinetic energy varies as  $1/r$                       B) angular momentum varies as  $\frac{1}{\sqrt{r}}$
- C) linear momentum varies as  $\frac{1}{\sqrt{r}}$                       D) frequency of revolution varies as  $\frac{1}{r^{3/2}}$
7. A double star system consists of two stars having masses  $M$  and  $2M$ . The distance between their centres is equal to  $r$ . They revolve under their mutual gravitational interaction. Then, which of the following statements are not correct.
- A) Heavier star revolves in circular orbit of radius  $2r/3$
- B) Both the stars revolve with same angular speed, period of which is equal to  $\frac{2\pi}{\sqrt{3GM}} r^{3/2}$
- C) Kinetic energy of the heavier star is twice that of the other star.
- D) Kinetic energy of the heavier star is  $\frac{GM^2}{3r}$

8. A satellite of mass  $m$  orbits around the Earth in an elliptical path of semi-major and semi-minor axis  $2a$  and  $a$  respectively such that the Earth is located at one of the foci of the elliptical path. At an instant when the satellite is located at a distance of  $r$ , its velocity is  $u$  at an angle of  $60^\circ$  with respect to the position vector of the satellite from the centre of the Earth. Consider only the gravitational force between the satellite and the Earth and assume that the Earth is a perfect sphere.

(Assume that mass of the satellite is negligible as compared to mass of the Earth)

- A) Orbital angular momentum of the satellite with respect to centre of the Earth is

$$\frac{\sqrt{3}}{2} mur$$

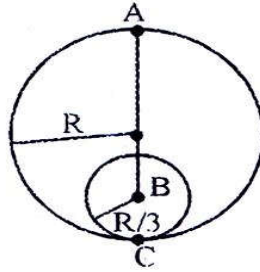
- B) Areal velocity of position vector of the satellite with respect to centre of the Earth

$$\text{is } \frac{\sqrt{3}}{4} ur$$

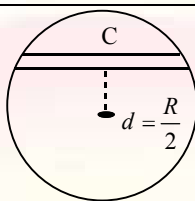
- C) Time period of orbital motion of the satellite around the Earth is  $\frac{8\pi a^2}{\sqrt{3} ur}$

- D) Time period of satellite's orbital motion is independent of mass of the Earth

9. Inside a uniform sphere of mass  $M$  ( $M$  is mass of complete sphere) and radius  $R$ , a cavity of radius  $\frac{R}{3}$  is made in the sphere as shown. Then



- A) Gravitational field inside the cavity is uniform.  
 B) Gravitational field inside the cavity is non-uniform  
 C) The escape velocity of a particle projected from point A is  $\sqrt{\frac{88GM}{45R}}$   
 D) Escape velocity is defined for earth and particle system only.
10. A tunnel is dug along a chord of non-rotating earth (hypothetical) at a distance  $d$  as shown. A small block is released from rest in the tunnel from the surface of earth. The block comes to rest at the centre C of tunnel. If  $\mu$  is the coefficient of friction between block and tunnel then



- A) The work done by the friction on the block is  $\frac{-3GMm}{8R}$
- B)  $\mu = \frac{\sqrt{3}}{2}$
- C)  $\mu = \frac{1}{2}$
- D) The gravity does +ve work on block
11. Assuming Earth, an uniform sphere of mass  $M$  and radius  $R$ . At a point  $P$  inside Earth at a distance  $r$  measured from centre of Earth ( $r < R$ )
- A) Pressure ( $p$ ) caused by gravitational compression is  $\frac{3}{8} \left[ 1 + \frac{r^2}{R^2} \right] \frac{Gm}{\pi R^3}$
- B) Pressure ( $p$ ) caused by gravitational compression is  $\frac{3}{8} \left[ 1 - \frac{r^2}{R^2} \right] \frac{GM^2}{\pi R^4}$
- C) Gravitational field Intensity at the centre is  $\frac{3}{2} \frac{GM}{R^2}$
- D) Gravitational field Intensity at the centre is zero.

12. If a body is projected with speed lesser than escape velocity:
- A) the body can reach a certain height and may fall down following a straight line path
  - B) the body can reach a certain height and may fall down following a parabolic path
  - C) the body may orbit the earth in a circular orbit
  - D) the body may orbit the earth in an elliptic orbit
13. Orbital velocity of a satellite revolving around a planet in a circular orbit depends on
- A) Mass of the satellite
  - B) Mass of the planet
  - C) Radius of the orbit
  - D) angular velocity of rotation of the planet

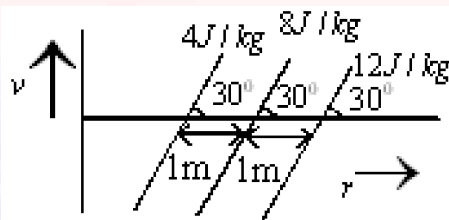
**SECTION – III**  
**(INTEGER ANSWER TYPE)**

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

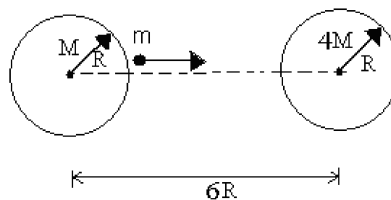
**Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases.**

14. Assuming  $V$  as the gravitational potential at the centre of earth, the potential at a distance half the radius from its centre will be  $\frac{11V}{10+x}$ . The value of  $x$  is

15. Gravitational potential versus distance  $r$  graph is represented in fig. The magnitude of gravitational field intensity is equal to  $X$  (N/Kg). Then find  $X$



16. A particle is projected vertically upwards from the surface of the earth (radius  $R_e$ ) with a speed equal to one fourth of escape velocity. The maximum height attained by it from the surface of the earth is  $\frac{R_e}{5N}$ . Then 'N' is
17. Two uniform solid spheres of equal radii  $R$ , but mass  $M$  and  $4M$  have a centre to centre separation.  $6R$ , as shown in the figure.



The two spheres are held fixed. A projectile of mass 'm' is projected with minimum velocity from the surface of the sphere of mass M directly towards the centre of the second sphere so that it reaches the surface of the second sphere. If the speed with which the projectile reaches the surface of the sphere of mass 4M is  $3\sqrt{\frac{xGM}{5R}}$  find x

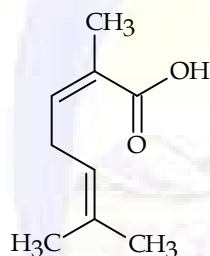
18. A bullet is fired vertically upwards with velocity v from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is 1/4th of its value at the surface of the planet. If the escape velocity from the planet is  $v_{\text{esc}} = v\sqrt{N}$ , then the value of N is (ignore energy loss due to atmosphere)

**SECTION – I**  
**(SINGLE CORRECT ANSWER TYPE)**

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.**

19. The IUPAC name of the following compound is



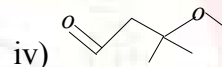
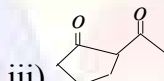
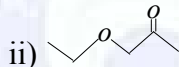
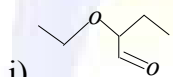
- A) 2,6-dimethylhepta-2,5-dienoic acid.  
 B) 3,7-dimethylhepta-2,5-dienoic acid.  
 C) 1-Hydroxy-2,6-dimethylhepta-2,5-dienone  
 D) 3-Hydroxy -2,6-dimethylhepta-2,5-dienone.
20. Some statements about structural isomers of  $C_5H_{11}Br$  are given below.
- It has four primary bromides
  - It has three secondary bromides
  - It has two tertiary bromides.

iv) It has one tertiary bromide.

Select the correct ones..

- A) i, ii                      B) i, ii, iv                      C) ii, iii                      D) iii

21. Which of the following compounds have both aldehydic and etheric functional groups?



- A) i, iii                      B) ii, iii                      C) i, iv                      D) i, ii, iii

22. Which of the following pairs are correctly matched ?

- A) n-butylchloride and iso-butyl chloride are positional isomers.  
 B) n-pentylalcohol and neo-pentyl alcohol are functional isomers.  
 C) n-propylmethanoate and ethylethanoate are metamers  
 D) ortho-dichlorobenzene and meta-dichlorobenzene are chain isomers

23. Identify the correct statement among the following

- A) The number of monochloroderivatives for propane (ignore stereoisomers) is three.  
 B) The number of dichloroderivatives for propane(ignore stereoisomers) is three.  
 C) The number of trichloroderivatives for propane(ignore stereoisomers) is five.  
 D) The number of monochloroderivatives for isobutane(ignore stereoisomers) is three.

**SECTION – II**  
**(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

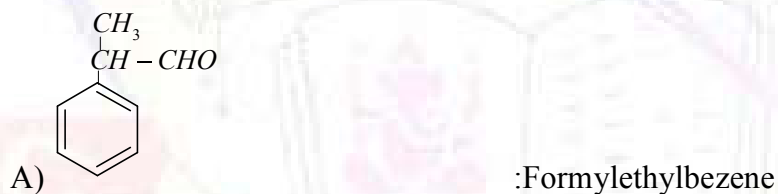
24. Which of the following pairs are functional isomers?

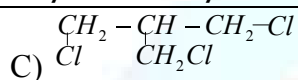
- A) Acetaldehyde and acetone
- B) Propanenitrile and ethyl isocyanide
- C) diethylamine and Butanamine
- D) Ortho-methylphenol and Benzylalcohol

25. Which of the following compounds having four structural isomers?

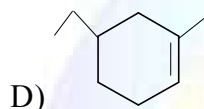
- A)  $C_4H_9Cl$       B)  $C_4H_{10}O$       C)  $C_4H_{11}N$       D)  $C_3H_9N$

26. Which of the following given IUPAC names are not correct?





:1,3-dichloro-2-chloromethylpropane



:1-methyl-3-ethylcyclohexene

27. Metamerism can be shown by

- A) Ketones      B) Esters      C) Ethers      D) tertiary amines

28. The correct names of functional isomers of butanoic acid is/ are

- A) Ethylethanoate      B) Methylpropanoate  
C) Propylmethanoate      D) Isopropylmethanoate

29. Identify the correct statement/s among the following is/are

- A) N,N-dimethylaniline and Phenylmethyl amine are functional isomers.  
B) N-methylaniline and phenylethanamine are functional isomers.  
C) iso-butylamine and tertiarybutyl amine are chain isomers.  
D) n-butylamine and iso-butylamine are chain isomers.

30. Which of the following names are correct for the compound containing molecular formula is  $C_7H_8O$ ?

- A) Phenylmethanol      B) 2-methylphenol  
C) 3-methylphenol      D) methoxybenzene.

31. Select the correct combination/s among the following is/are

- A) Ortho-cresol : 2-methylphenol.  
 B) Catechol : Benzene-1,3-diol.  
 C) Resorcinol : Benzene-1,2-diol.  
 D) Hydroquinone : Benzene -1,4-diol

**SECTION – III**  
**(INTEGER ANSWER TYPE)**

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

**Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases**

32. No of metamers from the following molecules whose molecular formula is  $C_6H_{12}O$  is/are.....

|                                                                  |                                                                                              |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| $CH_3 - CH_2 - CO - CH_2 - CH_2 - CH_3$                          | $CH_3 - CO - \underset{\substack{  \\ CH_3}}{CH} - CH_3$                                     |
| $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - CHO$                         | $CH_3 - CH_2 - CO - \underset{\substack{  \\ CH_3}}{CH} - CH_3$                              |
| $CH_3 - \underset{\substack{  \\ CH_3}}{CH} - CH_2 - CH_2 - CHO$ | $CH_3 - CO - CH_2 - CH_2 - CH_2 - CH_3$                                                      |
| $CH_3 - CO - \underset{\substack{  \\ CH_3}}{CH} - CH_2 - CH_3$  | $CH_3 - CO - CH_2 - \underset{\substack{  \\ CH_3}}{CH} - CH_3$                              |
| $CH_3 - CO - \underset{\substack{  \\ CH_3}}{C} - CH_3$          | $\underset{\substack{  \\ CH_3}}{CH_3} - C - CO - \underset{\substack{  \\ CH_3}}{C} - CH_3$ |

33. The number of structural isomers of  $C_4H_{10}O$  is 'X' and for  $C_2H_6O$  is 'Y'.

Then the X + Y is..

34. The number of correct IUPAC names for the given compounds is..

i)  $CH_3 - CH = CH - CH_2 - C \equiv C - CH_3$  : hept-2-en-5-yne.

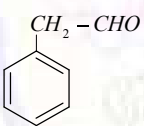
ii)  $CH_2 = CH - CH_2 - C \equiv CH$  : pent-1-en-4-yne.

iii)  $CH_3 - CH = CH - CH_2 - C \equiv CH$  : hex-4-en-1-yne.

iv)  $CH_2 = CH - CH_2 - C \equiv C - CH_3$  : hex-5-en-2-yne.

v)  $(CH_3)_3C - CH = CH - C(CH_3)_3$  : 1,2-di(3°-butyl)ethene.

vi)  $NC - CH_2 - \underset{\substack{| \\ CN}}{CH} - CH_2 - CN$  : propane -1,2,3-tricarbonitrile.

vii)  : benzeneacetaldehyde.

35. The number of structural isomers possible for  $C_6H_6O_3$  containing benzene ring (Do not consider peroxybond containing isomers) is X.

The number of structural isomers possible for  $C_7H_9N$  containing benzene ring is Y.

Then X + Y is....

36. The number of cyclic isomers for  $C_5H_{10}$  is 'X' and the number of open chain structural isomers for  $C_5H_{10}$  is 'Y'. Then  $X - Y$  is..

**SECTION – I**  
**(SINGLE CORRECT ANSWER TYPE)**

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.**

37. If  $3x + 2y = 1$  acts as a tangent to  $y = f(x)$  at  $x = 1/2$  and if

$$p = \lim_{x \rightarrow 0} \frac{x(x-1)}{f\left(\frac{e^{2x}}{2}\right) - f\left(\frac{e^{-2x}}{2}\right)} \text{ then, } \sum_{r=1}^{\infty} p^r = \underline{\hspace{2cm}}$$

- A)  $1/2$                       B)  $1/3$                       C)  $1/6$                       D)  $1/7$

38. If the equation  $x^{5/3} - 5x^{2/3} = K$  has exactly one positive root, then, the complete solution set of K is,

- A)  $(-\infty, \infty)$               B)  $(-\infty, 0)$               C)  $(3, \infty)$               D)  $(0, \infty)$

39. A sphere of radius 10cm shrinks to radius 9.8cm. The approximate decrease in volume is \_\_\_\_\_ cu.cm

- A)  $80\pi$                       B)  $-80\pi$                       C)  $16\pi$                       D)  $-16\pi$

40. If the point on  $y = x \tan \alpha - \frac{ax^2}{2u^2 \cos^2 \alpha}$ ,  $0 < \alpha < \frac{\pi}{2}$  where the tangent is parallel to  $y=x$  has

an Ordinate  $\frac{u^2}{4a}$  then the value of  $\alpha$  is

- A)  $\frac{\pi}{2}$                       B)  $\frac{\pi}{6}$                       C)  $\frac{\pi}{3}$                       D)  $\frac{\pi}{4}$

41. A horse runs along a circle with a speed of 20km/h. A lantern is at the centre of the circle. A fence is there along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse moves along the fence at the moment when it covers  $1/8$  of the circle in km/h is

- A) 20                      B) 40                      C) 30                      D) 60

### SECTION – II

#### (MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

42. Which of the following curves are orthogonal

- A)  $\frac{x^2}{21} + \frac{y^2}{17} = 1, \frac{x^2}{3} - \frac{y^2}{1} = 1$                       B)  $\frac{x^2}{47} + \frac{y^2}{4} = 1, \frac{x^2}{21} + \frac{y^2}{23} = 1$   
 C)  $\frac{x^2}{49} + \frac{y^2}{4} = 1, \frac{x^2}{23} - \frac{y^2}{22} = 1$                       D)  $\frac{x^2}{21} + \frac{y^2}{9} = 1, \frac{x^2}{42} + \frac{y^2}{30} = 1$

43. Points on the curve  $f(x) = \frac{x}{1-x^2}$ , where the tangent is inclined at an angle of  $\frac{\pi}{4}$  to x axis are
- A) (0,0)      B)  $\left(\sqrt{3}, \frac{\sqrt{3}}{2}\right)$       C)  $\left(-2, \frac{2}{3}\right)$       D)  $\left(-\sqrt{3}, \frac{\sqrt{3}}{2}\right)$
44. The family of curves defined by the equations  $y = ax$  and  $x^2 + y^2 = c^2$  are perpendicular for
- A)  $a = 2, c = 4$       B)  $a = -2, c = 3$       C)  $a = 3, c = 2$       D)  $a = 3, c = -2$
45. The normal to the curve  $y^2 = 5x - 1$  at the point (1, -2) is of the form  $ax - 5y + b = 0$  then
- A)  $a = 4$       B)  $b = -14$       C)  $a + b = -10$       D)  $ab = 40$
46. Two motor cars start with the same velocity 'v' km/hour from distances 'a' km and 'b' km from the junction of two roads inclined at  $90^\circ$  and travel towards the junction and after 2 hours, they are nearing each other at the rate of
- A)  $\frac{4v^2 - (a+b)v}{\sqrt{a^2 + b^2 - 4(a+b)v + 8v^2}} \text{ km / hour}$       B)  $\frac{2v^2 - (a+b)v}{\sqrt{a^2 + b^2 - 2(a+b)v + 8v^2}} \text{ km / hour}$
- C)  $\frac{6v^2 - (a+b)v}{\sqrt{a^2 + b^2 - 6(a+b)v + 8v^2}} \text{ km / hour}$       D)  $\frac{8v^2 - (a+b)v}{\sqrt{a^2 + b^2 - 8(a+b)v + 8v^2}} \text{ km / hour}$

47. The radius of a right circular cylinder increases at the rate of 0.1 cm/min, and the height decreases at the rate of 0.2 cm/min. The rate of change of the volume of the cylinder, in  $\text{cm}^3/\text{min}$ , when the radius is 2 cm and the height is 3 cm, is
- A)  $-2\pi$       B)  $-\frac{8\pi}{5}$       C)  $-\frac{3\pi}{5}$       D)  $\frac{2\pi}{5}$
48. A point on the parabola  $y^2 = 18x$  at which the ordinate increases at twice the rate of the abscissa is
- A) (2, 6)      B) (2, -6)      C)  $\left(\frac{9}{8}, -\frac{9}{2}\right)$       D)  $\left(\frac{9}{8}, \frac{9}{2}\right)$
49. An error of 0.01cm is made while measuring the radius 5cm of a circle.
- A) The error in circumference will be  $0.02\pi \text{ cm}$
- B) The error in area will be  $0.1\pi \text{ sq.cm}$
- C) The relative error in circumference will be 0.002
- D) The percentage error in area will be 0.004

**SECTION – III**  
**(INTEGER ANSWER TYPE)**

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

**Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases**

50. Water is dripping out from a conical funnel of semi vertical angle  $\frac{\pi}{4}$  at the uniform rate of  $2\text{cm}^3 / \text{s}$ . Through a tiny hole at the vertex at the bottom when the slant height of the water is 4cm. If  $\lambda\text{cm} / \text{s}$  be the rate of decreases of the slant height of the water, then the value of  $4\pi\sqrt{2}\lambda$  must be
51. The rate of change of the volume of a sphere with respect to its surface area, when the radius is 2 cm, is
52. If  $y = 4x - 5$  is tangent to the curve  $y^2 = px^3 + q$  at the point (2, 3) then  $p - q =$
53. If the normal to the curve  $y = f(x)$  at  $x = 0$  be given by the equation  $3x - y + 3 = 0$ , then the value of  $\lim_{x \rightarrow 0} x^2 \left\{ f(x^2) - 5f(4x^2) + 4f(7x^2) \right\}^{-1}$  is  $\frac{-1}{k}$  then  $k =$
54. A man 1.6m high walks at the rate of 30m per minute away from a lamp which is 4m above ground. If  $k$  is the rate at which the man's shadow is lengthening, then the value of  $k/4$  is \_\_\_\_\_



# Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. SUPER60-I

JEE ADVANCED

DATE : 14-10-17

TIME : 07:30 AM to 10:30 AM

WAT-21

MAX MARKS : 186

## KEY SHEET 2016-P1

### PHYSICS

|    |     |    |      |    |     |    |    |    |     |
|----|-----|----|------|----|-----|----|----|----|-----|
| 1  | C   | 2  | B    | 3  | A   | 4  | D  | 5  | A   |
| 6  | ACD | 7  | AC   | 8  | ABC | 9  | AC | 10 | ABD |
| 11 | BD  | 12 | ABCD | 13 | BC  | 14 | 2  | 15 | 8   |
| 16 | 3   | 17 | 3    | 18 | 2   |    |    |    |     |

### CHEMISTRY

|    |     |    |      |    |     |    |      |    |      |
|----|-----|----|------|----|-----|----|------|----|------|
| 19 | A   | 20 | B    | 21 | C   | 22 | C    | 23 | C    |
| 24 | BCD | 25 | AD   | 26 | ABD | 27 | ABCD | 28 | ABCD |
| 29 | D   | 30 | ABCD | 31 | AD  | 32 | 6    | 33 | 9    |
| 34 | 4   | 35 | 8    | 36 | 0   |    |      |    |      |

### MATHS

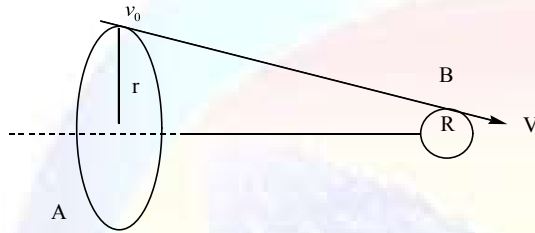
|    |    |    |    |    |      |    |     |    |   |
|----|----|----|----|----|------|----|-----|----|---|
| 37 | A  | 38 | D  | 39 | A    | 40 | C   | 41 | B |
| 42 | AC | 43 | AD | 44 | ABCD | 45 | ABC | 46 | A |
| 47 | D  | 48 | CD | 49 | ABCD | 50 | 2   | 51 | 1 |
| 52 | 9  | 53 | 3  | 54 | 5    |    |     |    |   |

**SOLUTIONS****PHYSICS**

1. Total potential energy is  $V = \sum U_{ij}$

$$= \frac{-Gm^2}{a} \left[ 6 \times \left( \frac{1}{a} \right) \right] + 6 \left( \frac{1}{\sqrt{3}a} \right) + 3 \left( \frac{1}{2a} \right) = \frac{-6Gm^2}{a} \left( 1 + \frac{1}{\sqrt{3}} + \frac{1}{4} \right)$$

2.



LCAM  $mv_0 r = mVR$

$$V = \frac{v_0 r}{R}$$

L.C.E  $\frac{1}{2}mv_0^2 = \frac{-GMm}{R} + \frac{1}{2}mv^2$

$$\frac{v_0^2}{2} + \frac{GM}{R} = \frac{1}{2} \frac{v_0^2 r^2}{R^2}$$

$$r^2 = \frac{(v_0^2 R + 2GM) R}{v_0^2}$$

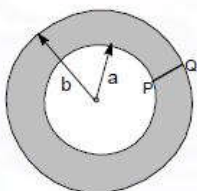
$$\text{Area} = \pi r^2 = \frac{\pi R^2}{v_0^2} \left( v_0^2 + \frac{2GM}{R} \right)$$

3. As in case of elliptic orbit of a satellite mechanical energy  $E = -(GMm/2a)$  remains constant, at any position of satellite in the orbit,

Therefore  $\frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}, i.e., v^2 = GM \left[ \frac{2}{r} - \frac{1}{a} \right]$

4. Potential inside a shell is constant

5.



$$V_Q = -\frac{Gm}{b} = -\frac{G}{b} \frac{4}{3} \pi (b^3 - a^3) \rho ; dV_p = -\frac{Gdm}{x} ; dm = 4\pi x^2 (dx) \rho ; V_p = -\int_a^b \frac{G(4\pi x^2 dx) \rho}{x}$$

$$= -4\pi G \rho \left( \frac{x^2}{2} \right)_a^b ; = -4\pi G \rho \left( \frac{b^2 - a^2}{2} \right) ; \frac{V_Q}{V_P} = \frac{\frac{G}{b} \frac{4}{3} \pi (b^3 - a^3) \rho}{+4\pi G \rho \left( \frac{b^2 - a^2}{2} \right)} ; = \frac{2(b^3 - a^3)}{3b(b^2 - a^2)}$$

6. Kinetic energy  $\frac{1}{2}mv^2 = \frac{GmM}{2r}$  or  $KE \propto \frac{1}{r}$ . Thus choice (A) is correct.

Angular momentum =  $mvr = mx\sqrt{\frac{GM}{r}}xr = m\sqrt{GM}r$ , which is proportional to  $\sqrt{r}$ .

Hence choice (B) is wrong.

Linear momentum  $mv = m\sqrt{\frac{GM}{r}}$ , which is proportional to  $\frac{1}{\sqrt{r}}$ . Hence choice (C) is

correct. The frequency of revolution is  $v = \frac{1}{T} = \frac{1}{2\pi} \sqrt{\frac{GM}{r^3}}$  i.e.  $v \propto \frac{1}{r^{3/2}}$ .

Hence the correct choices are (A), (C) and (D)

7. A) The centre of mass of the double star system remains stationary and both the stars revolve around in circular orbits which are concentric with the centre of mass

The distance of centre of mass from the heavier star is equal to  $\left( \frac{Mr + 2M \cdot 0}{M + 2M} \right) = \frac{r}{3}$

Hence, the heavier star revolves in a circle of radius  $r/3$  while the lighter star in a circle of radius  $2r/3$ . Hence option(a) is wrong.

B)  $(2M) \left( \frac{r}{3} \right) \omega^2 = 2GM^2$

$$(2M) \frac{r}{3} \omega^2 = \frac{2GM^2}{r^2} \Rightarrow \omega = \left( \frac{3GM}{r^3} \right)^{1/2} = \frac{2\pi}{T} = \left( \frac{3GM}{r^3} \right)^{1/2} \Rightarrow T = 2\pi \frac{r^{3/2}}{(3GM)^{1/2}}$$

Where  $r$  is distance between two stars. Hence, option (b) is correct.

C) Kinetic energy of a star will be equal to  $\frac{1}{2}mv^2$  where  $v$  is speed of the star which is equal to (radius of its circular orbit)  $\times \omega$ .

Hence, K.E. of heavier star is

$$E_1 = \frac{1}{2}(2M) \left( \frac{r}{3} \omega \right)^2 = \frac{Mr^2 \omega^2}{9} \frac{3GM}{r^3} = \frac{GM^2}{3r}$$

And that of lighter star,  $E_2 = \frac{1}{2}M \left( \frac{2r}{3} \omega \right)^2$ .

It means, KE of the lighter star is twice that of the heavier star.

Hence, option (C) is wrong.

$$8. \quad A) \quad L = m(u \cos 30^\circ)r = \frac{\sqrt{3}mur}{2}$$

$$B) \quad \frac{dA}{dt} = \frac{L}{2m} = \frac{\sqrt{3}ur}{4} \quad C) \quad T = \frac{\text{total area}}{\text{areal velocity}} = \frac{2\pi a^2}{\frac{\sqrt{3}ur}{4}} = \frac{8\pi a^2}{\sqrt{3}ur}$$

$$9. \quad M_{\text{cavity}} = \frac{M}{\frac{4}{3}\pi R^3} \times \frac{4}{3}\pi \left(\frac{R}{3}\right)^3 = \frac{M}{27}$$

$$\text{At 'A' } V_{\text{total}} = V_{\text{cavity}} + V_{\text{remaining}}$$

$$-\frac{GM}{R} = -\frac{G \frac{M}{27}}{\frac{5R}{3}} + V_{\text{remaining}}$$

$$V_R = -\frac{GM}{R} + \frac{GM}{45R} = -\frac{44GM}{45R}$$

For escape velocity

$$\frac{1}{2}mv^2 = \frac{44GMm}{45R}$$

$$V = \sqrt{\frac{88}{45} \frac{GM}{R}}$$

10. Work done due to friction is given by

$$W = \left[ \frac{-GMm}{2R^3} \left( 3R^2 - \frac{R^2}{4} \right) \right] - \left[ \frac{-GMm}{R} \right]$$

$$= \frac{-GMm}{2R^3} \frac{11R^2}{4} + \frac{GMm}{R} = \frac{-3GMm}{8R}$$

$$dW = -\mu \frac{GMm}{R^3} \sqrt{x^2 + R^2} \frac{d}{\sqrt{x^2 + R^2}} dx$$

$$= -\mu \frac{GMm}{2R^2} d(x)$$

$$W = -\mu \frac{GMm}{2R^2} (x)_0^{\sqrt{3R/2}}$$

$$= -\mu \frac{GMm}{2R^2} \frac{\sqrt{3R}}{2} = -\mu \frac{GMm}{4R} \sqrt{3}$$

$$\Rightarrow \mu \frac{GMm}{4R} \sqrt{3} = \frac{3GMm}{8R} \quad \Rightarrow \mu = \frac{\sqrt{3}}{2}$$

11. Inward gravitational pull on the layer

$$df = \left( \frac{GM}{R^3} \cdot r \right) (4\pi r^2 p \cdot dr), \quad P = \frac{3M}{4\pi r^3}$$

$$dp = \frac{df}{4\pi r^2}, \quad P = \int_r^R dp = \frac{3}{8} \left[ 1 - \frac{r^2}{R^2} \right] \frac{GM^2}{\pi R^4}$$

12. (A) If it is projected radially it will go up and then move down in a straight line

B) If it is projected with a small velocity near the earth's surface, g will be almost constant. So its path will be almost parabolic. (Projectile Motion)

(C) If the body is projected tangentially with orbital speed  $\left( V_0 = \sqrt{\frac{GM}{r}} \right)$  then it will revolve in circular orbit

(D) If the body is projected with a velocity V ( $V_0, V_e$ ) it may revolve in an elliptical orbit.

13. Conceptual

$$14. \quad v = \frac{-3GM}{2R}, \quad v_{atr} = \frac{-GM}{2R^3} (3R^2 - r^2)$$

$$\left( \text{at } r = \frac{R}{2} \right) = \frac{-11GM}{8R} = \frac{11}{8} \left( \frac{2V}{3} \right) = \frac{11V}{12}$$

$$15. \quad E = \frac{-dv}{dx} \quad \text{here } dx = dr \sin \theta$$

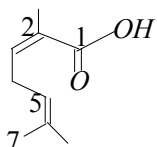
$$16. \quad \frac{1}{2} m \cdot \frac{1}{4} \left( \frac{2GM}{R} \right) - \frac{GMm}{R} = -\frac{GMm}{R+h} \quad \therefore h = \frac{R}{15}$$

$$17. \quad -\frac{GMm}{2R} - \frac{GMm}{4R} = -\frac{GMm}{5R} - \frac{GMm}{R} + \frac{1}{2} mV^2 \Rightarrow V = 3\sqrt{\frac{3GM}{5R}}$$

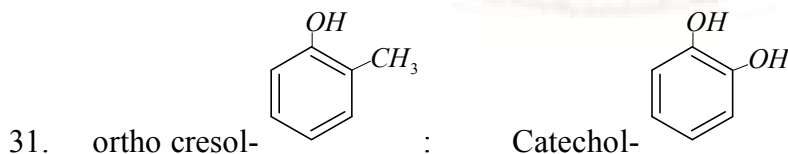
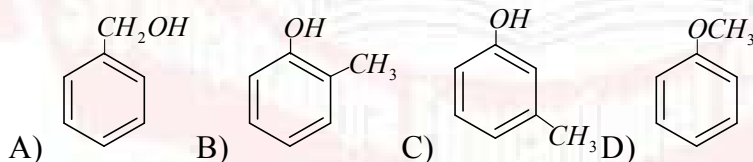
18. At height R from the surface of planet acceleration due to planet's is  $\frac{1}{4}$  th in comparison to the value at the surface

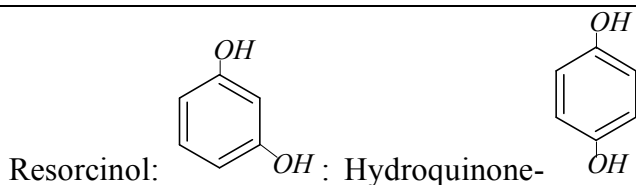
$$\text{So, } -\frac{GMm}{R} + \frac{1}{2} mv^2 = -\frac{GMm}{R+R} \quad \text{and} \quad -\frac{GMm}{R} + \frac{1}{2} mv_{esc}^2 = 0$$

$$\therefore v_{esc} = v\sqrt{2}$$

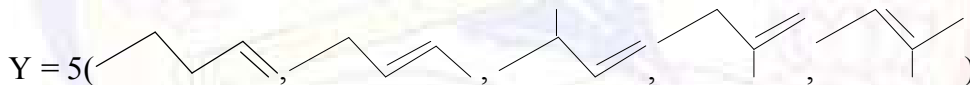
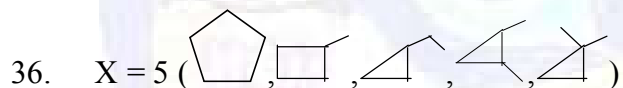
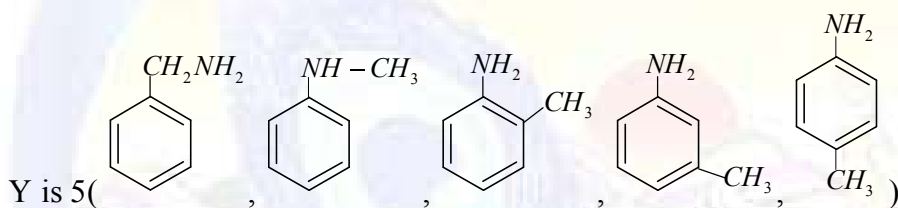
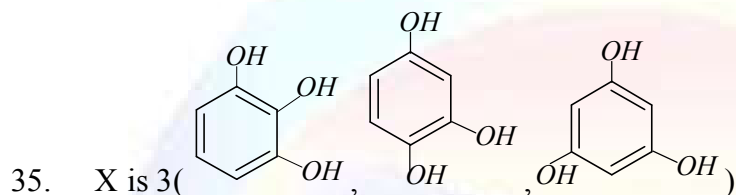
**CHEMISTRY**

- 19.
20. It has 4 primary, 3 secondary and one tertiary bromides.
21. i and iv have aldehydic and etheric groups
22. Different alkyl groups connected to same functional groups are called metamers.
23. For A is 2, for B is 4, for D is 2
24. Acetaldehyde and acetone are non isomers (molecular formula is not same)
25. A has four isomers (n-butyl, sec. butyl, isobutyl and ter. butyl chlorides)  
 B has seven structural isomers (four alcoholic and three etheric)  
 C has 8 structural isomers (primary amines = 4, sec. Amines = 3, ter. Amine = 1)  
 D has 4 structural isomers (propanamine - 1, propanamine-2, N-methylethanamine and trimethyl amine.)
26. A (2-phenyl propanal)    B) pent-2-en-1,5-dial    D) 4-ethyl-2-methylcyclohexene-1
27. all can show metamerism
28. A)  $\text{CH}_3\text{COOC}_2\text{H}_5$     B)  $\text{CH}_3\text{CH}_2\text{COOCH}_3$   
 C)  $\text{H}-\text{COOCH}_2\text{CH}_2\text{CH}_3$     D)  $\text{H}-\text{COOCH}(\text{CH}_2)\text{CH}_3$
29. A) are not isomers  
 C) are positional isomers
30. All are correct





32. ii, x are not isomers for the given compound and iii and v are aldehydes:  
 33.  $X = 7$  and  $Y = 2$   
 34. i, ii, iii, vi are correct.



### MATHS

37. Sol. slope of  $3x + 2y = 1$  is  $-\frac{3}{2}$

$$\Rightarrow f^1\left(\frac{1}{2}\right) = -\frac{3}{2}$$

$$p = \lim_{x \rightarrow 0} \frac{x(x-1)}{f\left(\frac{e^{2x}}{2}\right) - f\left(\frac{e^{-2x}}{2}\right)} \left(\frac{0}{0}\right) = \frac{-1}{f^1\left(\frac{1}{2}\right) + f^1\left(\frac{1}{2}\right)} = \frac{1}{3}$$

$$\therefore \sum_{r=1}^{\infty} p^r = \frac{1}{3} + \frac{1}{3^2} + \dots \infty = \frac{1/3}{1-1/3} = \frac{1/3}{2/3} = \frac{1}{2}$$

38. Sol. Sketch  $y = x^{5/3} - 5x^{2/3}$  and  $y = K$

40. Sol. Given  $m=1 \Rightarrow \tan \alpha - \frac{ax}{u^2 \cos^2 \alpha} = 1 \Rightarrow x = \frac{(\tan \alpha - 1)}{a} u^2 \cos^2 \alpha$  substitute x and

y values in given equation  $\frac{u^2}{4a} = \frac{u^2}{a} \left[ \sin^2 \alpha - \frac{1}{2} \right] \Rightarrow \alpha = \frac{\pi}{3}$

42. Conceptual

53. Put  $x^2 = t$ , and  $f'(0) = -\frac{1}{3}$



# Success



## JEE ADVANCED WEEKEND TEST



### Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

**Sec: Jr. Super60-I**

**Time: 07:30 AM to 10:30 AM**

**Date: 21-10-17**

**Max.Marks:180**

Name of the Student: \_\_\_\_\_

I.D. NO:

|  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |
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## 2013\_PAPER-II

### IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

### OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your roll number.

### DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

**21-10-17\_Jr.Super60-I\_Jee Adv\_WAT-22\_Weekend Syllabus**

**MATHS:** SYLLABUS COVERED FROM 16-10-17 TO 20-10-2017

**PHYSICS:** SYLLABUS COVERD FROM 16-10-17 TO 20-10-2017

**CHEMISTRY:** Geometrical isomerism, Conformational isomerism of ethane, butane, derivatives of ethane and cyclohexane

Sri Chaitanya IIT Academy

21-10-17\_Jr.Super60-I\_JEE-Adv\_2013\_P2\_WAT-22\_QP

**IIT-JEE-2013-P2-Model**

Time: 07:30 AM to 10:30 AM

**IMPORTANT INSTRUCTIONS**

Max Marks: 180

**PHYSICS**

| Section                  | Question Type                                                           | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|-------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 1 – 8)     | Questions with Multiple Correct Choice                                  | 3         | -1         | 8         | 24          |
| Sec – II(Q.N : 9 – 16)   | Questions with Comprehension Type<br>(4 Comprehensions – 2 +2+2+2 = 8Q) | 3         | -1         | 8         | 24          |
| Sec – III(Q.N : 17 – 20) | Matrix Matching Type                                                    | 3         | -1         | 4         | 12          |
| <b>Total</b>             |                                                                         |           |            | <b>20</b> | <b>60</b>   |

**CHEMISTRY**

| Section                  | Question Type                                                           | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|-------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 21 – 28)   | Questions with Multiple Correct Choice                                  | 3         | -1         | 8         | 24          |
| Sec – II(Q.N : 29 – 36)  | Questions with Comprehension Type<br>(4 Comprehensions – 2 +2+2+2 = 8Q) | 3         | -1         | 8         | 24          |
| Sec – III(Q.N : 37 – 40) | Matrix Matching Type                                                    | 3         | -1         | 4         | 12          |
| <b>Total</b>             |                                                                         |           |            | <b>20</b> | <b>60</b>   |

**MATHEMATICS**

| Section                  | Question Type                                                           | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|-------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 41 – 48)   | Questions with Multiple Correct Choice                                  | 3         | -1         | 8         | 24          |
| Sec – II(Q.N : 49 – 56)  | Questions with Comprehension Type<br>(4 Comprehensions – 2 +2+2+2 = 8Q) | 3         | -1         | 8         | 24          |
| Sec – III(Q.N : 57 – 60) | Matrix Matching Type                                                    | 3         | -1         | 4         | 12          |
| <b>Total</b>             |                                                                         |           |            | <b>20</b> | <b>60</b>   |

**SEC: Jr. Super60-I****space for rough work****Page 2**

**PHYSICS****Max.Marks:60****SECTION – I****(MULTIPLE CORRECT CHOICE TYPE)**

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

1. The physical quantities having same units are
  - A) mass and moment of inertia per unit area
  - B) pressure and energy density
  - C) Moment of force and torque
  - D) Linear momentum and angular momentum
2. If U, m L and G denote energy, mass, angular momentum and universal gravitational constant, respectively. The dimensions of  $\frac{m^5 G^2}{UL^2}$  are same as
  - A) co-efficient of restitution
  - B) refractive index
  - C) solid angle
  - D) distance

3. If  $y = a^3 \sin^{-1}\left(\frac{x}{b} - 1\right)$ , where  $y$  is in metre. Then,
- A) the dimensions of  $a$  is  $\left(M^0 L^{\frac{1}{3}} T^0\right)$
  - B) the unit of  $\sin^{-1}\left(\frac{x}{b} - 1\right)$  is radian
  - C) the dimensions of  $x$  are same as those of  $b$
  - D) the unit of  $a$  is metre
4. The speeds of two particles are measured to be  $v_1 = (10.20 \pm 0.02) \text{ms}^{-1}$  and  $v_2 = (6.40 \pm 0.01) \text{ms}^{-1}$ . Then, mark the correct option(s).
- A) The sum of speeds of particles is  $(16.60 \pm 0.03) \text{ms}^{-1}$
  - B) The difference of speeds of particles is  $(3.80 \pm 0.03) \text{ms}^{-1}$
  - C) The difference of speeds of particles is  $(6.40 \pm 0.01) \text{ms}^{-1}$
  - D) The sum of speeds of particles is  $16.60 \text{ms}^{-1}$

5. If  $y = \sin \theta$ , the error in measurement of  $\theta$  is  $1^\circ$ . The fractional error in  $y$
- A) may be zero
  - B) may be infinity
  - C) will have any value from zero to infinity
  - D) must be zero
6. A student uses a simple pendulum of exactly 1 m length to determine  $g$ , the acceleration due to gravity. He uses a stopwatch with the least count of 1 s for this and records 40 s for 20 oscillations. For the observation, which of the following statements (s) is/are correct?
- A) Error  $\Delta T$  in measuring  $T$ , the time period is 0.05 s
  - B) Error  $\Delta T$  in measuring  $T$ , the time period is 1 s
  - C) Percentage error in the determination of  $g$  is 5%
  - D) Percentage error in the determination of  $g$  is 2.5%

7. Out of the following expressions for couple per unit twist (C), time period (T) of a compound pendulum, time period ( $T^1$ ) of a simple pendulum and frequency (f) of a stretched wire respectively, the dimensionally consistent equations are ( $\ell$  = length, F = Force, A = area,  $\theta$  = angle, r = radius)
- A)  $C = \frac{\pi \eta r^4}{2\ell}$  (here  $\eta = F / A \times \theta$ )
- B)  $T = 2\pi \sqrt{\frac{K^2 + \ell}{g}}$  here K = radius of gyration
- C)  $T^1 = 2\pi \sqrt{\frac{\ell}{g}}$
- D)  $f = \frac{1}{\ell} \sqrt{\frac{T_0}{\mu}}$  here  $\mu$  = linear density,  $T_0$  = tension in string
8. If the dimension of the length are expressed as  $G^x c^y h^z$ , where G, c and h are the universal gravitational constant, speed of light and Planck's constant respectively, then
- A)  $x = 1/2, y = 1/2$                       B)  $x = 1/2, z = 1/2$
- C)  $y = 1/2, z = 3/2$                       D)  $y = -3/2, z = 1/2$

**SECTION - II**  
**(COMPREHENSION TYPE)**

This section contains **4 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

**Paragraph for Questions 9 and 10**

A small spherical ball suspended from a rigid support using inextensible and light string and free to oscillate without friction is called a simple pendulum. The time

period of small oscillation of simple pendulum is given by  $T = 2\pi \sqrt{\frac{L}{g}}$ .



Here,  $L$  = Distance between suspension point  $O$  and centre  $C$  of the ball,

$g$  = Acceleration due to gravity.

In the determination of value of acceleration due to gravity  $g$  by simple pendulum, the time period is measured by a stopwatch whose least count is 0.5 s and the length of string is measured with a meter scale, having least count of 1 mm. The diameter of the

bob is measured with the vernier calipers with least count 0.01 cm. The following observations are recorded.

- i. Length of the string = 105.3 cm      ii. Diameter of the bob = 2.45 cm  
iii. Time period = 2.07 s                  iv. Number of oscillations = 10

9. Find percentage error in calculation of acceleration due to gravity.

- A) 3%                  B) 4.9%                  C) 9.8%                  D) 0.28%

10. If we increase the number of oscillations from 10 to 20, find the new percentage error in calculation of g.

- A) 2.6%                  B) 4.9%                  C) 9.8%                  D) 6%

**Paragraph for Question 11 and 12**

The time period of simple pendulum is given by  $T = k\ell^x g^y$  where k is the constant of proportionality,  $\ell$  is the length of pendulum and g, the acceleration due to gravity, x and y are dimensionless constants.

11. In the above equation, dimensions of  $\ell$  and g are respectively

- A)  $[L], [MLT^{-2}]$                   B)  $[L], [L^{-2}T]$   
C)  $[L], [LT^{-2}]$                   D)  $[L^{-1}], [LT^{-2}]$

12. If the given equation is dimensionally correct then

- A)  $x = 1, y = \frac{1}{2}$                       B)  $x = \frac{1}{2}, y = -\frac{1}{2}$   
 C)  $x = -\frac{1}{2}, y = \frac{1}{2}$                       D)  $x = -1, y = \frac{1}{2}$

**Paragraph for Questions 13 and 14**

In a new system of units, let the units of force, energy and velocity be 10 N, 0.1 J and 100 m/s respectively. In this new system,

13. the unit of length is

- A) 0.50 m                      B) 1.00 m                      C)  $10^{-3}$  m                      D)  $10^{-2}$  m

14. the unit of mass is

- A)  $10^{-5}$  kg                      B)  $10^{-3}$  kg                      C)  $10^{-4}$  kg                      D)  $10^{-2}$  kg

**Paragraph for Questions 15 and 16**

The energy 'E' of a particle varies with time t according to the equation

$E = E_0 \sin(\alpha t) e^{\frac{-\alpha t}{\beta x}}$ ; where x is displacement from mean position.  $E_0$  is energy at infinite position and  $\alpha$  &  $\beta$  are constants of appropriate dimensions.

15. Dimensional formula of  $\alpha$  is

- A)  $[M^0 L^0 T^{-1}]$                       B)  $[M^{-1} L^0 T^0]$                       C)  $[M^0 L^{-1} T^0]$                       D)  $[M^0 L^0 T^0]$

16. Dimensional formula of  $\beta$  is

- A)  $[M^{-1}L^0T^0]$     B)  $[M^0L^{-1}T^0]$     C)  $[M^0L^0T^{-1}]$     D)  $[M^0L^0T^0]$

**SECTION – III**  
**(MATRIX MATCH TYPE)**

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C), and (D) out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

17. Considering Force (F), Velocity (V) and Energy (E) as fundamental quantities, match the correct dimensions of following quantities

|      | Column I                             |    | Column II          |
|------|--------------------------------------|----|--------------------|
| I.   | Mass                                 | P. | $[F^{-1} V^0 E^1]$ |
| II.  | Light year                           | Q. | $[F^1 V^1 E^{-1}]$ |
| III. | Frequency $\left(\frac{1}{T}\right)$ | R. | $[F^3 V^0 E^{-2}]$ |
| IV.  | Pressure                             | S. | $[F^0 V^{-2} E^1]$ |

Codes

- |    |   |    |     |     |    |     |     |     |     |
|----|---|----|-----|-----|----|-----|-----|-----|-----|
|    | I | II | III | IV  |    | I   | II  | III | IV  |
| A) | S | P  | Q   | S,R | B) | S   | P   | Q   | R   |
| C) | P | Q  | P,Q | P,S | D) | P,Q | R,S | R,S | R,S |

18. Match the Column I with Column II and mark the correct option from the given below.

|      | Column I                                                                                                                                         |    | Column II                                                                     |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------|
| I.   | If $m, g, \ell$ and $I$ denote mass, acceleration due to gravity, length and moment of inertia, respectively. The formula of $B = \frac{mgl}{I}$ | P. | In the dimensional formula of defined quantity, the dimension of mass is 1    |
| II.  | If $\ell, F$ and $\mu$ denote length, force and linear mass density, respectively. The formula of $C = \sqrt{\frac{F}{\mu \ell^2}}$              | Q. | In the dimensional formula of defined quantity, the dimension of time is (-2) |
| III. | $G$ = universal gravitational constant. $M_e, R_e$ are mass and                                                                                  | R. | In the dimensional formula of defined quantity, the dimension of              |

|     |                                                                                                                      |    |                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------|----|------------------------------------------------------------|
|     | radius of earth. The formula of<br>$V = \frac{GM_e}{R_e}$                                                            |    | length is zero                                             |
| IV. | If U and V represent potential energy and velocity of a particle respectively, The formula of<br>$R = \frac{U}{V^2}$ | S. | The dimensional formula of defined quantity is $[ML^0T^0]$ |

Codes

- |    | I     | II  | III | IV  |    | I   | II  | III | IV  |
|----|-------|-----|-----|-----|----|-----|-----|-----|-----|
| A) | Q,R   | R   | Q   | P,S | B) | P,R | Q,R | Q,S | P,S |
| C) | P,Q,R | P,Q | Q,R | P,S | D) | Q,R | R   | Q,S | P,R |

19. Match the following columns and mark the correct option from the codes given below.

|      | Column I            |    | Column II |
|------|---------------------|----|-----------|
| I.   | $\frac{1}{3}$       | P. | 1         |
| II.  | $\frac{1.00}{3.00}$ | Q. | 0.3       |
| III. | $2 - 0.3$           | R. | 0.33      |
| IV.  | $2 - 0.6$           | S. | 2         |

Codes

- |    |     |     |     |     |    |   |    |     |    |
|----|-----|-----|-----|-----|----|---|----|-----|----|
|    | I   | II  | III | IV  |    | I | II | III | IV |
| A) | Q   | R   | S   | P   | B) | P | Q  | R   | S  |
| C) | Q,R | Q,R | P,S | P,S | D) | Q | R  | P   | S  |

20. During determination of acceleration due to gravity by simple pendulum in repeated experiments, following observations are obtained.

| S. No. of observations | Acceleration due to gravity (in $\text{ms}^{-2}$ ) |
|------------------------|----------------------------------------------------|
| 1                      | 9.85                                               |
| 2                      | 9.76                                               |
| 3                      | 9.91                                               |
| 4                      | 9.79                                               |
| 5                      | 9.78                                               |

Match the following columns and mark the correct option from the codes given below.

|      | Column I                                                         |   | Column II |
|------|------------------------------------------------------------------|---|-----------|
| I.   | Mean value of acceleration due to gravity (in $\text{ms}^{-2}$ ) | P | 0.05      |
| II.  | Mean absolute error (in $\text{ms}^{-2}$ )                       | Q | 9.82      |
| III. | Relative error                                                   | R | 0.51      |
| IV.  | Percentage error                                                 | S | 0.0051    |

Codes:

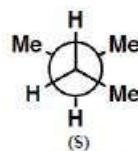
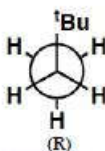
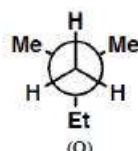
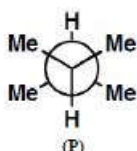
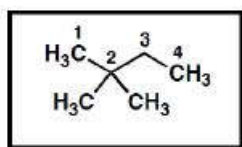
|    |   |    |     |    |    |   |    |     |    |
|----|---|----|-----|----|----|---|----|-----|----|
|    | I | II | III | IV |    | I | II | III | IV |
| A) | Q | P  | S   | R  | B) | Q | P  | R   | S  |
| C) | Q | R  | S   | P  | D) | P | Q  | S   | R  |

**CHEMISTRY****Max.Marks:60****SECTION – I****(MULTIPLE CORRECT CHOICE TYPE)**

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

21. Identify the structure (s) that represent a conformation of **2,2-dimethylbutane** sighting along any C-C bond



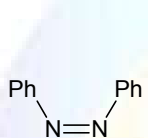
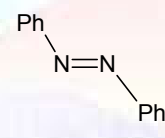
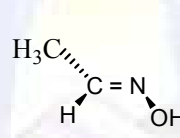
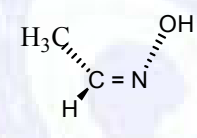
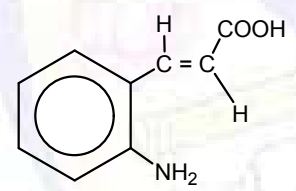
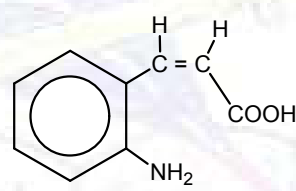
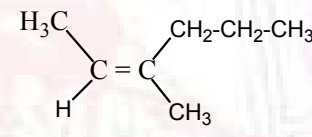
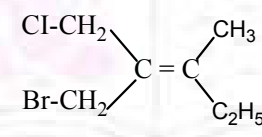
A) P

B) S

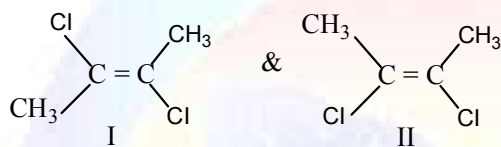
C) Q

D) R

22. For which of the following pairs of compounds are the CORRECT notation given?

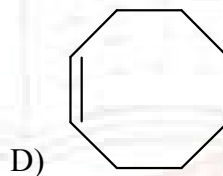
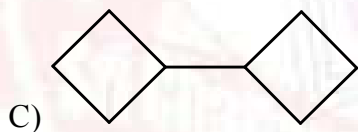
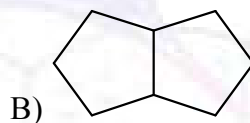
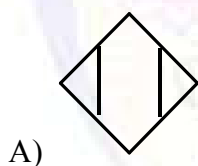
- A)    
 A) Z-azobenzene E-azobenzene
- B)    
 B) E-acetaldoxime Z-acetaldoxime
- C)    
 C) Trans-o-amino cinnamic acid Cis-o-amino cinnamic acid
- D)    
 D) Z-isomer E-isomer

23. Which statement is/are right regarding

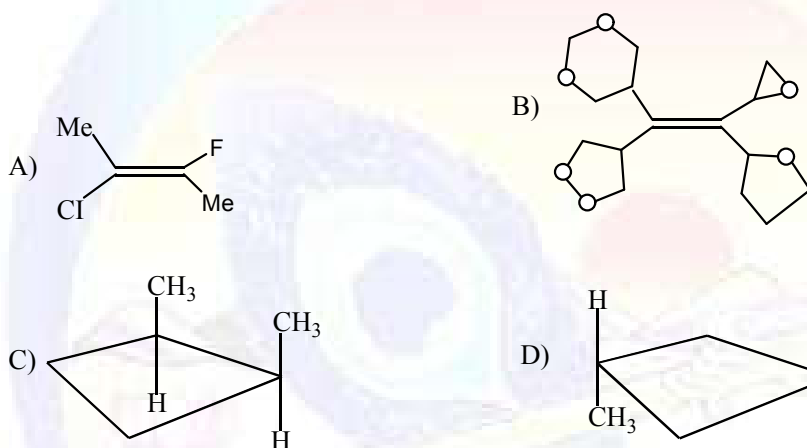


- A) They are geometrical isomers
- B) They are diastereoisomers
- C) I is E-isomer and II is Z-isomer
- D) They are conformers

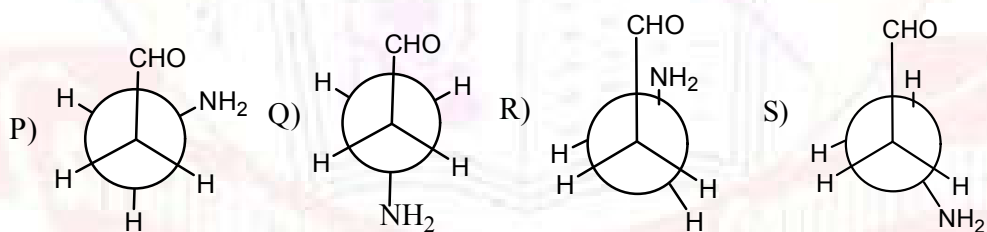
24. Which of the compounds shown below are isomers?



25. Which of the following molecules are E-isomer?



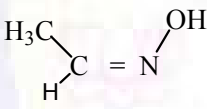
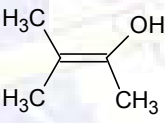
26. Consider the following conformations of 3-Aminopropanal, amongst the given conformations (P,Q,R,S) one of them is most stable.



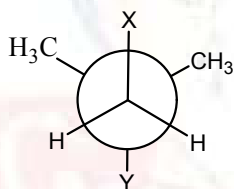
The correct statements for the above are:

- A) H-bonding is present in the most stable conformer
- B) Gauche conformation is the most stable conformer
- C) Anti conformation is the most stable conformer
- D) larger groups being separated by maximum distance in the conformer

27. Which of the following compounds can exhibit geometrical isomerism?

- A)  $H_3C-CH_2-CH=CH-CH_3$
- B) 
- C) 
- D) 

28. In the Newmann projection for 2,2- dimethylbutane



X and Y can respectively be

- A) H and H      B) H and  $C_2H_5$       C)  $C_2H_5$  and H      D)  $CH_3$  and  $CH_3$

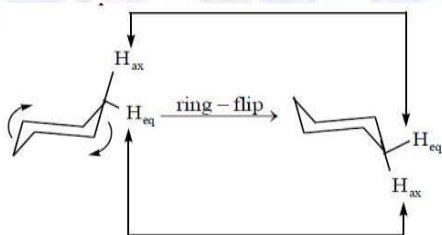
**SECTION - II**  
**(COMPREHENSION TYPE)**

This section contains **4 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

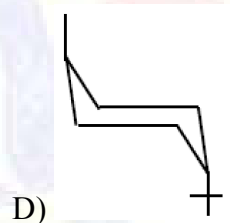
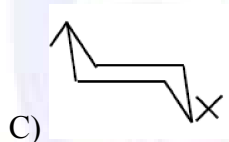
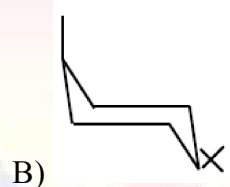
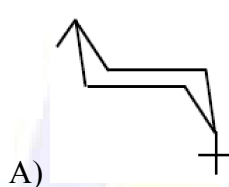
**Paragraph for Questions 29 and 30**

Cyclohexane exist as two chair conformations in rapid equilibrium at room temperature. Each carbon atom on a cyclohexane ring has one axial and one equatorial hydrogen. Ring flipping converts axial H's to equatorial H's and vice-versa.

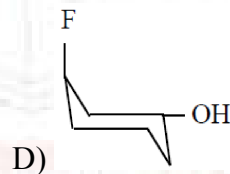
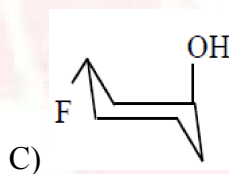
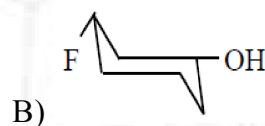
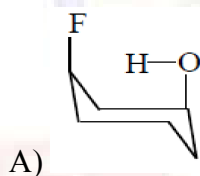


In substituted cyclohexane, groups larger than hydrogen are more stable in the equatorial position. The cis isomers has two groups on the same side of the ring, either both up or both down. The trans isomer has two groups on opposite side of the ring, one up and one down.

29. Find out most stable substituted cyclohexane among the following:



30. Find out most stable conformer of cis-3-fluorocyclohexanol



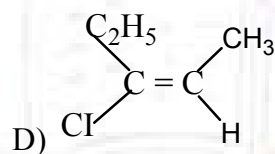
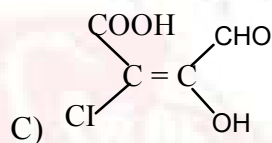
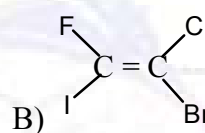
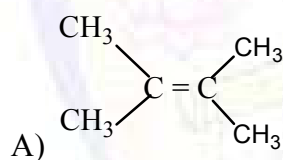
**Paragraph for Questions 31 and 32****Geometrical Isomerism**

Molecules having same molecular formula but differing in the orientation of atoms in a space due to restricted rotation.

31. Compound which do not exhibit geometrical isomerism

- A) diethyl ether                      B) 2-butene  
C) 3,4- dimethyl-3-hexene              D) 2-chloro-3-methyl-2-pentene

32. Which molecule is in "E" configuration



**Paragraph for Questions 33 and 34**

Stereoisomers, which can be interconverted simply by rotation about sigma bonds, are conformational isomers. While those, which can be converted only by breaking and remaking of bonds and not simply by rotation, are called configurational isomers.

The angle between C – C and C – H bonds on adjacent carbon atoms in any conformation is called dihedral angle. The cyclic compounds most commonly found in nature containing six membered rings can exist in a conformation that is almost completely free of strain. The most stable conformation of cyclohexane is chair form. According to Bayer strain theory, the greater is the deviation from the normal tetrahedral angle, greater is the angle strain or torsional strain and hence lesser is the stability of the cycloalkane.

33. Dihedral angle between two methyl groups of n-butane in the gauche and anti forms respectively are:

- A)  $60^\circ, 0^\circ$       B)  $60^\circ$  and  $180^\circ$       C)  $0^\circ, 60^\circ$       D)  $180^\circ, 60^\circ$

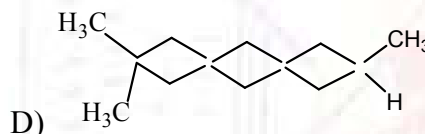
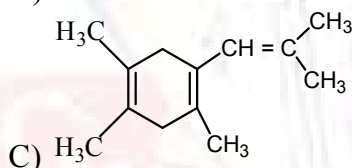
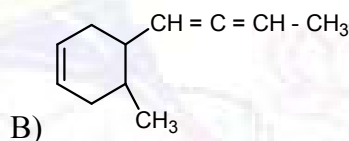
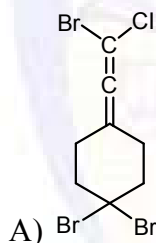
34. Which among the following conformation of cyclohexane is the most stable form?

- A) Chair form      B) Half chair form  
C) Twist boat form      D) Boat form

**Paragraph for Questions 35 and 36**

Those stereoisomers are called configurational isomers if they differ in spatial arrangement of atom or group and are not interconvertible without cleaving a bond. Geometrical isomerism arises due to restricted rotation about a bond or centre. The number of G.I. is  $2^n$  where  $n$  = number of double bonds or rings or both. In some cases the number of G.I. is less than  $2^n$ , where either side nomenclature will be possible.

35. Which of the following compounds can show geometrical isomerism?



36. The number of G.I. of the compound  $C_2H_5 - CH = CH - CH = CH - CH = CH - CH_3$  is

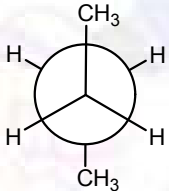
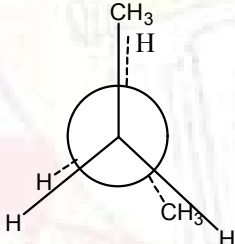
A) 4                      B) 3                      C) 8                      D) 6

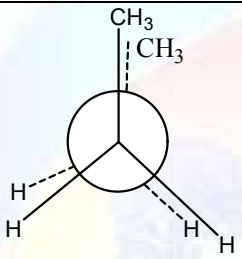
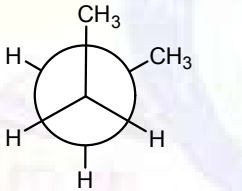
**SECTION – III**  
**(MATRIX MATCH TYPE)**

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C), and (D) out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

- 37.** Match the conformers of n-Butane along  $C_2 - C_3$  bond in Column I with torsional energies given in Column II

|     | Column I                                                                          |    | Column II          |
|-----|-----------------------------------------------------------------------------------|----|--------------------|
| I.  |  | P. | 18.4 – 25.5 KJ/mol |
| II. |  | Q. | 14.2 KJ/mol        |

|      |                                                                                   |    |            |
|------|-----------------------------------------------------------------------------------|----|------------|
| III. |  | R. | 3.3 KJ/mol |
| IV.  |  | S. | Zero       |

- |    |   |    |     |    |
|----|---|----|-----|----|
|    | I | II | III | IV |
| A) | P | Q  | S   | R  |
| B) | Q | R  | P   | S  |
| C) | S | Q  | P   | R  |
| D) | P | Q  | R   | S  |

38. Match the following compounds from Column I with number of Geometrical Isomers

from Column II

|      | Column I                                      |    | Column II |
|------|-----------------------------------------------|----|-----------|
| I.   | $CH_3 - CH = C = CH - CH_3$                   | P. | 4         |
| II.  | $CH_3 - CH = CH - CH = N - OH$                | Q. | 8         |
| III. | $CH_3 - CH = C = C = CH - CH_3$               | R. | 0         |
| IV.  | $CH_3 - CH = CH - CH = CH - CH = CH - C_2H_5$ | S. | 2         |

- |    |   |    |     |    |
|----|---|----|-----|----|
|    | I | II | III | IV |
| A) | S | P  | R   | Q  |
| B) | R | P  | S   | Q  |
| C) | Q | R  | S   | P  |
| D) | P | Q  | R   | S  |

39. Match the following compounds in Column I with type of isomerism in Column II

| Column I<br>(Molecule) |                                     | Column II<br>(Most stable conformer) |                |
|------------------------|-------------------------------------|--------------------------------------|----------------|
| I                      | Ethane                              | P                                    | Gauche         |
| II                     | Ethylene Glycol                     | Q                                    | Anti           |
| III                    | 3-Fluoropropanal about<br>$C_2-C_3$ | R                                    | Staggered      |
| IV                     | n-Pentane about $C_2-C_3$           | S                                    | Fully eclipsed |

- |    |   |    |     |    |
|----|---|----|-----|----|
|    | I | II | III | IV |
| A) | R | S  | P   | Q  |
| B) | Q | R  | P   | S  |
| C) | R | P  | P   | Q  |
| D) | P | Q  | R   | S  |

40. Match the following compounds in Column I with their properties from Column II

|      | Column I |    | Column II        |
|------|----------|----|------------------|
| I.   |          | P. | Torsional strain |
| II.  |          | Q. | Melting point    |
| III. |          | R. | Dipole moment    |
| IV.  |          | S. | Boiling point    |

|    |   |    |     |    |    |   |    |     |    |
|----|---|----|-----|----|----|---|----|-----|----|
|    | I | II | III | IV |    | I | II | III | IV |
| A) | P | Q  | R   | S  | B) | Q | P  | R   | S  |
| C) | S | Q  | P   | R  | D) | R | S  | P   | Q  |

## SECTION – I

## (MULTIPLE CORRECT CHOICE TYPE)

This section contains 8 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

41. Let  $f : (0, \infty) \rightarrow (0, \infty)$  be a derivable function and  $F(x)$  is the primitive of  $f(x)$  such that  $2(F(x) - f(x)) = f^2(x)$  for any real positive  $x$ .

- A)  $f$  is strictly increasing  
B)  $\lim_{x \rightarrow \infty} \frac{f(x)}{x} = 1$   
C)  $f$  is strictly decreasing  
D)  $f$  is non-monotonic

42. Let  $f : R \rightarrow R$  be defined as  $f(x) = \sin x + ax + b$  ( $a, b \in R$ ). Then the equation  $f(x) = 0$  has :

- A) only one real root which is positive if  $a > 1, b < 0$   
B) only one real root which is negative if  $a > 1, b > 0$   
C) only one real root which is negative if  $a < -1, b < 0$   
D) only one real root which is positive if  $a < -1, b < 0$

43. Let  $f$  be a real-valued function on  $\mathbb{R}$  defined as  $f(x) = x^4(1-x)^2$ , then which of the following statement (s) is (are) correct ?
- A)  $f'(c) = 0$  for some  $c \in (0,1)$
- B)  $f''(x)$  vanishes exactly twice in  $\mathbb{R}$ .
- C)  $f(x)$  is an even function
- D) Monotonic increasing in  $\left(0, \frac{2}{3}\right) \cup (1, \infty)$
44. For which of the following functions Rolle's theorem is applicable ?
- A)  $f(x) = \frac{x}{2} + \frac{2}{x}, x \in [1, 4]$
- B)  $f(x) = x + 1 - x^{3/2}, x \in [0, 1]$
- C)  $f(x) = |x+1|^3, x \in [-2, 0]$
- D)  $f(x) = \operatorname{sgn}(x) + \operatorname{sgn}(-x), x \in \left[-\frac{5}{2}, \frac{1}{2}\right]$  ( $\operatorname{sgn}(x)$  denotes signum functions) of  $x$ )

45. Let  $f(x)$  be a twice differentiable function such that  $f''(x) > 0$  in  $[0, 2]$ . Then :

A)  $f(0) + f(2) = 2f(c)$  for atleast one  $c \in (0, 2)$

B)  $f(0) + f(2) < 2f(1)$

C)  $f(0) + f(2) > 2f(1)$

D)  $2f(0) + f(2) > 3f\left(\frac{2}{3}\right)$

46. Let  $h(x) = f(x) - (f(x))^2 + (f(x))^3$  for every real number  $x$ . Then

A)  $h$  is increasing whenever  $f$  is increasing

B)  $h$  is increasing whenever  $f$  is decreasing

C)  $h$  is decreasing whenever  $f$  is decreasing

D)  $h$  is decreasing whenever  $f$  is increasing

47. In  $[0, \pi]$  Rolle's theorem is applicable to

A)  $f(x) = \sin x$

B)  $f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$

C)  $f(x) = \cos 2x$

D)  $f(x) = \sin^2 x + \sin x$

48. if  $\frac{a_0}{n+1} + \frac{a_1}{n} + \frac{a_2}{n+1} + \dots + \frac{a_{n-1}}{2} + a_n = 0$  then  $a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = 0$  has
- A) no solution in  $(0,1)$                       B) at least one solution in  $(0,1)$
- C) exactly one solution in  $(0,1)$                       D) at least one solution in  $(2,3)$

**SECTION - II**  
**(COMPREHENSION TYPE)**

This section contains **4 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.  
**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

**Paragraph for Questions 49 and 50**

Consider,  $f(x) = (\lambda^4 - 2\lambda^2)x + 3\cos x, \lambda \in R$ .

49. The least positive integral value of  $\lambda$  for which  $f(x)$  is monotonically increasing for all  $x \in R$ .
- A) 2                      B) 3                      C) 4                      D) 5
50. For  $\lambda = -2$ , if  $f(\pi x - x^2) > f(\sin x)$ , then the sum of all integral values of  $x$ , is:
- A) 3                      B) 5                      C) 6                      D) 10

**Paragraph for Questions 51 and 52**

Consider a cubic,  $f(x) = ax^3 + bx^2 + cx + 4, a, b, c \in R$  and  $f''\left(\frac{-2}{3}\right) = 0$  and tangent

drawn to the graph of the function  $y = f(x)$  at  $x = \frac{-2}{3}$  is  $y = \frac{5x}{3} + \frac{100}{27}$

51. The value of  $(a + b + c)$  is equal to:

- A) 4                      B) 6                      C) 7                      D) 10

52. If  $g$  is the inverse of  $f$ , then  $\frac{d}{dx}(g(x) \cdot f(g(x)))$  at  $x = 4$  is equal to :

- A)  $\frac{1}{4}$                       B)  $\frac{4}{7}$                       C)  $\frac{3}{4}$                       D)  $\frac{4}{3}$

**Paragraph for Questions 53 and 54**

Consider,  $f(x) = \cos 2x + 2x\lambda^2 + (2\lambda + 1)(\lambda - 1)x^2, \lambda \in R$ .

53. For  $\lambda = 1$ , if  $f(3x^2 - 2x + 1) < f(x^2 - 2x + 9)$ , then number of integral values of  $x$  in  $[-10, 10]$ :

- A) 3                      B) 5                      C) 16                      D) 18

54. If  $f(x)$  is increasing for all  $x \in R$ , then number of values of  $\lambda$

- A) 1                      B) 2                      C) 3                      D)  $\infty$

**Paragraph for Questions 55 and 56**

Let  $f'(\sin x) < 0$  and  $f''(\sin x) > 0 \forall x \in \left(0, \frac{\pi}{2}\right)$  and  $g(x) = f(\sin x) + f(\cos x)$

55. which of the following is true in  $\left(0, \frac{\pi}{2}\right)$ ?

- A)  $g'$  is increasing                      B)  $g'$  is decreasing  
C)  $g'$  is non-increasing                      D)  $g'$  is neither increasing nor decreasing

56. which of the following is true ?

- A)  $g(x)$  is decreasing in  $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$   
B)  $g(x)$  increasing in  $\left(0, \frac{\pi}{4}\right)$   
C)  $g(x)$  is monotonically increasing  
D) None of these

**SECTION – III**  
**(MATRIX MATCH TYPE)**

This section contains **4 multiple choice questions**. Each question has matching lists. The codes for the lists have choices (A), (B), (C), and (D) out of which **ONLY ONE** is correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.**

**57.** Match the column

For the function  $f(x) = ax^2 - b|x|$

Column – I

Column – II

(A)  $f(x)$  has local max. at  $x = 0$

(p) When  $a > 0, b > 0$

(B)  $f(x)$  has local min at  $x = 0$

(q) When  $a > 0, b < 0$

(C)  $f(x)$  has local extremum at  $x = \frac{b}{2a}$

(r) When  $a < 0, b < 0$

(D)  $f(x)$  is not diff. at  $x = 0$

(s) When  $a < 0, b > 0$

|    | A     | B   | C   | D       |
|----|-------|-----|-----|---------|
| A) | Q,R   | S,R | P,S | P,Q,R,S |
| B) | P,S   | Q,R | P,R | P,Q,R,S |
| C) | P,R,S | Q,R | P,Q | P,Q,R,S |
| D) | P,S   | P,Q | Q,R | P,Q,R,S |

58. Observe the following list

|    | Column I                                                          |    | Column II                       |
|----|-------------------------------------------------------------------|----|---------------------------------|
| A. | $f(x) = x^2 - 2x + 5$ is increasing in                            | P. | $\phi$                          |
| B. | $f(x) = e^{-x}$ is increasing in                                  | Q. | $(-\infty, 1) \cup (2, \infty)$ |
| C. | $f(x) = \log_e x$ increasing in                                   | R. | $(1, \infty)$                   |
| D. | $f(x) = \frac{x^3}{3} - \frac{3x^2}{2} + 2x + 5$ is increasing in | S. | $(0, \infty)$                   |
|    |                                                                   | T. | R                               |

|    | A | B | C | D |
|----|---|---|---|---|
| A) | R | P | S | Q |
| B) | P | R | Q | S |
| C) | Q | R | S | P |
| D) | S | Q | P | R |

59. Observe the following lists

|    | Column I                                  |    | Column II                     |
|----|-------------------------------------------|----|-------------------------------|
| A. | $f(x) = x + \frac{1}{x}$ is decreasing in | P. | R                             |
| B. | $f(x) = 1 - x^3$ is decreasing in         | Q. | $\left(0, \frac{1}{e}\right)$ |
| C. | $f(x) = xe^{-x}$ is decreasing in         | R. | $(-1, 0) \cup (0, 1)$         |
| D. | $f(x) = x^x$ is decreasing in             | S. | $(1, \infty)$                 |

The correct matching for list -I from list -II

|    |   |   |   |   |
|----|---|---|---|---|
|    | A | B | C | D |
| A) | R | S | Q | P |
| B) | Q | R | P | S |
| C) | R | P | S | Q |
| D) | Q | R | S | P |

60. Let  $f(x) = (2^x - 1)(2^x - 2)$  and  $g(x) = 2 \sin x + \cos 2x$  in  $[0, \pi]$

|    | Column I       |    | Column II                          |
|----|----------------|----|------------------------------------|
| A. | f increases on | P. | $(\log_2(3/2), \infty)$            |
| B. | f decreases on | Q. | $(-\infty, \log_2(3/2))$           |
| C. | g decreases on | R. | $\left(0, \frac{\pi}{6}\right)$    |
| D. | g increases on | S. | $\left(\frac{5\pi}{6}, \pi\right)$ |

|    |   |   |   |   |
|----|---|---|---|---|
|    | A | B | C | D |
| A) | R | P | Q | S |
| B) | Q | S | R | P |
| C) | P | Q | S | R |
| D) | S | R | Q | P |



# Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR - HYD

Sec: Jr. Super60-I

Time: 07:30 AM – 10:30 AM

WAT-22

2013\_P2

Date: 21-10-17

Max.Marks:180

## KEY SHEET

### PHYSICS

|    |     |    |     |    |     |    |    |    |     |
|----|-----|----|-----|----|-----|----|----|----|-----|
| 1  | ABC | 2  | ABC | 3  | ABC | 4  | AB | 5  | ABC |
| 6  | AC  | 7  | ACD | 8  | BD  | 9  | B  | 10 | A   |
| 11 | C   | 12 | B   | 13 | D   | 14 | A  | 15 | A   |
| 16 | B   | 17 | B   | 18 | A   | 19 | A  | 20 | A   |

### CHEMISTRY

|    |    |    |     |    |     |    |     |    |     |
|----|----|----|-----|----|-----|----|-----|----|-----|
| 21 | CD | 22 | ABC | 23 | ABC | 24 | BCD | 25 | ABD |
| 26 | AB | 27 | ABC | 28 | BD  | 29 | C   | 30 | A   |
| 31 | A  | 32 | D   | 33 | B   | 34 | A   | 35 | B   |
| 36 | C  | 37 | C   | 38 | B   | 39 | C   | 40 | D   |

### MATHS

|    |    |    |     |    |    |    |      |    |    |
|----|----|----|-----|----|----|----|------|----|----|
| 41 | AB | 42 | ABC | 43 | AD | 44 | ABCD | 45 | BD |
| 46 | AC | 47 | ACD | 48 | B  | 49 | A    | 50 | C  |
| 51 | B  | 52 | D   | 53 | A  | 54 | A    | 55 | A  |
| 56 | D  | 57 | B   | 58 | A  | 59 | C    | 60 | C  |

## SOLUTIONS PHYSICS

1. a) MASS  $\rightarrow kg = I \Rightarrow mr^2 = kgm^2 \quad I / A = \frac{kgm^2}{m^2} = kg$
- b) Pressure =  $F / A (N / m^2)$ ,  $\frac{\text{energy}}{\text{volume}} = \frac{kgm^2 / \text{sec}^2}{m^3} = kg / m \text{sec}^2$
- $$= \frac{kgm / \text{sec}^2}{m^2} = \frac{kg}{m \text{sec}^2}$$
- c) moment of force = force x distance = time
- d)  $p = kgm / \text{sec} \quad L = kg (m / \text{sec}) m = kgm^2 / \text{sec}$

2.  $e = \frac{V_{rel}}{u_{rel}}, \mu = \frac{\sin i}{\sin r}, \Omega = A / r^2$

$$\begin{aligned} \frac{m^5 G^2}{UL^2} &= \frac{kg^5 \left[ \frac{Nm^2}{kg^2} \right]^2}{N - m \times [(kg)(m / \text{sec})(m)]^2} \\ &= \frac{kg^5 N^2 m^4 \times \text{sec}^2}{kg^4 \times N - m \, kg^2 \times m^2 \times m^2} \\ &= \frac{kg \, kg \, m / \text{sec}^2}{kg^2 m^2} = \text{dimensionless} \end{aligned}$$

3. a) Here,  $\sin^{-1} \left( \frac{x}{b} - 1 \right)$  is dimensionless

b) Let  $\sin^{-1} \left( \frac{x}{b} - 1 \right) = \theta$

$$\therefore \sin \theta = \left( \frac{x}{b} - 1 \right)$$

Here,  $\theta$  should be in radian.

So, the unit of  $\sin^{-1} \left( \frac{x}{b} - 1 \right)$  is radian

- c) Dimensions of  $\frac{x}{b}$  = Dimensions of 1 = dimensionless

$\therefore$  Dimension of  $x$  = Dimension of  $b$

4. a)  $Sum = v_1 + v_2 = (10.20 \pm 0.02) + (6.40 \pm 0.01)$

$$= (16.60 \pm 0.03) ms^{-1}$$

b) Difference  $= v_1 - v_2 = (10.20 \pm 0.02) - (6.40 \pm 0.01)$

$$= 3.80 \pm (0.02 + 0.01)$$

$$= (3.80 \pm 0.03) ms^{-1}$$

5.  $y = \sin \theta$  or  $\frac{dy}{d\theta} = \cos \theta$

Or  $dy = \cos \theta d\theta$

Or  $\Delta y = \cos \theta \Delta \theta$

Or  $\frac{\Delta y}{y} = \frac{\cos \theta \Delta \theta}{y} = \frac{\cos \theta \Delta \theta}{\sin \theta}$

Fractional error in  $y = \frac{\Delta y}{y} = \frac{\Delta \theta}{\tan \theta}$

But  $\tan \theta$  has any value from zero to infinity. Hence, fractional error will have any value from infinity to zero. Hence, options (a), (b), (c) are correct.

6.  $T = \frac{40s}{20} = 2s$

Further,  $t = nT = 20T$  or  $\Delta t = 20\Delta T$

$$\therefore \frac{\Delta t}{t} = \frac{\Delta T}{T}$$

Or  $\Delta T = \frac{T}{t} \Delta t = \left(\frac{2}{40}\right)(1) = 0.05s$

Further,  $T = 2\pi \sqrt{\frac{l}{g}}$  or  $T \propto g^{-1/2}$

$$\therefore \frac{\Delta T}{T} \times 100 = -\frac{1}{2} \times \frac{\Delta g}{g} \times 100$$

Percentage error in determination of  $g$  is

$$\frac{\Delta}{g} \times 100 = -200 \times \frac{\Delta T}{T} = \frac{200 \times 0.05}{2} = -5\%$$

$\therefore$  Correct options are (a) and (c).

7. a) is consistent, C is torque per unit twist,  $\eta$  is modulus of rigidity, r is radius,  $\ell$  is length  
 b) is inconsistent, T is time period, k is the radius of gyration, g is acceleration due to gravity  
 c) is consistent  
 d) is consistent, f is frequency, T is tension  $\mu$  is mass per unit length

8.  $L = (M^{-1}L^3T^{-2})^x (LT^{-1})^y (ML^2T^{-1})^z$   
 $-x + z = 0, 3x + y + 2z = 1, -2x - y - z = 0$   
 $\Rightarrow x = 1/2, y = -3/2, z = 1/2$

9.  $T = 2\pi\sqrt{\frac{L}{g}} \Rightarrow g = \frac{4\pi^2 L}{T^2}$   
 $= 105.3 + \frac{2.45}{2} = 106.5 \text{ 25cm}$

$$T = 2.07s$$

$$\therefore g = 4\pi^2 \frac{L}{T^2}$$

$$\therefore \frac{\Delta g}{g} = \frac{\Delta L}{L} + 2 \frac{\Delta T}{T} = \frac{0.1 + 0.005}{106.525} + \frac{2 \times 0.5}{2.07 \times 10}$$

$$= 0.00098 + 0.048 = 0.04898 = 0.049$$

$$\therefore \text{Percentage error in } g = \frac{\Delta g}{g} \times 100 = 0.049 \times 100 = 4.9\%$$

10.  $\frac{\Delta g}{g} = 0.00098 + \frac{2 \times 0.5}{2 \times 20.07} = 0.026$

$$\therefore \text{percentage error in calculation of } g \text{ is } \frac{\Delta g}{g} \times 100 = 0.026 \times 100 = 2.6\%$$

11.  $\ell$  and g are the length pendulum and the acceleration due to gravity. Hence, their respective dimensions are L and  $LT^{-2}$

12. We have,

$$T = k l^x g^y \quad \dots (I)$$

Writing dimensions on both sides of eq. (I), we get

$$T = L^x (LT^{-2})^y$$

$$\text{Or } M^0 L^0 T = M^0 L^{x+y} T^{-2y}$$

Equating exponents of M, L and T on both sides, we get

$$x + y = 0$$

$$\text{And } -2y = 1$$

$$\text{Solving, } y = -\frac{1}{2} \text{ and } x = \frac{1}{2}.$$

13. force = 10N energy = 0.1J velocity = 100m/sec

Limit of surface

$$L = F^x E^y v^z = [MLT^{-2}]^x [ML^2T^{-2}]^y [LT^{-1}]^z$$

$$x + y = 0 \quad x + 2y + z = 1 \quad -2x - 2y - z = 0$$

$$x = -y \Rightarrow -2(-y) - 2y = z \quad 2y - 2y = z \Rightarrow$$

$$-y + 2y = 0 = 1 \quad L = 10^{-1} \times 0.1^1 \times 100^0 = 10^{-2}$$

14.  $M = [MLT^{-2}]^x [MLT^{-2}]^y [LT^{-1}]^z$

$$x + y = 1 \quad x + 2y + z = 0 \quad -2x - 2y - z = 0 \quad -x = 0$$

$$M = 10^0 \times (0.1)^1 \times 100^{-2}$$

15. LT should be dimensionless  $[M^0 L^0 T^{-1}][T] = M^0 L^0 T^0$

16.  $\beta x$  should be dimensionless  $[M^0 L^0 T^{-1}][L] = M^0 L^0 T^0$

17. Conceptual

18. (i)  $[B] = \left[ \frac{mgl}{l} \right] = \frac{[MLT^{-2}] \times [L]}{[ML^2]} = [T^{-2}]$

Thus, (i)  $\rightarrow (q, r)$

(ii)  $[C] = \sqrt{\frac{F}{\mu l^2}} = \sqrt{\frac{[MLT^{-2}]}{[ML^{-1}][L^2]}} = [T^{-1}]$

Thus, (ii)  $\rightarrow (r)$

(iii)  $[V] = \frac{[M^{-1} L^3 T^{-2}]}{L} = M^{-1} L^2 T^{-2}$ , Thus, (iii)  $\rightarrow (Q)$

$$(iv) [R] = \frac{[ML^2T^{-2}]}{(LT^{-1})^2} = ML^0T^0$$

Thus, (iv)  $\rightarrow (p, s)$

19. a) (i)  $\frac{1}{3} = 0.333 = 0.3$

Thus, (i)  $\rightarrow (q)$

(ii)  $\frac{1.00}{3.00} = 0.333 = 0.3$

Thus, (ii)  $\rightarrow (r)$

(iii)  $2 - 0.3 = 2 - 0 = 2$

Thus, (iii)  $\rightarrow (s)$

(iv)  $2 - 0.6 = 2 - 1 = 1$

Thus, (iv)  $\rightarrow (p)$

20. (i)  $\frac{g}{g} = \frac{g_1 + g_2 + g_3 + g_4 + g_5}{5}$   
 $= \frac{9.85 + 9.76 + 9.91 + 9.79 + 9.78}{5} = 9.82ms^{-2}$

Thus, (i)  $\rightarrow (q)$

(ii)  $\Delta g_1 = |g_1 - \bar{g}| = 0.032ms^{-2}$

$\Delta g_2 = |g_2 - \bar{g}| = 0.060ms^{-2}$

$\Delta g_3 = |g_3 - \bar{g}| = 0.090ms^{-2}$

$\Delta g_4 = |g_4 - \bar{g}| = 0.030ms^{-2}$

$\Delta g_5 = |g_5 - \bar{g}| = 0.040ms^{-2}$

$\therefore \Delta \bar{g} = \frac{\Delta g_1 + \Delta g_2 + \Delta g_3 + \Delta g_4 + \Delta g_5}{5} = 0.05ms^{-2}$

Thus, (ii)  $\rightarrow (p)$

(iii)  $Relative\ error = \frac{\Delta \bar{g}}{g} = \frac{0.05}{9.82} = 0.0051$

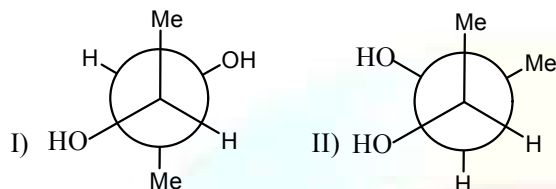
Thus, (iii)  $\rightarrow (s)$

(iv)  $Percentage\ error = \frac{\Delta \bar{g}}{g} \times 100 = 0.51$

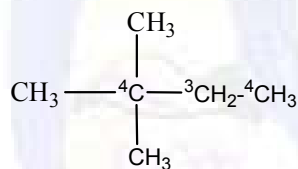
Thus, (iv)  $\rightarrow (r)$

**CHEMISTRY**

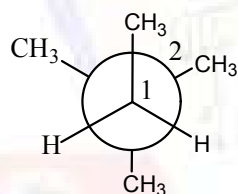
21. CONCEPTUAL  
 22. CONCEPTUAL  
 23. Geometrical isomers are diastereomers



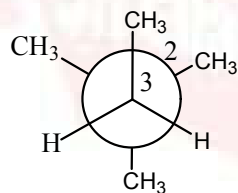
24.  
 25. As per CIP same priority groups on opposite gives E-isomer and same side Z-isomers  
 26. 1) Stable due to H-bonding 2) P is most stable due to H-bonding in it's gauche conformations.  
 27. CONCEPTUAL  
 28. Structural formula 2,2-dimethylbutane is



(I) Newman projection using  $C_1 - C_2$  bond



(II) Newman projection using  $C_3 - C_2$  bond



29. In stable conformation, larger group occupies equatorial position  
 30. Due to H-bonding, structure (A) is most stable  
 31. To exhibit geometric isomerism, compound must have unsaturation or compound must be cyclic with at least two different groups  
 32. Same priority groups are on opposite side

33.  $60^\circ$  and  $180^\circ$
34. Chair form is the most stable form
35. Allenes with even number of carbons are not planar & do not show G.I.
36.  $2^n = 2^3 = 8$
37. CONCEPTUAL
38. No of geometrical isomers  $= 2^n$
39. CONCEPTUAL
40. trans - compound has higher melting point than cis - compound. cis - compound is more polar and hence has higher boiling point

**MATHS**

41.  $f(n) > 0$   $2(f(x) - f(x)) = (f(x))^2$   $F(x) = \frac{(f(x))^2}{2} + f(x)$

$$\Rightarrow F'(x) = f(x)f'(x) + f'(x)$$

But  $f'(x) = f(x)$

$$\therefore f'(x) = \frac{f(x)}{1+f(x)} = 1 - \frac{1}{1+f(x)} > 0$$

$\therefore f$  is strictly increasing

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{f'(x)}{1} = \lim_{x \rightarrow \infty} \frac{f(x)}{1+f(x)} = 1$$

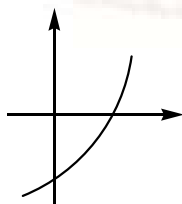
As  $\lim_{x \rightarrow \infty} f(x) \rightarrow \infty$

42.  $f: R \rightarrow R$  and  $f(x) = \sin x + ax + b$   $f'(x) = \cos x + a$

if  $a > 1$ , then  $f$  is increasing

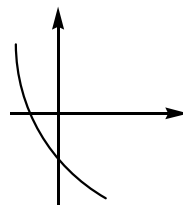
If  $a < -1$  then  $f$  is decreasing

If  $a > b, f(0) = b$  and  $b < 0 \Rightarrow f(0) < 0$



y shown in the diagram

$\therefore f(x) = 0$  has only the +ve root hence a is correct



If  $f(0) > 0$  again  $f$  has only one negative root

If  $a < -1, b < 0$  then  $f$  is decreasing

43.  $f(x), x(1-x)^2$

$$f'(x) = 2x^3(x-1)(3x-2)$$

$$f'(x) = 0 \Rightarrow x = 0, 2/3, 1$$

$$f''(x) = 2x^2(15x^2 - 20x + b)$$

$$f''(0) = 0 \text{ and } 15ax^2 - 20x + b = 0 \text{ has two roots}$$

44. CONCEPTUAL

45.  $X \in [0, 2]$  and  $f''(x) > 0$

$\Rightarrow f'(x)$  is increasing in  $[0, 2]$

$\Rightarrow$  LMVT is applicable let  $c_1 \in (0, 1)$  and  $c_2 \in (1, 2)$

$$\text{By LMVT, } f'(c_1) = \frac{f(1) - f(0)}{1 - 0}$$

$$f'(c_2) = \frac{f(2) - f(1)}{2 - 1}$$

$f'(x)$  is increasing and  $c_1 < c_2$

$$\Rightarrow f'(c_1) < f'(c_2)$$

$$\Rightarrow f(1) - f(0) < f(2) - f(1)$$

$$f(0) + f(2) > 2f(1)$$

Let  $c_1 \in (0, 2/3)$  and  $c_2 \in (2/3, 2)$

$$f'(1) < f'(2)$$

$$\Rightarrow \frac{f(2/3) - f(0)}{\frac{2}{3} - 0} < \frac{f(2) - f(2/3)}{2 - \frac{2}{3}}$$

$$\Rightarrow 2f(0) + f(2) > 3f(2/3)$$

46.  $h(x) = f(x) - (f(x))^2 + (f(x))^3$

$$h'(x) = (3(f(x))^2 - 2f(x) + 1)f'(x)$$

$$3(f(x))^2 - 2f(x) + 1 > 0, \forall x \in R$$

$$h'(x) > 0 \text{ if } f'(x) > 0$$

$\therefore h$  is increasing whenever  $f$  is increasing

$$h'(x) < 0 \text{ if } f'(x) < 0$$

$\therefore h$  is decreasing whenever  $f$  is decreasing

47. CONCEPTUAL

48. CONCEPTUAL

PASSAGE I  $f(x) = (\lambda^4 - 2\lambda^2)x + 3$  is  $x, \lambda \in R$

49.  $f'(x) = \lambda^4 - 2\lambda^2 - 3\sin x \geq 0$

$$-1 \leq \sin x \leq 1 \Rightarrow \lambda^4 - 2\lambda^2 - 3 \geq 0$$

$$\Rightarrow x \leq \sqrt{3} \text{ or } \lambda \geq \sqrt{3}$$

Least positive integer  $\lambda = 2$

$$\Rightarrow f'(x) = 16 - 8 - 3\sin x = 8 - 3\sin x > 0$$

$\Rightarrow f$  is increasing

50.  $f(\pi x - x^2) > f(\sin x)$

$$\Rightarrow \pi x - x^2 > \sin x$$

$$\Rightarrow x = 1, 2, 3 \text{ respectively}$$

$$\sin x = 1 + 2 + 3 = 6$$

PASSAGE II

$$f(x) = ax^3 + bx^2 + cx + 4$$

$$f'(-2/3) = 5/3, f''\left(-\frac{2}{3}\right) = 0$$

Degree of  $f'(x)$  is two  $f''(x)$  is one

$$\Rightarrow f'(x) = k(x + 2/3)^2 + c$$

$$\Rightarrow f'(-2/3) = c = 5/3$$

$$f'(x) = \frac{k}{2} \left( x + \frac{2}{3} \right)^2 + 5/3$$

$$f(x) = \frac{k}{2} \left( x + \frac{2}{3} \right)^3 + \frac{5x}{3} + b$$

$$\text{But } f\left(\frac{-2}{3}\right) = 0 - \frac{10}{9} + b$$

$$\text{And } y\left(\frac{-2}{3}\right) = \frac{5}{3}(-2/3) + \frac{100}{27}$$

$$\Rightarrow b = \frac{100}{27}$$

$$\text{But coefficient } x^3 = 1$$

$$\therefore f(x) = \left(x + \frac{2}{3}\right)^3 + \frac{5}{3}x + \frac{100}{27}$$

$$= x^3 + 2x^2 + 3x + 4$$

$$a=1, b=2, c=3$$

$$51. \quad a+b+c=6$$

$$52. \quad g = f^{-1} \text{ and } f(g(x)) = x$$

$$\frac{d}{dx}(g(x) \times (g(x))) = \frac{d}{dx}(g(x) \cdot x)$$

$$= x \cdot g'(x) + g(x)$$

$$\text{at } x = 4, \text{ the derivative}$$

$$= 4 \cdot g'(x) + g(x)$$

$$= 4 \cdot \frac{1}{3} + 0$$

$$\text{Hence } f^{-1}(4) = 0 = g(x) \text{ and } g'(x) = \frac{1}{f'(0)}$$

PASSAGE III

$$f(x) = \cos 2x + 2x\lambda^2 + (2\lambda + 1)(\lambda - 1)x^2, \lambda \in R$$

$$53. \quad \lambda = 1 \Rightarrow f(x) = \cos 2x + 2x$$

$$f'(x) \geq 0 \Rightarrow f \text{ is increasing}$$

$$\Rightarrow 3x^2 - 2x + 1 \leq x^2 - 2x + 9$$

$$\Rightarrow -2 < x < 2$$

$$\Rightarrow x = -1, 0, 1$$

54.  $f$  is increasing  $\forall x \in R \Rightarrow f'(x) = -2\sin 2x + 2\lambda^2 + (2\lambda + 1)(\lambda - 1)/2x > 0$

at  $\lambda = \frac{-1}{2}, f'(x) < 0$

$\therefore$  this is valid only at  $\lambda = 1$

55.  $f'(\sin x) < 0$  and  $f''(\sin x) > 0 \forall x \in [0, \pi/2]$

$$g(x) = f(\sin x) + f(\cos x)$$

$$g'(x) = f'(\sin x)\cos x + f'(\cos x)(-\sin x)$$

$$g''(x) = f''(\sin x) \cdot \cos^2 x - \sin x + (\sin x)$$

+      +      +      -

$$g''(x) > 0 \Rightarrow g''(x) \text{ is increasing}$$

56.  $g'\left(\frac{\pi}{4}\right) = 0 \Rightarrow g'(x) > 0, \text{ for } x \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

and  $g'(x) < 0 \text{ for } x \in \left(0, \frac{\pi}{4}\right)$

$g$  is increasing in  $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$g$  is decreasing in  $\left(0, \frac{\pi}{4}\right)$

57. CONCEPTUAL

58. CONCEPTUAL

59. CONCEPTUAL

60. CONCEPTUAL

# Success



## JEE ADVANCED WEEKEND TEST



### Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

**A right Choice for the Real Aspirant**

ICON CENTRAL OFFICE, MADHAPUR - HYD

**Sec: Jr. Super60-I**

**Time: 07:30 AM to 10:30 AM**

**Date: 28-10-17**

**Max.Marks:264**

### 2015\_PAPER-I

#### IMPORTANT INSTRUCTIONS:

- 1) This booklet is your Question Paper.
- 2) Use the Optical Response Sheet (ORS) provided separately for answering the questions
- 3) Blank spaces are provided within this booklet for rough work.
- 4) Write your name, roll number and sign in the space provided on the back cover of this booklet.
- 5) You are allowed to take away the Question Paper at the end of the examination.

#### OPTICAL RESPONSE SHEET:

- 6) Darken the appropriate bubbles on the ORS by applying sufficient pressure. This will leave an impression at the corresponding place on the Candidate's sheet.
- 7) The ORS will be collected by the invigilator at the end of the examination.
- 8) Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
- 9) Write your name, roll number and code of the examination center, and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else** on the ORS. Darken the appropriate bubble under each digit of your roll number.

#### DARKENING THE BUBBLES ON THE ORS

- 10) Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
- 11) Darken the bubble **COMPLETELY**.
- 12) The correct way of darkening a bubble is as:
- 13) The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
- 14) Darken the bubbles **ONLY** IF you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.

### 28-10-17\_Jr.Super60-I\_Jee Adv\_WAT-23\_Weekend Syllabus

**MATHS:** SYLLABUS COVERED FROM 23-10-17 TO 27-10-2017

**PHYSICS:** SYLLABUS COVERD FROM 23-10-17 TO 27-10-2017

**CHEMISTRY:** Optical isomerism, Inductive effect

## JEE-ADVANCE-2015-P1-Model

Time: 3:00 Hours

**IMPORTANT INSTRUCTIONS**

Max Marks: 264

**PHYSICS:**

| Section                  | Question Type                              | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|--------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 1 – 8)     | Questions with Integer Answer Type         | 4         | 0          | 8         | 32          |
| Sec – II(Q.N : 9 – 18)   | Questions with Multiple Correct Choice     | 4         | -2         | 10        | 40          |
| Sec – III(Q.N : 19 – 20) | Matrix Matching<br>(+2/-1 for every match) | 8         | -4         | 2         | 16          |
| <b>Total</b>             |                                            |           |            | <b>20</b> | <b>88</b>   |

**CHEMISTRY:**

| Section                  | Question Type                              | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|--------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 21 – 28)   | Questions with Integer Answer Type         | 4         | 0          | 8         | 32          |
| Sec – II(Q.N : 29 – 38)  | Questions with Multiple Correct Choice     | 4         | -2         | 10        | 40          |
| Sec – III(Q.N : 39 – 40) | Matrix Matching<br>(+2/-1 for every match) | 8         | -4         | 2         | 16          |
| <b>Total</b>             |                                            |           |            | <b>20</b> | <b>88</b>   |

**MATHEMATICS:**

| Section                  | Question Type                              | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|--------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 41 – 48)   | Questions with Integer Answer Type         | 4         | 0          | 8         | 32          |
| Sec – II(Q.N : 49 – 58)  | Questions with Multiple Correct Choice     | 4         | -2         | 10        | 40          |
| Sec – III(Q.N : 59 – 60) | Matrix Matching<br>(+2/-1 for every match) | 8         | -4         | 2         | 16          |
| <b>Total</b>             |                                            |           |            | <b>20</b> | <b>88</b>   |

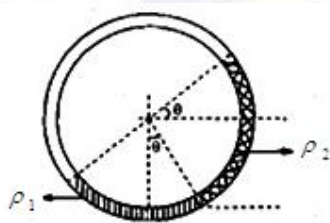
**Sec: Jr. Super60-I****space for rough work****Page 2**

**SECTION – I**  
**(SINGLE INTEGER ANSWER TYPE)**

This section contains 8 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

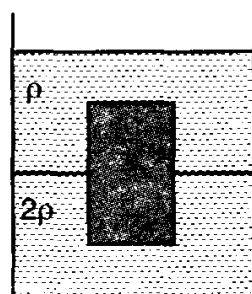
**Marking scheme +4 for correct answer, 0 if not attempted and 0 in all other cases.**

1. A circular tube of uniform cross-section is filled with two liquids of densities  $\rho_1$  and  $\rho_2$  such that half of each liquid occupies a quarter of volume of the tube. If the line joining the free surface of the liquids makes an angle  $\theta$  with horizontal, then  $\tan \theta = \frac{\rho_1 - K_1 \rho_2}{K_2 \rho_1 + \rho_2}$ , where  $K_1$  and  $K_2$  is integer. Find the numerical value  $K_1 / K_2$ ?



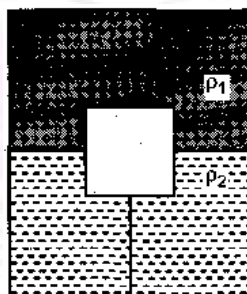
2. A wooden stick of length  $L$ , radius  $R$  and density  $\rho$  has a small metal piece of mass  $m$  (of negligible volume) attached to its one end. The minimum value for the mass  $m$  (in terms of given parameters) that would make the stick float vertically in equilibrium in a liquid of density  $\sigma$  is  $\pi K R^2 L \rho \left[ \sqrt{\frac{\sigma}{\rho}} - 1 \right]$ . Find the value of  $K$ .

3. A cylinder of uniform cross-section is immersed into two liquids as shown in Fig. Equal volume of cylinder is submerged in both the liquids and density of cylinder is  $\rho$ . The magnitude of acceleration of the cylinder at the shown instant is  $g/K$ . Where equal volume of cylinder in two liquids. Find the value of  $K$ .



4. A liquid is kept in a cylindrical vessel which is rotated about its axis. The liquid rises at the side. If the radius of the vessel is  $0.05\text{m}$  and the speed of rotation is  $2\text{ rev/sec}$ . Find the difference in the height of the liquid at the centre of the vessel and its sides (in cm). ( $g = 9.86\text{m/s}^2$ )

5. A spherical tank of 1.2 m radius is half filled with oil of relative density 0.8. If the tank is given a horizontal acceleration of  $10 \text{ m/s}^2$ , the maximum pressure at any point on the tank is  $\sqrt{2}(100 - X)100$  Pascal. Find the value of X. ( $g = 10 \text{ m/s}^2$ )
6. A vessel contains two immiscible liquids of density  $\rho_1 = 1000 \text{ kg/m}^3$  and  $\rho_2 = 1500 \text{ kg/m}^3$ . A solid block of volume  $V = 10^{-3} \text{ m}^3$  and density  $d = 800 \text{ kg/m}^3$  is tied to one end of a string and the other is tied to the bottom of the vessel as shown in fig. The block is immersed with  $2/5^{\text{th}}$  of its volume in the liquid of higher density and  $3/5^{\text{th}}$  in the liquid of lower density. The entire system is kept in an elevator which is moving upwards with an acceleration of  $a = g/2$ . Find the tension (in newtons) in the string. ( $g = 10 \text{ m/s}^2$ )



7. A sphere is just immersed in a liquid. Find the ratio of hydrostatic force acting on bottom half and top half of the sphere.
8. The vertical limbs of a U shaped tube are filled with a liquid of density  $\rho$  upto a height  $h$  on each side. The horizontal portion of the U tube having length  $2h$  contains a liquid of density  $2\rho$ . The U tube is moved horizontally with an accelerator  $g/2$  parallel to the horizontal arm. The difference in heights in liquid levels in the two vertical limbs, at steady state is  $\frac{nh}{7}$ . Find  $n$ .

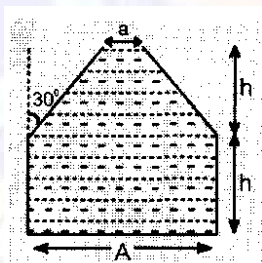
**SECTION – II****(ONE OR MORE CORRECT ANSWER TYPE)**

This section contains 10 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.**

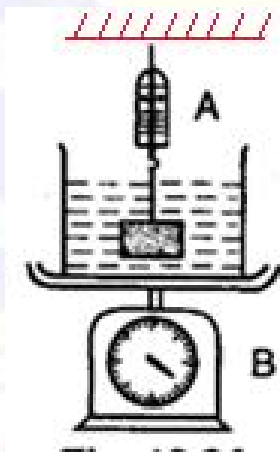
9. A solid completely immersed in a liquid. The force exerted by the liquid on the solid will
- A) Increase if it is pushed deeper inside the liquid
  - B) Change if its orientation is changed
  - C) Decrease if it is taken partially out of the liquid
  - D) Be in the vertically upward direction.

10. The vessel shown in the figure has two sections. The lower part is a rectangular vessel with area of cross-section  $A$  and height  $h$ . The upper part is a conical vessel of height  $h$  with base area ' $A$ ' and top area ' $a$ ' and the walls of the vessel are inclined at an angle  $30^\circ$  with the vertical. A liquid of density  $\rho$  fills both the sections upto a height  $2h$ . Neglecting atmospheric pressure ( $F$ = force exerted by the liquid on the base of the vessel,  $w$  weight of liquid)



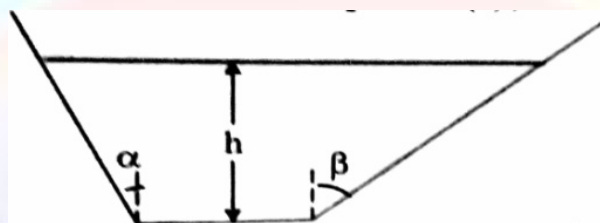
- A) the force  $F$  exerted by the liquid on the base of the vessel is  $2h\rho g \frac{(A+a)}{2}$
- B) the pressure  $P$  at the base of the vessel is  $2h\rho g \frac{A}{a}$
- C) the weight of the liquid  $W$  is greater than the force exerted by the liquid on the base
- D) the walls of the vessel exert a downward force  $(F-W)$  on the liquid

11. The spring balance A reads 2 kg with a block suspended from it. A balance B reads 5 kg when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in Fig. In this situation:



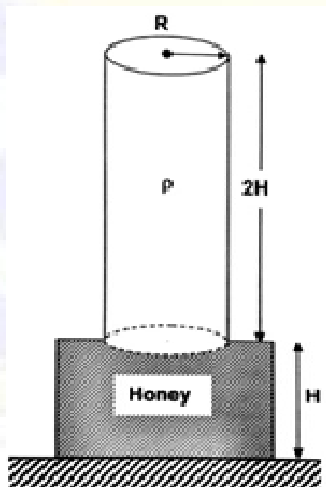
- A) the balance A will read more than 2 kg
- B) the balance B will read more than 5 kg
- C) the balance A will read less than 2 kg and B will read more than 5 kg
- D) the balances A and B will read 2 kg and 5 kg respectively

12. A liquid is filled up to height  $h$  in a vessel, as shown. Find correct (s);



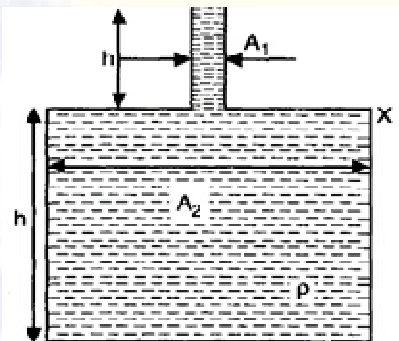
- A) If  $\alpha = \beta$ , horizontal component of forces on left and right side of inclined faces will be equal and opposite.
- B) If  $\alpha \neq \beta$ , horizontal component of forces on left and right side of inclined faces will be equal and opposite.
- C) If  $A$  is the area of the base of the vessel, then force exerted by liquid on walls of the vessel is greater than  $(P_{atm} + \rho gh)A$
- D) As above, the force exerted by liquid on walls is equal to  $(P_{atm} + \rho gh)A$
13. A bottle is kept on the ground as shown in the figure. The bottle can be modeled as having two cylindrical zones. The lower zone of the bottle has a cross-sectional radius of  $R\sqrt{2}$  and is filled with honey of density  $2\rho$ . The upper zone of the bottle is filled with the water of density  $\rho$  and has a cross-sectional radius  $R$ . The height of the lower

zone is  $H$  while that of the upper zone is  $2H$ . If now the honey and the water parts are mixed together to form a homogeneous solution: (Assume that total volume does not change)



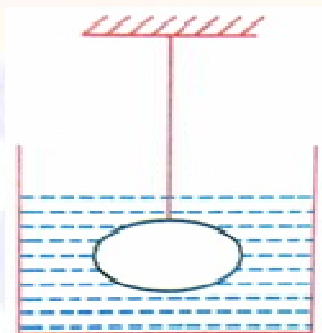
- A) The pressure inside the bottle at the base will remain unaltered
- B) The normal reaction on the bottle from the ground will remain unaltered
- C) The pressure inside the bottle at the base will increase by an amount  $(1/2) \rho g H$
- D) The pressure inside the bottle at the base will decrease by an amount  $(1/4) \rho g H$

14. The vessel shown in the Fig. has two sections of areas of cross-section  $A_1$  and  $A_2$ . A liquid of density  $\rho$  fills both the sections, up to a height  $h$  in each. Neglect atmospheric pressure.



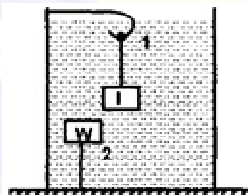
- A) the pressure at the base of the vessel is  $2h\rho g$   
 B) the force exerted by the liquid on the base of the vessel is  $2h\rho g A_2$   
 C) the weight of the liquid is  $< 2h\rho g A_2$   
 D) the walls of the vessel at the level  $X$  exert a downward force  $h\rho g(A_2 - A_1)$  on the liquid
15. A solid sphere of mass  $m$ , is suspended by means of a string in a liquid as shown. The string has some tension. Magnitudes of net force due to liquid on upper hemisphere

and that on lower hemisphere are  $F_A$  and  $F_B$  respectively. Which of the following is/are true.



- A) Density of material of the sphere is greater than density of liquid
  - B) Difference of  $F_B$  and  $F_A$  is dependent of atmospheric pressure
  - C)  $F_B - F_A = mg$
  - D)  $F_B - F_A < mg$
16. When a body is weighed in a liquid, the loss in its weight is equal to:
- A) weight of liquid displaced by the body
  - B) weight of water displaced by the body
  - C) the difference in weights of body in air and liquid
  - D) the up thrust of liquid on the body

17. An iron block and a wooden block are positioned in a vessel containing water as shown in the figure. The iron block (I) hangs from a massless string with a rigid support from the top while the wooden block floats being tied to the bottom through a massless string. If now the vessel starts accelerating upwards



- A) tension in the string 1 will increase.  
B) tension in both the strings will increase.  
C) tension in both the strings will decrease.  
D) tension in the string 2 will remain constant.
18. A piece of ice is floating in a liquid( water soluble liquid) in a container. What happens to the level of liquid when all ice melts?
- A) Level remains same if liquid is water  
B) Level will fall if liquid is water  
C) Level will rise if liquid is denser than water  
D) Level will rise if liquid is lighter than water

**SECTION - III****(MATRIX MATCHING ANSWER TYPE)**

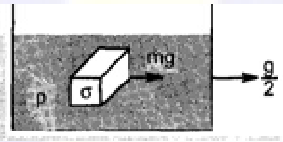
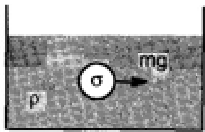
This section contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and four statements (P, Q, R and S) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question darken the bubbles corresponding to Q and R in the OMR sheet. For each correct matching **will be awarded +2 marks ONLY and 0 if not attempted and -1 in all other cases.**

19. Match the following columns

Bucket A contains only water, an identical bucket B contains water, but also contains a solid object in the water. Match the Column-I with Column-II.

|                                                                                             |                                                                 |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| (A) The object floats in bucket B, and the buckets have the same water level.               | (P) Bucket A has more weight.                                   |
| (B) The object floats in bucket B and the buckets have the same volume of water.            | (Q) Both buckets have the same weight.                          |
| (C) The object sinks completely in buckets B, and the buckets have the same water level.    | (R) Bucket B has more weight.                                   |
| (D) The object sinks completely in bucket B, and the buckets have the same volume of water. | (S) The answer cannot be determined from the information given. |

20. Match the following columns ( liquid is non viscous)

|                                                                                                                                                                                                                                                                                                                                                                                                 |                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <p>(A) A cube of mass <math>m</math> and density <math>\sigma</math> is pulled by a force <math>\vec{F}_x = mg\hat{i}</math> in an accelerating liquid of density <math>\rho</math>. The value of <math>\frac{a_x}{a_y}</math> is</p> <p>(<math>\sigma &gt; \rho, a_x, a_y</math> with respect to ground)</p>  | <p>(P) <math>\frac{\rho}{\sigma}</math></p>          |
| <p>(B) The sphere of mass <math>m</math> is pulled horizontally by a force <math>F_x = mg</math> in a non accelerating liquid.</p> <p>The value of <math>\frac{a_x}{a_y}</math> is (<math>\sigma &gt; \rho</math>)</p>                                                                                         | <p>(Q) 1 : 1</p>                                     |
| <p>(C) A solid cube of side <math>X</math> is placed in a tube of rotating liquid such that its closer surface is at a distance of <math>X</math> from the</p>                                                                                                                                                                                                                                  | <p>(R) <math>\frac{\sigma}{\sigma - \rho}</math></p> |